RAPID-CYCLE PSA

INSTALLATION, OPERATION AND MAINTENANCE MANUAL

2’’-14” VESSEL DIAMETER “G2”
16”-20” VESSEL DIAMETER “G3”
24’’-60” VESSEL DIAMETER “G4”

Customer: Inentec
Model Number: PSA3H-G2-330A-480
Xebec Job Number: 30784

Xebec Document Number: Revision: 0 Date of Issue: March 2018

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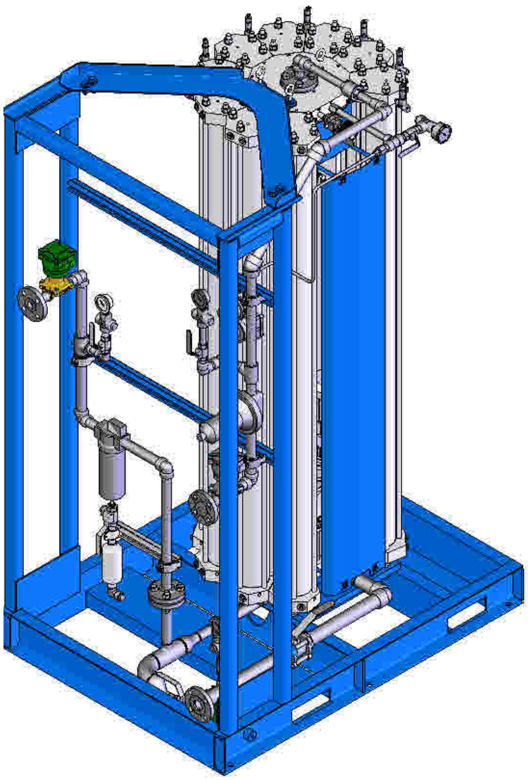
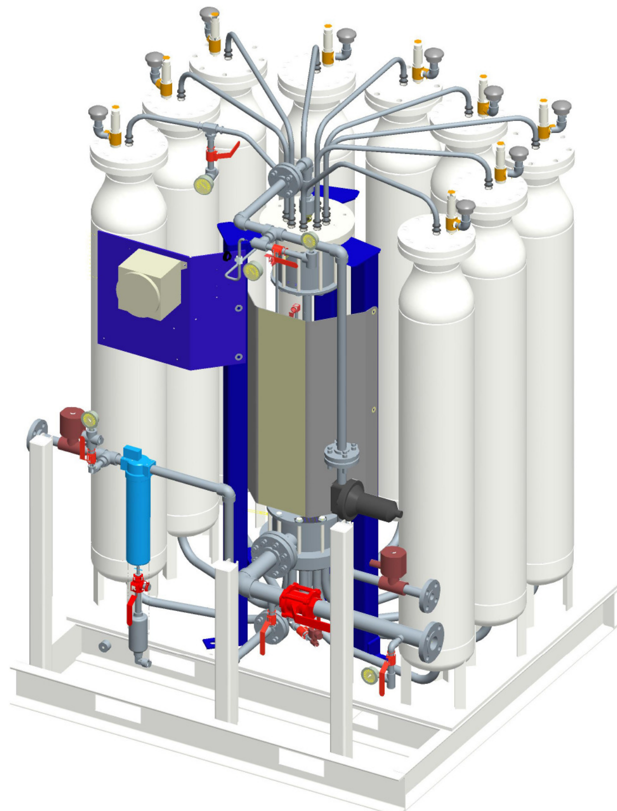
Your PSA

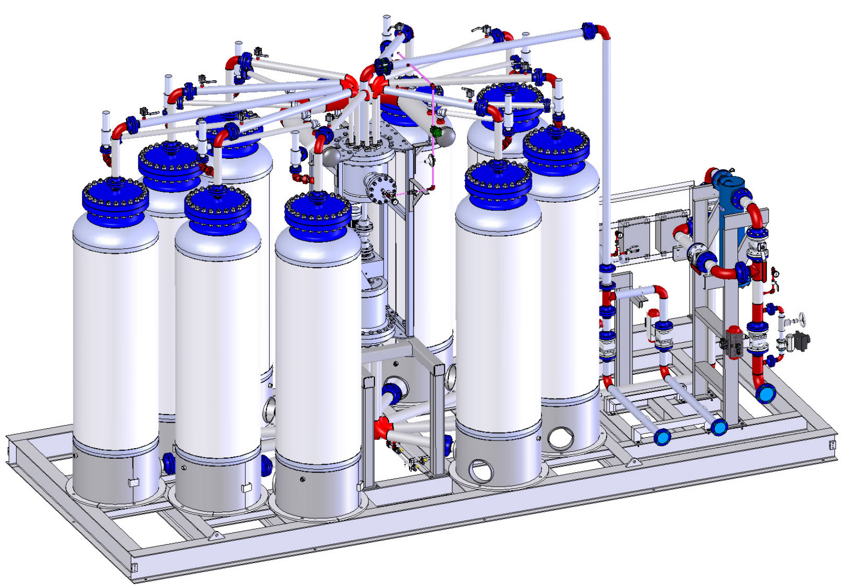
Xebec offers various models of PSA systems. The instructions in this manual apply equally to all models unless indicated otherwise. The 3 most common configurations are illustrated below.



G2 (bed diameter 6" or less)



G2 (bed diameter 8" to 14")



G3 or G4 (bed diameter 16" or more)

Important Message

Thank you and congratulations on your purchase of a Xebec™ Rapid-Cycle PSA Gas Separation System. This piece of equipment has been fabricated to the highest standards with each component carefully checked for quality and safety. By reading this manual prior to operation, it will ensure the highest level of performance and extended life of the equipment.

ATTENTION:

This manual should be thoroughly read and understood before moving, installing or operating the Xebec Rapid-Cycle PSA. Failure to do so can result in serious personal injury and permanent damage to the equipment.

The PSA process is extremely dependant on stable operating conditions and process gas that closely matches the composition specified during the design of your PSA. Any deviation from the specified operating conditions may result in less than optimal performance and possible hazards. Of particular importance for the life of the machine, is to not allow liquids to enter the PSA. Proper care should be taken to ensure moisture in the feed gas is removed through upstream knockout systems.

All customer enquiries should be directed to the Xebec Adsorption Product Support Department by:

Phone:

1-800-GO-XEBEC

(1-800-469-3232)

Email:

service@xebecinc.com

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1-450-979-7869

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c/o Product Support Department

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Preface

This manual will introduce the PSA system, outline safety considerations, define recommended facility requirements and describe installation, operation and maintenance procedures.

This manual contains information for:

- **Warranty**
- **Safety considerations**
- **Product overview**
- **Xebec & Customer requirements**
- **Nomenclature**
- **Shipping, receiving & storage**
- **Installation**
- **Commissioning**
- **Operation**
- **Inspection & Maintenance**
- **Troubleshooting**
- **Spare parts recommendations**
- **Equipment documentation**

THIS DOCUMENT IS INTENDED FOR OPERATING PERSONNEL FAMILIAR WITH THE PRINCIPLES OF PRESSURE SWING ADSORPTION (PSA), AWARE OF THE SAFE OPERATING LIMITS OF THE PROCESS, AND EXPERIENCED IN THE HANDLING OF COMPRESSED FLAMMABLE GASES AND INDUSTRIAL MACHINERY.

Xebec would like to thank its customers for their feedback over the years and we encourage you to continue. We hope this IOM manual provides a better understanding of the PSA process and our equipment.

Warranty

Standard Agreement

Xebec warrants the Goods against defects in workmanship or materials under normal use for twelve months from date of commissioning or eighteen months from date of shipment, whichever is earlier. All replacements or repairs necessitated by inadequate preventive maintenance, or by normal wear and usage, or by fault of the Purchaser, or by unsuitable power sources or by attack or deterioration under unsuitable environmental conditions, or by abuse, accident, alteration, misuse, improper installation, modification, repair, storage or handling, or any other cause not the fault of Xebec ("Non-warranty Repairs") are not covered by this Limited Warranty, and shall be at Purchaser's expense. All costs of dismantling, reinstallation and freight and the time and expenses of Xebec's personnel for site travel and non-warranty repairs shall be borne by Purchaser unless accepted in writing by Xebec. Xebec's sole responsibility shall be, at its option, to repair or replace and install free of charge any parts or equipment found to be defective, up to a maximum of 50% of the purchase price, or refund the full purchase price paid by the Purchaser. Goods repaired and parts replaced during the warranty period shall be in warranty for the remainder of the original warranty period or ninety (90) days, whichever is longer. THE FOREGOING WARRANTIES AND REMEDIES ARE EXCLUSIVE AND EXPRESSLY IN LIEU OF ALL OTHER WARRANTIES EXPRESSED OR IMPLIED; AND XEBEC MAKES NO REPRESENTATION OR WARRANTY, EXPRESS OR IMPLIED, AS TO THE FITNESS OF A PARTICULAR PURPOSE OR MERCHANTABILITY OF THE GOODS.

XEBEC SHALL NOT BE LIABLE FOR DAMAGES CAUSED BY DELAY OR PERFORMANCE. THE SOLE ANDEXCLUSIVE REMEDY FOR BREACH OF WARRANTY HEREUNDER SHALL BE LIMITED TO REPAIR, CORRECTION OR REPLACEMENT UNDER THE LIMITED WARRANTY CLAUSE ABOVE. IN NO EVENT, REGARDLESS OF THE FORM OF THE CLAIM OR CAUSE OF ACTION (WHETHER BASED IN CONTRACT, INFRINGEMENT, NEGLIGENCE, STRICT LIABILITY, OTHER TORT OR OTHERWISE), SHALL XEBEC'S LIABIL ITY TO PURCHASER EXTEND TO INCLUDE INCIDENTAL, CONSEQUENTIAL OR PUNITIVE DAMAGES. THE TERM "CONSEQUENTIAL DAMAGES" SHALL INCLUDE, BUT NOT BE LIMITED TO, LOSS OF ANTICIPATED PROFITS, LOSS OF USE, LOSS OF REVENUE, AND COST OF CAPITAL.

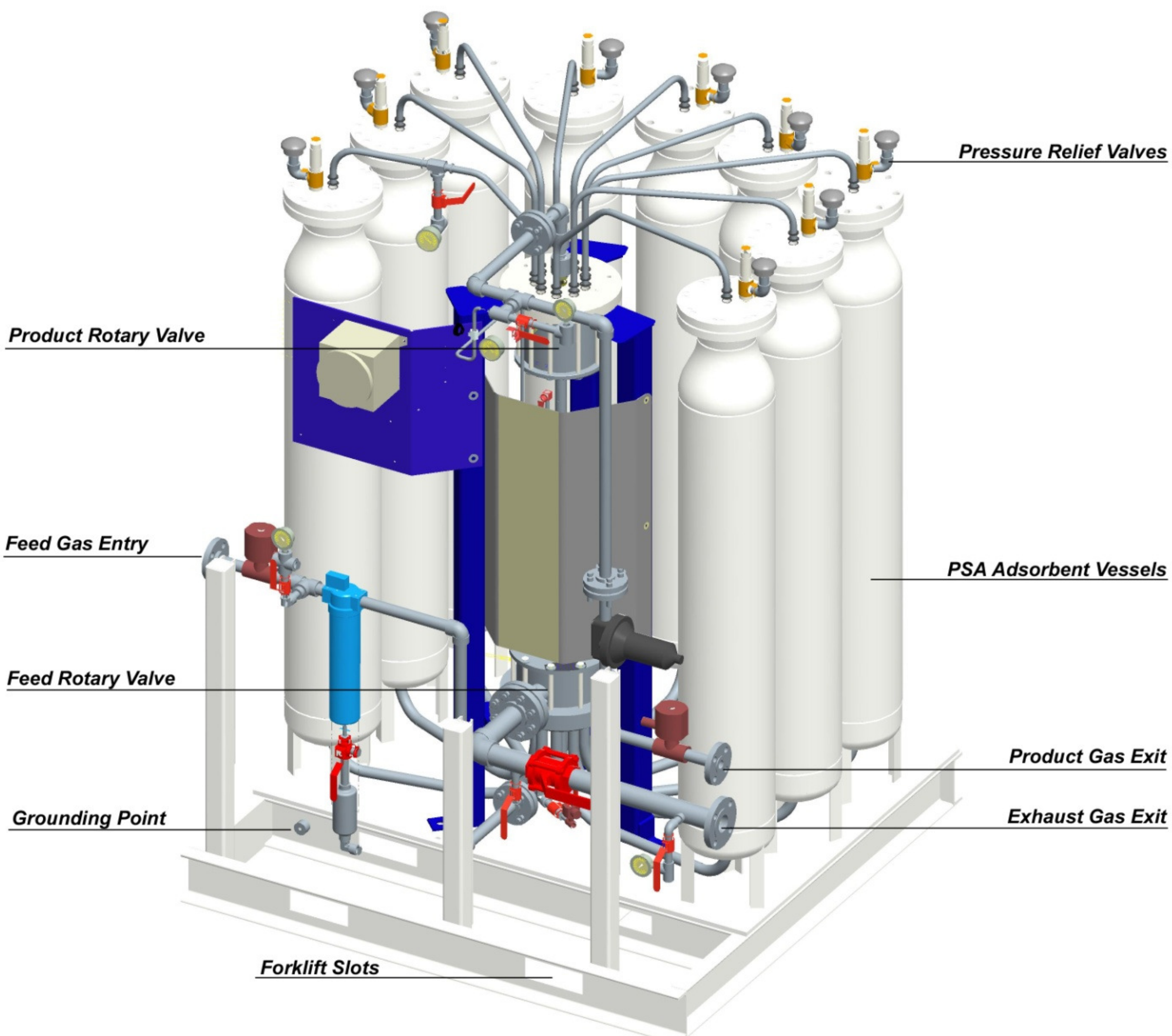
1.0 Safety Considerations

1.1 Included equipment

Please note that the **Rapid-Cycle PSA** will contain equipment that will vary for each installation and application. The references and information made in this manual are meant to cover the included equipment for a full scope **Rapid-Cycle PSA**.

For the equipment that is included for your application, please refer to the Piping and Instrumentation Diagram (P&ID), Application Specification, electrical wiring diagrams and General Assembly drawing that are included in your PSA documentation package.

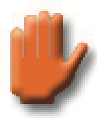
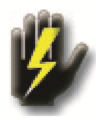
The P&ID and Application Specification will specifically reference installation conditions (equipment, safe location of drains, vents, etc.) that the Owner/Operator must ensure are present to ensure proper and safe operation of the **Rapid-Cycle PSA**. The diagram below shows equipment on a full scope **Rapid-Cycle PSA**.



Major components of a Rapid-Cycle PSA

1.2 Safety Conventions

In this Manual the following pictograms are used to alert the reader to items of particular importance.

	**REMARK:** Gives you tips or suggestions to perform certain tasks more easily.
	**ATTENTION:** Remark with additional information; draws your attention to potential problems.
	**ELECTRICAL WARNING:** You may contact live electrical parts when not applying the procedures carefully.
	**WARNING:** DO NOT. You can damage the machine when not applying the procedures carefully.
	**DANGER WARNING:** You can put yourself or others at risk when not applying the procedures carefully.

Xebec has endeavoured to identify safety hazards that would lead to harm and conditions that would compromise operability of the PSA unit. However, warnings provided in this Manual are not intended to be exhaustive.

In all situations, good safe working practices, the use of appropriate personal protective equipment and training of operating and maintenance personnel must be followed in accordance with industry practice, site safety policy, and local codes and standards and laws.

Responsibility for personal safety and equipment protection lies with the Owner/Operator of the PSA and its associated equipment. Xebec accepts no responsibility, nor damage to the equipment through improper operation, negligence, or operator error.

1.3 Safety Measures Implemented by Xebec

The Rapid-Cycle PSA comes equipped with several safety and separation performance degradation protection features:

ATTENTION:

It is highly recommended that the Owner/Operator implement signal loss detection that results in an **E-STOP** and isolation of the PSA if the motor rotation stops. This is a safety measure for the adsorbent media.

Mechanical safety devices:

Pressure relief devices have been installed on all surge tanks and on adsorbent beds which are designed for overpressure events including shut-in adsorbed gas with external fire.

Emergency stop:

Control logic for an Emergency Stop is present, but an actual E-Stop button(s) is recommended if not supplied by Xebec. This E-stop should be readily accessible in the event of an emergency. We recommend placing one E-stop at a safe distance from the PSA in the event of a fire and one E-stop on or very near the PSA skid.

Isolation valves:

Automatic isolation valves are installed on the feed and product line to preserve the integrity of the PSA adsorbent and valves during emergency shut down, unless your order has called for manual isolation valves. These should close immediately on a loss of run signal or E-Stop being pressed. In the case of manual shut down, an operator will have to be on hand to shut off the manual valves.

Grounding Points:

Grounding locations have been provided but need to be connected upon installation. By grounding the PSA skid you ensure no static charges are release during operation or maintenance.

Drains:

A feed filter and drain assembly as well as a low point drainare provided on the feed line to ensure no liquid enters the PSA. An exhaust drain point is provided to ensure no liquid enters from a possible reverse flow. These drains should be manually opened, or if so equipped, programmed to be opened every 1hr for 3 seconds.

1.4 Pre-Startup Safety Considerations

1.4.1 Intended Use

Each PSA is designed for a specific application. Although the machined parts and components are generally the same for each PSA, the adsorbent and process speeds are highly dependent on the feed gas composition. For this reason, the process gas must closely match the specifications either on the APPLICATION SPECIFICATION SHEET or the Design Basis that was created during the design stage. If you require a copy of this, please contact your Xebec representative. For the PSA to perform as expected and produce the desired recovery and purity, all operating conditions must be met.

It is very important that the feed gas have no oxygen (air) in the feed stream, either through the supply source or leaks in the piping. This may cause O_2 separation and concentration in the PSA valve or vessels which can, in rare occasions, spontaneously combust.

DANGER WARNING:

For applications involving flammable gases such as hydrogen (H_2) or methane (CH_4), the area around the equipment is considered a hazardous area. Any potential **source of ignition** (e.g. smoking, static electricity, arcing tools, welding, etc.) shall be avoided. **Local safe practice regulations and proceduresshall be followed.**

Leaks resulting from failure to observe these precautions could result in noxious exposure, fire or explosion and may result in personal injury or equipment damage.

1.4.2 Intended Users

The PSA is intended to be used by trained or experienced operators only. Once the commissioning process is complete, under the supervision of a Xebec representative, the customer should feel confident in operating the equipment unsupervised. A sound understanding of the operating principals and potential hazards of operation should be demonstrated.

If additional training is required, Xebec can provide onsite courses for PSA operation, maintenance, and theory behind PSA. Covering topics from chemical adsorption to valve maintenance, we can certify your company to start and maintain future PSA installations.

Additional information can be requested to service@xebecinc.com

The **Rapid-Cycle PSA** gas separation system may contain **flammable and non-flammable gases under pressure**. Operators and other personnel shall follow the procedures as laid out in this manual, augmented by the Owner/Operator's safe practice and procedures, as well as applicable local laws and regulations in the safe operation and maintenance of this equipment.

ATTENTION:

Commissioning and Tuning are to be performed by **Xebec technicians or other Xebec trained and certified personnel** only. Failure to abide by this can result in equipment damage, personal injury and loss of warranty.

1.4.3 Limits of the PSA

The PSA should be operated within its limits like any other piece of equipment. The Application Specification/Design Basis should be consulted for limitations.

DANGER WARNING:

Operating the PSA unit outside of prescribed procedures or the limits of the machine may lead to performance degradation or hazardous conditions for personnel and equipment.

Speed: The speed of the motor rotation has a minimum value that can be achieved based on the motor manufacturer. This varies according to the gear box arrangement and should be discussed with Xebec staff during commissioning.

Flow Rate: The PSA can be thought of as a simple orifice. It is only capable of handling up to a certain flow before it will "choke". However, permanent damage can be caused by excessive flow rates through the vessels. The adsorbent will decay due to attrition (vibrating) and the PSA performance will suffer. In addition to this, the dust from the broken adsorbent can enter the valves and cause premature wear.

Purity: The purity of the product gas is highly dependent on the rotation (cycle) speed of the PSA valves and the feed composition. The speed will be adjusted accordingly during commissioning and composition should not vary from the original design.

Pressure: The designed operating pressure of the PSA is indicated on your Application Specification/ Design Basis Sheet. Operation should not exceed this level. Pressure safety relief valves have been installed on the top of the vessels and will vent if pressure is excessive.

1.4.4 Personal Protective Equipment (PPE)

Standard PPE for working around machinery and high pressure gas should be used. This includes, but is not limited to:

- Safety glasses
- Hard hat
- Ear protection
- Safety shoes
- Fire retardant coveralls (ex. Nomex)
- Personal gas monitor

1.4.5 Pressurized Gas

The Xebec PSAs operate with different gases, hydrogen (H_2) and methane (CH_4) being the two most common. Both of these gases are flammable and great care should be taken while working around them. No smoking, open flame, grinding or welding nearby.

Hydrogen is combustible at levels as low as 4% in air while methane is combustible between 5-15%. In addition to these two primary gasses, high levels of CO, CO_2 and SO_x can be present.

1.5 Design and Installation

- For those systems supplied without pressure relief at the request of the Owner/Operator, the Owner/Operator shall ensure adequate relief of system overpressure as required for safe operation and to meet local pressure equipment regulations.

- For those systems employed in the processing of flammable or noxious gases, it is recommended that the Owner/Operator consider on-site **gas leak** and **fire detection and extinguishing equipment** that will shut down the system and render it safe.

- To limit risk of propagating an internal fire should it occur within the equipment, it is recommended that **flame arrestors** be installed within the piping system upstream and downstream of the PSA.

- If a **fire** does occur, all applicable laws and regulations shall be followed for the inspection and testing of vessels, piping, electrical systems, etc. before returning the equipment to operation.
- The PSA system is not to be commissioned or otherwise put into operation before verifying the correct operation of all **alarms, trips** and **interlocks**.

- Should the Owner/Operator choose to connect any additional equipment to the **Rapid-Cycle PSA**, the Owner/Operator must ensure that its design pressure is equal to or greater than the design pressure shown in the Application Specification.

- Extend all drain and vent lines to a **safe location** allowing for the periodic expulsion of system gases at high pressure without harm to personnel or equipment. Installation of **warning signs** for high pressure and flammable gas are recommended.

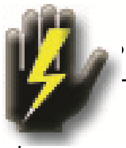
- <u>Pressure Safety Valve</u> (PSV) warnings:
  - Ensure that there are **no obstructions** downstream of any Pressure Safety Devices that would impede the safe operation and system pressure relief through these devices.
  - Inadequate venting may cause backpressure in the vent line and cause the PSV to leak at the spring chamber. **The PSVs are not designed to operate with pressure downstream of the device. Downstream pressure will potentially cause leakage through the spring chamber of the PSV.**
  - Ensure that the vent line is tied into a vent header that creates **no back pressure**.

- Ensure that **ventilation** of the area is sufficient to ensure concentration of flammable gases cannot occur.

- The centre of gravity of the unit may be slightly off-centre. Please take this into consideration when **lifting** the unit.

- The **Rapid-Cycle PSA** is supplied pressurized with dry inert gas. Storage of the system within an enclosed space may lead to build-up of asphyxiant gases. Ensure adequate ventilation.

- The **Rapid-Cycle PSA** is not designed for service in environments where high vibration can occur. Should high vibration from nearby equipment be present, the Owner/Operator should include the appropriate vibration-dampening connections and other mechanical supports as needed to minimize vibration at the **Rapid-Cycle PSA**.

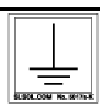


ELECTRICAL WARNING:

Ensure that adequate **electrical grounding** of the PSA system is provided before start up in accordance with local regulations and using a dedicated grounding boss.

1.6 Warning Labels

There are many labels found on the PSA skid and its associated control panel. These include but are not limited to:

Sign	Description	Location on the equipment
	WARNING: Electrical hazard. Access only by certified personnel.	Motor junction box. Junction Box JB1 and JB2.
	WARNING: "HOT"	On the insulation of the heat tracing if equipped
EXHAUST	PIPE MARKER "EXHAUST" Directional Flow Arrow Tape	On the Exhaust Line, at Battery Limits
FEED	PIPE MARKER "FEED" Directional Flow Arrow Tape	On the Feed Line, at Battery Limits
HYDROGEN*	PIPE MARKER "HYDROGEN" Directional Flow Arrow Tape	On the Product Line, at Battery Limits
	MARKER "GROUND"	At the 3 grounding bosses

*The sign on the product line will change according to the chemical composition of the product (hydrogen, methane, etc.)

2.0 Tool & Disposables List

The following is a basic list of tools and disposable items that are required for proper installation, commissioning and operation of the standard PSA models. This list is not exhaustive: additional specialty items will be required for larger PSAs and for certain activities including maintenance, commissioning and troubleshooting.

2.1 Installation

Installation		
Item	**Size / Type**	**Purpose**
Fork Lift	9500 kg (21000lb) lift	Depending on the installation, a forklift may be required to move and position the PSA skid
Crane	9500 kg (21000lb) lift	Depending on the installation, a crane may be required to move and position the PSA skid
Adjustable Wrenches	1/2"-1.5"	All bolts and fittings should be checked after shipping
Gaskets	150# and 300#	Tie-in points
Teflon Tape	Standard-yellow	Various fittings may require sealing
Thread Sealant	Loctite 757	Various fittings may require sealing
Multi-meter	Standard	Electrical checks of all components must be completed
Multi-bit Screwdriver	phillips, flat	

2.2 Commissioning

Commissioning		
Item	**Size**	**Purpose**
Tuning Valve Tools	n/a	Custom tools used for removing and adjusting internal valves
DAQ	n/a	Data collection system
Adjustable Wrenches	1/2"-1.5"	All bolts and fittings should be checked after shipping
Gaskets	150# and 300#	Tie-in points
Teflon Tape	Standard-yellow	Various fittings may require sealing
Thread Sealant	Loctite 757	Various fittings may require sealing
Pressure Transmitters	(-1 to +35 bar) -14 to 500 psi	To be installed during commissioning and removed after
Allan Keys	Imperial	Removal of bean plate to access tuning valves
Gas Monitor	n/a	Personal gas monitor for safety
Inert Gas	N2	Bottled inert gas for purging the system
Multi-meter	n/a	Electrical checks of all components must be completed
Multi-bit Screwdriver	phillips, flat	
Portable Gas analyzer (GEM or other brand)	n/a	Take gas sample at different point (inlet, product, exhaust)

2.3 Operation

Operation		
Item	**Size**	**Purpose**
Gas Analyzer	n/a	Analyzer for monitoring gas composition on feed, product and exhaust
Flow Analyzer	n/a	Feed and product flow measurements
Spare Parts	n/a	A recommended spare parts list will be provided

3.0 Quickstart Guide

3.1 Starting for the first time

Starting your Xebec PSA for the first time since manufacturing should be done only under the supervision of a Xebec representative. This will ensure proper operation, allow a chance to answer any questions you may have, and guarantee the equipment for the stated warranty period. Scheduling of this supervision can be done by emailing service@xebecinc.com.

We strongly recommend supervision to ensure that all equipment on the PSA skid has arrived in working order and that all upstream/downstream components are in place. Once the PSA is running, the service personnel will optimize the internal settings and speed for the best performance possible and train your plant operators on proper operation.

3.2 Startup after long term shutdown

A long term shutdown is defined as a PSA system stop of greater than 24 hours where the system was depressurized of all process gas and/or padded with 0.3 to 0.7 bar (5-10PSI) of inert gas (this is the same condition during the first start up of the PSA). This shutdown should have closed the feed and product isolation valves while allowing the motor to rotate until all internal gas and pressure was vented through the exhaust line. The exhaust line should then have been closed.

To start the PSA:

1. Ensure that all upstream and downstream components are ready for the PSA to operate
2. Start the PSA motor at the recommended speed for the load you will be supplying
3. Open the exhaust hand isolation valve and product line isolation valve
4. Gradually pressurize the PSA with feed gas by following step 4a or 4b

   4a. Open the gradual pressurization line if equipped and allow the PSA to pressurize at a rate of 7%/minute (ex. If your system operates at 10 bar, pressurize at a rate of 0.7 bar/minute)

   4b. If your PSA is not equipped with a gradual pressurization line, ensure the manual hand valves on the feed line are closed. Open the feed isolation valve and use the hand valve to gradually pressurize the PSA at a rate of 7%/minute (ex. If your system operates at 10 bar, pressurize at a rate of 0.7 bar/minute)

3.3 Startup after short term shutdown

A short term shutdown is defined as a PSA system stop of less than 24 hours where the system was maintained at pressure. This will allow you to skip the gradual pressurization stage. If your operating sequence is correct, this shutdown should have isolated the feed and product lines and stopped the PSA motor. If any of these steps did not take place, your shutdown procedure should be reviewed.

To start the PSA:

1. Ensure that all upstream and downstream components are ready for the PSA to operate
2. Start the PSA motor at the recommended speed for the load you will be supplying
3. Open the exhaust hand isolation valve
4. Open the product and feed solenoid operated isolation valves

Your PSA should now be back in full operation mode. Please note that it may take up to 3 hours for a CH_4 system and 8 hours for an H_2 system to produce quality gas.

3.4 Start up after emergency shutdown

An emergency shutdown is defined as a PSA stop when an E-stop button was pressed or a power failure occurred. Before starting the PSA, identify and resolve the cause of the shutdown. Depending on the PLC logic or customer shut down sequence, it could have resulted in the PSA being depressurized. If this is the case, follow the Start up after long term stop procedure. If the system was maintained at pressure, follow the Start up after short term shut down procedure.

4.0 Shutdown Guide

Because the adsorbent filling the PSA vessels is moisture sensitive, there are various ways of shutting down and storing the PSA depending on the length of shutdown.

4.1 Short term shutdown procedure

Short term shut down is defined as a PSA System shutdown for less than 24 hours. The PSA can either remain at pressure or be drained of all gas. Keeping the system at pressure will allow you to start faster when ready; there is no need for gradual pressurization.

If the system is relieved of pressure, it will have to be started using the gradual pressurization procedure (see section 3.2).

During each of these types of shutdowns, all inlet and outlet lines to the PSA beds should be closed. This is done either with hand valves or isolation valves. If air is allowed to enter the PSA, the adsorbent can become deactivated and may need to be replaced.

4.2 Long term shutdown procedure

A long term shut down is when your PSA is planned to be out of operation for an extended period of time, greater than 24 hours. Precautions have to be taken to avoid deactivation of the moisture sensitive adsorbent in the beds.

The system should be run with the feed and product isolation valves closed. The remaining gas in the system will be flow out the exhaust until the system is at atmospheric pressure. The system should then be "padded" with an inert gas such as N_2 according to the procedure in section 11.2.

4.3 Emergency Shutdown

DANGER WARNING:

Personnel should not attempt to approach equipment under unsafe conditions. The equipment is designed to relieve itself in the event of over-pressurization or fire.

When investigating the root cause following an emergency shutdown, exercise caution. Surfaces may be hot, and vessels, piping, and tubing could be at high pressure with combustible gas.

If an emergency shutdown is required or is initiated, follow the steps below:

1. Divert feed gas away from the PSA either by manually closing the feed valve if it is safe to do so. Another option is to shut down the source (compressor, reformer etc.) and have the PLC close the isolation valves.
2. If your operation is automated through a PLC, the initialization of the emergency stop should have caused the motors to stop. In a manual system, this should be done after the feed gas supply has been stopped.
3. Manually close the exhaust hand isolation valve. The system will still be at pressure and this is only a temporary shutdown procedure to resolve immediate problems.
4. Once the cause of the shutdown is determined and repaired, resume normal operation. If the system cannot be run, proceed to depressurization and long term storage steps in Section 9.10.2

5.0 Introduction

5.1 Nomenclature

PSA:	*Pressure Swing Adsorption*; a pressure varying process that uses a porous media to separate gasses. Throughout this manual we use PSA to refer to the Xebec Pressure Swing Adsorber machine.
Rotary Valves:	Xebec's two proprietary valves (Lights/Product and Heavies/Feed) that distribute the process gases to and from the adsorbent beds.
Lights/Product Valve:	The proprietary rotary valve found at the top of the **Rapid-Cycle PSA** that collects the Product (the "light" gas) from the adsorbent beds
Heavies/Feed Valve:	The proprietary rotary valve found at the bottom of the system that distributes the Feed and collects the Exhaust gas (the "heavy" gases) to and from the adsorbent beds.
Adsorption:	Adsorption is a chemical bonding process where molecules of gas adhere to the surface of an adsorbent such as Zeolite.
Feed:	Un-purified raw process gas coming into the PSA for processing.
Product:	Purified product gas exiting from the PSA to downstream consumers or equipment.
Exhaust:	Waste gas exiting from the PSA to a "safe location" (e.g. flare or burner). Also known as "tail gas".
Automatic Drain Trap:	A device connected to the bottom side of the feed filter in the Feed Line that automatically releases any collected condensate through the drain line.
Product Bypass:	A take-off upstream of the product isolation valve intended to divert product gas away from sensitive downstream users during start-up until product purity can be established.
Vent Line:	Line connected to the discharge side of safety devices, to direct relief gas to battery limits tie pint in the event of over-pressurization.
Timing:	The relative rotational position of the ports in the Lights and Heavies Valve. Any change in this setting will affect separation performance.
Manual Control:	A Control option whereby the **Rapid-Cycle PSA** is operated and adjusted manually: manual isolation valves and product back-pressure regulator are supplied.
Automatic Control:	A Control option whereby the **Rapid-Cycle PSA** is operated remotely and automatically through the customer's PLC. Typically this option includes an automated isolation valve for both PSA feed and Product. It also include a VFD (signal by customer) to adjust the PSA rotation speed. It may also include other control instrument and automated or control valves depending on "customized option" the end user may want to have. This option is also known as "Auto Controls".
Advanced Control:	A Control option whereby the **Rapid-Cycle PSA** is supplied with "Auto Controls" (see above), operating remotely and automatically through a PLC control and human-machine interface (HMI) supplied by Xebec Adsorption.

Cold Weather Protection: Heat tracing and insulation provided on all wetted process lines and Heavies Valve to allow for process start-up and operation in sub-freezing ambient environments.

Load: The load refers to the amount of feed gas you are applying to the PSA. This can be give in terms of volume or percentage of the maximum design load for your PSA.

Turndown: The ability of the PSA to operate at flows below design capacity. Typically 40% to 100% of design capacity is required.

Fluidization: The lifting or floating of adsorbent beads caused by excessive upward flow forces exceeding the downward force of gravity on a bead. Design is to avoid fluidization due to the negative impacts of layer mixing and breakdown (attrition) of the beads.

Process Gas: The raw or operational gas to be purified by the PSA.

PSA cycle: The period for any one bed to complete one full pressure cycle. For Rapid-Cycle PSA this is one revolution of the rotary valves.

Cycle Step: One nth of the cycle where n= no. of beds (ie. In a 9 bed system one step is 360/9 = 40 degrees)

Port: Gas flow passages internal to the rotary valves

EQ1: First Equalization (one bed exchanges gas pressure with another, both end up at the same pressure.)

EQ2, EQ3, …: Second Equalization, third and so on.

SP: Supply purge (internally supplied “clean” purge gas)

CCB: Counter current blowdown (depressurization out the bottom of the bed to exhaust)

BF: Backfill. Last step in the PSA cycle to bring the bed up to production pressure.

5.2 Components of the PSA System

Feed line isolation Valve –used to isolate the PSA from the upstream source of gas. This valve is opened only when the PSA motor is turning and a means of gradual pressurization has been confirmed. For the "Manual Control" option, this is a hand valve and it needs to be opened manually by an operator. For the "Auto Control" and "Advance Control" option, the valve is operated by a solenoid with a PLC or DCS output.

Gradual Pressurization Line (option) – this line is used to gradually pressurize the PSA with feed gas. If equipped, the solenoid operated isolation valve should be opened only when the PSA motor is turning.

Feed Line Hand Valve (option) – this hand valve can be used to gradually pressurize the PSA if a Gradual Pressurization line is not installed.

Feed Line Pressure Indicator – this pressure represents the feed pressure at the inlet to the PSA.

Feed Filter Bypass Line (Option) – this line is used to bypass the coalescing filter on the feed line. It should only be used if the filter is undergoing maintenance

Feed Line Coalescing Filter – This filter is used to removed small amounts of moisture and particulates from entering the PSA

Drain Trap – this float type drain trap is used to allow moisture in the feed line to exit. It remains open during regular operation and the outlet should be piped to a safe location. If water enters the trap, the float will activate allowing water to exit.

Low Point Drain – the solenoid operated valve on the feed line is used to blow out moisture from the lowest point in the PSA. It should be operated when the PSA is being pressurized and periodically during regular operation. The "on time" should be 2-3 seconds only, each hour. The frequency of opening for the valve could be readjusted from actual operating condition. It should be plumbed to a safe location.

PSA Heavies (Feed) Valve – this half of the PSA valve set is used for 2 things; allowing feed gas to be distributed to the vessels and allowing exhaust gas to exit the vessels. As the valve rotates via the motor/shaft assembly, ports align with the vessels and allow gas to enter and exit.

Drive Motor – the motor is used to turn both PSA valve halves. The motor speed is adjustable via a VFD (variable frequency drive) and is usually set by a PLC via an HMI. It controls the rotation speed of the PSA. The motor is linked to a gearbox and the output speed for the PSA valve rotation is expected to by very low; in the range of 0.08 RPM to 2-3 RPM.

Shaft Assembly – the shaft connects the Lights Valve (Product) and Heavies Valve (Feed) so they turn at the same rate. The components on the shaft are used for fine synchronization adjustments during manufacturing. These should not be adjusted except by Xebec staff.

Proximity Switch – the proximity switch is used to tell the PLC or control panel (depending on the customer application) if the motor has stopped rotating. If the motor stops rotating, the feed isolation valve(s) should close immediately. The proximity switch gives an on/off pulse signal by reading the teeth on the sprocket. It's also use to calculate (with the PLC) the approximate speed rotation of the PSA.

PSA Lights (Product) Valve – this half of the PSA valve set is used for 2 things; allowing the product gas to exit and transferring gas between vessels for process steps such as equalizations and purging. Similar to the Heavies Valve, as the valve rotates, ports align with the vessels and gas is transferred.

Product Purge Line (Option) – The product purge takes some clean product gas and flushes the PSA vessels during their exhaust step. This is an optional component and is set during commissioning if necessary.

PSA Vessels – these vessels are filled with adsorbent beads of different chemical composition depending on the process application.

Adsorbent – these chemicals fill the PSA vessels and purify the feed gas by means of Pressure Swing Adsorption. Microscopically they resemble volcanic rock (are porous) and allow the contaminants to bond to the surface at high pressure.

Back Pressure Regulator (BPR) – installed on the product line to maintain pressure in the PSA. This pressure setting is approx 0.7 bar less than the feed. The most common is a manual BPR but in cases where the feed pressure may fluctuate, a pressure control valve is used. Once the feed pressure drops below the BPR set pressure, it will close.

Product Line Isolation Valve – This valve is used to isolate the product line from the downstream components. . For the “Manual Control” option, this is a hand valve and it needs to be opened manually by an operator. For the “Auto Control” and “Advance Control” option, the valve is operated by a solenoid with a PLC or DCS output.

Exhaust Line Hand Valve – a manual valve used to isolate the exhaust line from downstream components. This valve should always be open during normal operation. There should be no valves or restrictions downstream of this valve.

Pressure Relief Valves – these valves installed at the top of the PSA vessels are used to vent pressure in access of the safe operating limits of the PSA system. These are sometimes vented to atmosphere or connected to a “vent header” which is connected to the exhaust line.

5.3 Operating principles and requirements

Following is a brief discussion of the **Rapid-Cycle PSA** process. Please refer to your P&ID for specific tag and instrument numbers.

Feed gas travels from Battery Limits through a coalescing filter to remove any water droplets. Feed flow enters the "Heavies Valve" and is directed toward one of the adsorbent beds. Product gas (the least adsorbed feed gas) flows through to the top of the bed and out through a check valve into the product line.

Simultaneously, the other beds are following a sequence of purging the adsorbed waste gases and regenerating in preparation for the next feed. Exhaust gas is released through the bottom of the bed and directed out of the Heavies Rotary Valve to Battery Limits.

As an introduction to the PSA (Pressure Swing Adsorption) process, please read carefully through the next few pages of this manual. There are simple operational requirements that, if not followed, can result in abnormal operation and possible damage to the PSA unit.

The Rapid-Cycle PSA gas separation system is designed to separate gas mixtures by means of the Pressure Swing Adsorption process. This process relies heavily on:

- A constant pressure drop across the Rapid-Cycle PSA
- The selective adsorption characteristics of gases on material when subject to varying pressures
- Constant feed, product and exhaust pressure and flow rate
- Gas composition very close to the initial design criteria (see Application Specification Sheet)

The gas mixture (Feed Gas) is made to flow vertically upwards through a tall column of adsorbent material at high pressure. The easily adsorbed components of the mixture ('heavy' components) are retained by the adsorbent at high pressure, while the less easily adsorbed gases ('light' components) pass through and are released from the top of the column as purified (product) gas. Before the adsorbent material becomes saturated with the 'heavy' gas components, feed gas delivery is halted and the pressure is released from the bottom of the adsorbent column ('bed'). The lower pressure allows the retained gas molecules to release and exit through exhaust. The adsorbent bed is then re-pressurized in steps and is ready to repeat the cycle.

The inherently intermittent nature of the process means that a single adsorbent bed can only deliver purified product gas for a portion of its cycle. To ensure a continuous supply of product gas, the Xebec **Rapid-Cycle PSA** is equipped with nine adsorbent beds which are cycled in a balanced, sequential manner through identical operating cycles. In addition to providing a continuous supply of product gas, the operating cycle also includes advanced features such as pressure equalization between beds to minimize energy losses and an advanced flow control system to allow adjustment for peak operating performance.

The system contains two rotary selector valves, which are central to its operation. The function of these valves is to perform the necessary switching functions to route the gases, in a sequential manner, to the nine adsorbent beds. One valve controls the gas flows into the bottom end of the adsorbent beds ('Heavies' or 'Feed') and the other valve controls the flows at the top end of the beds ('Lights' or 'Product'). These valves are driven by a gearmotor and a variable frequency drive.

The product, feed, and exhaust lines transport gas through the required valves and control equipment to the rotary valves. The PSA equipment has the option to come with instrumentation for process management as well as for shut down and start-up duties. These devices are easily connected to a supervisory and control PLC. A product and exhaust surge tank can also be used to aid separation performance and to minimize flow and pressure surges.

5.4 System Characteristics

The **Rapid-Cycle PSA** can be thought of as a controlled escapement mechanism (expander) for the flow of gas entering the unit. The main route out is through the exhaust flow, while the secondary route is through the product flow.

Exhaust flow is controlled primarily by cycle speed: faster cycle speed (i.e. higher speed values on the variable speed drive), permits greater exhaust flow. Product flow is controlled also by cycle speed and by a back-pressure regulating or pressure control valve, located on the product line. This control valve maintains a set pressure from the product receiver and hence controls the back-pressure at the top of the beds and flow rate through the product line.

For pressure control systems, increasing the backpressure setting on the control valve will decrease product flow and, likewise, decreasing the back-pressure setting will increase product flow. There will also be some effect on feed pressure as product pressure is adjusted (i.e. higher product back pressure at given valve speed will tend to increase system back-pressure and therefore feed pressure). Results in the system parameters by changing these two controls are:

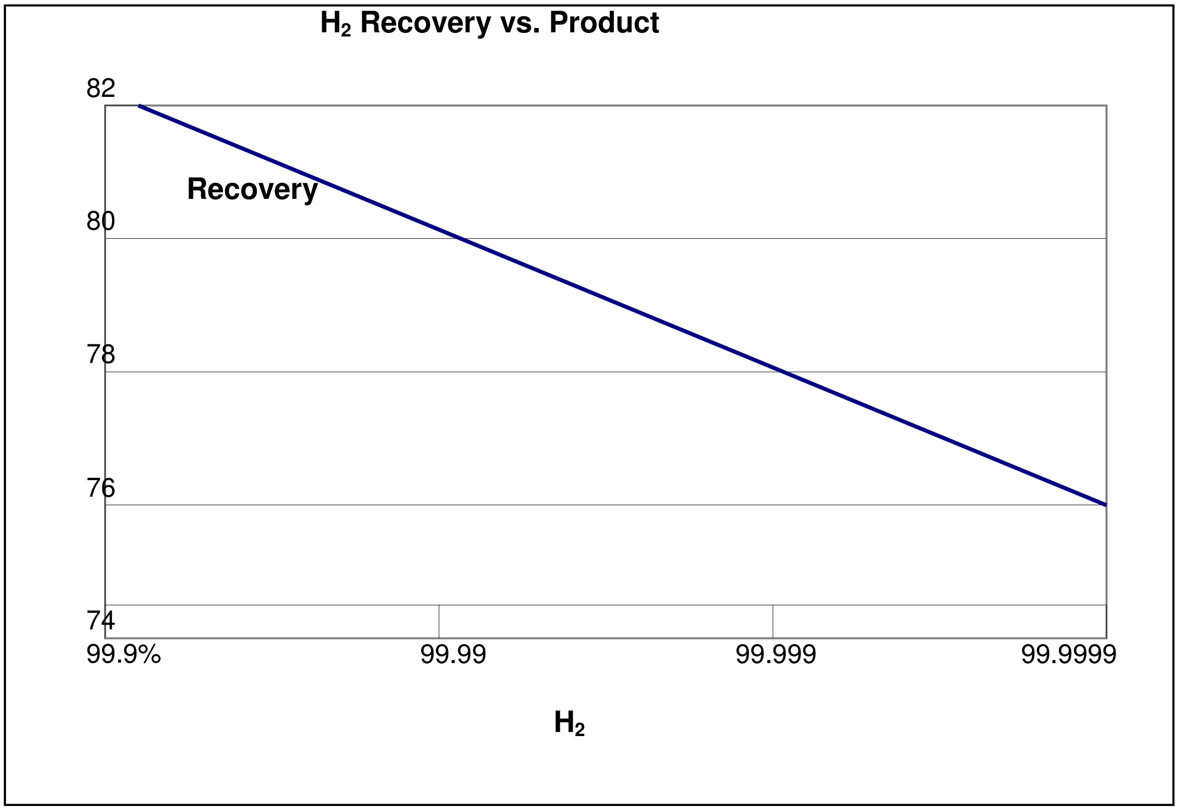
- **Cycle speed ⇑= product purity ⇑ + product flow ⇓ + peak bed pressure ⇓+ recovery ⇓**
- **Cycle speed ⇓= product purity ⇓ + product flow ⇑ + peak bed pressure ⇑+ recovery ⇑**
- **Product flow⇑ = product purity ⇓ +recovery ⇑**
- **Product flow⇓ = product purity ⇑ +recovery ⇓**

Note that this control methodology is not applicable for fixed supply flow systems.

These analogies will allow the operator to understand flow capacity and flow “resistance” in trouble-shooting the **Rapid-Cycle PSA** system.

For a **fixed flow supply** (eg. reformer feeding PSA only), care must be taken not to create too much back-pressure in the supply system. When the cycle speed is reduced without corresponding reduction in feed flow, the back-pressure to the source and peak bed pressures will increase. Output purity will DECREASE unless product flowrate is reduced. This is due to the higher feed and bed pressure creating too large a flow at the same product back-pressure setting. The reverse is also true when the cycle speed is increased. Reducing the product flowrate (without changing cycle speed) will also increase the back-pressure, although at a reduced amount compared with changing cycle speed.

Recovery vs. Purity – The direct tradeoff between recovery and purity for a given system is clear. If one is increased the other is lowered. A rule of thumb tradeoff has been determined in hydrogen systems as 2 points of recovery change for a change in significant digit of purity (ie. Recovery improves 2 points from say 78% to 80% for a reduction in product purity from 99.999% to 99.99% for a given system). For a methane system, the relationship is similar.



Cycle Speed – For a given flowrate, the cycle speed can be used to adjust product purity (and recovery). Increasing the cycle speed (reducing cycle time) for a given flowrate will cause the purity to increase as recovery drops. For a given system as flow is increased or decreased, cycle speed is adjusted to maintain the purity. This cycle speed vs. flowrate is confirmed at the time of commissioning and allows the operator to have automatic load following or cycle speed change to maintain purity.

NOTE: The PSA does not control the flowrate of feed or product gas. These are controlled through flow control valves (either on the feed or product lines) or back pressure regulation on the product line.

Although the design of the Xebec Rapid-Cycle PSA limits the flow and pressure fluctuations in the feed, product and exhaust lines, there is still some fluctuation (up to 10%). This may affect upstream (reformer, compressors, oil separators) or downstream (compressors, cracking towers, burners, etc) processes and should be thoroughly considered during the plant design. Typically less than 2% fluctuation of the feed stream is required for stable PSA operation.

Turndown – The ability of the PSA to operate at flows below design capacity. Typically a **Rapid-Cycle PSA** can maintain product purity with 40% to 100% of design feed flow rate.

Exhaust heating value stability (Wobbe Index) – Where exhaust stream is used in a burner system, the Wobbe Index has been used to specify the variation of fuel gas. Typically less than 5% fluctuation in Wobbe Number is required. The **Wobbe Index** (WI) is an indicator of the interchangeability of fuel gases. If V_C is the higher heating value, or calorific value, and G_S is the specific gravity, the Wobbe Index, I_W, is defined as

$$I_W = \frac{V_C}{\sqrt{G_S}}.$$

That is, Wobbe Index = higher heating value/ (square root of gas specific gravity).

Since the exhaust gas can varies in composition, it is highly recommended that an exhaust surge (mixing tank) be used if this gas is fueling a burner or turbine.

Pressure Drop – Is the permanent pressure loss from the feed gas line to the product gas line caused by the piping, valves, components and bed frictional losses. Typically this is 0.3 to 0.7 bars as measured across the PSA battery limits between feed and product. The PSA essentially operates similar to a restricting orifice, so with a constant flow to the PSA, the pressure drop stays constant.

6.0 Receiving, Storage and Shipping

6.1 Safety Considerations

A Xebec G2 PSA with 2 to 14” (51 to 355mm) diameter vessels will weigh approximately 2700kg (6000lbs) and come in a closed crate. The crate has clear indications which direction is up and it should not be tilted or tipped from this orientation. There are tip-indicators installed on the crate as well as the PSA skid. If these are activated, please contact Xebec before opening the crate. In addition to the main PSA skid, there may be loose parts that were shipped with the PSA such as the control panel. Use caution when opening the crate as these items may fall out.

If you have purchased a G3 or G4 PSA with vessels larger than 14” (355mm) diameter the weight will be substantially more but will likely arrive in pieces. The valve tower will arrive in a single crate and weigh approximately 1350kg (3000lbs). The beds will arrive empty and weigh approximately 900kg (2000lbs). Refer to the General Assembly drawing for the weight of your PSA.

The PSA was shipped with 0.2-0.7 bar of inert padding gas to preserve the adsorbent. Do not open any fittings, flanges or piping without first depressurizing the PSA. **The PSA should be stored with this padding gas at ALL TIMES to prevent moisture from contaminating the adsorbent media.**

6.2 “As Shipped”

A Xebec G2 PSA with 2 to 14” (51 to 355mm) diameter vessels will arrive in an enclosed wooden crate measuring 3x3x3 meters (8x8x8ft) or less. The crate is movable by forklift only and weighs approximately 2700kg (6000lbs).

ATTENTION:

Although the Rapid-Cycle PSA is manufactured for outdoor installations, it is recommended that you store the unit as shipped prior to installation.

To open the crate, use an appropriate socket wrench or crowbar to remove the top panel. Next, remove the wall screws using a Philips screw driver. Some assistance is recommended to brace the wall when all of the fasteners are removed. Each wall weighs approximately 27kg (60lbs).

The contents you will find inside are:

- 1 x Rapid-Cycle PSA system
- 1 x electrical control panel found in a separate box, if ordered.
- Miscellaneous parts as requested (see P&ID for details)

Once the walls are removed you will find the PSA base bolted to the wooden shipping base. These bolts can be removed using the 9/16th inch socket.

When it is received, the pressure indicators located on the Rapid-Cycle PSA should read between 0.2-0.7 bar (2-10PSI). Prior to shipment, the PSA is pressurized with inert gas to preserve the adsorbent. If these gauges are reading below 0.2 bar (2 PSI), follow the purge instructions in section 11.2 and contact Xebec.

6.3 Receiving and Delivery Inspection

As mentioned in the previous section, the space required for the PSA crate will measure approximately 3x3x3 m (8x8x8ft). A forklift will be required to move the crate. If a forklift is not available, the crate will need to be removed and a crane will be used and attached to the lifting points on the PSA skid. These are shown on your General Arrangement drawing.

Before opening the crate, any damage to the outside should be noted. If possible, pictures should be taken and sent to service@xebecinc.com for insurance purposes.

Once the crate is removed, the PSA should be visually inspected.

6.4 Storage and Preservation

To preserve the adsorbent, dry inert gas should be kept in the PSA beds. A weekly inspection of the gauges should be completed until the PSA is installed. If a pressure decrease (pressure <0.2 bar or <2 PSI) is observed, supply dry inert gas such as nitrogen at 0.3 bar (5 PSI) to the capped connection at the feed line pressure gauge, and electrically rotate the valves at 0.2 RPM for one complete revolution to ensure each bed receives gas. For more details on the proper purging procedure, refer to section 11.2.

Note that manual revolution of the rotary valve is not possible. The operator needs to supply power to the motor to turn the valve.

ATTENTION:

Inert Padding gas of approximately 5 PSI must be maintained in the PSA during storage to preserve the adsorbent beads.

7.0 Installation

Preparations should be made for installation prior to receiving the PSA. Depending on your site/plant, there are several different considerations for the installation of your Xebec PSA. Some questions you should ask yourself:

- Will the delivery truck fit in our unloading area?
- Are there any special directions for the delivery driver?
- Are there any special permits I need?
- How will I get the PSA off the truck?
- Is there a clear path to where I will store or install the PSA?

7.1 Lifting

A forklift is the preferred method to remove the PSA from the delivery truck. The PSA is bolted to a wooden skid with slots for the forklift.

Care should be taken not to tip or jar the PSA crate during transport.

Refer to the General Assembly drawing for the weight of your PSA.

DANGER WARNING:

When moving the **Rapid-Cycle PSA**, great care must be taken not to bend or vibrate the PSA tanks and base. This can put stress on their support and the surrounding components which could lead to cracked welds. If using a forklift to move the PSA, choose a flat, even route and operate the forklift at idle speed. If using a crane, use smooth transitions between directions to avoid stressing the lifting points.

Forklift and crane operation should be done using your common workplace safety practices.

If a forklift is not available, the crate can be removed from around the PSA and a crane can be used. There are specific lifting points on each PSA skid as shown in the General Assembly drawing.

7.2 Step by Step Installation

The following steps should be followed for complete and proper installation of your Xebec PSA.

STEP 1: PSA Inspection

Inspect the **Rapid-Cycle PSA** for any shipping / handling damage prior to installation. If any damage has occurred, photograph it and inform Xebec and shipper immediately.

STEP 2: Skid Placement

Using the proper lifting technique outlined in section 7.1, place the skid on a flat, level, stable foundation in a safe location. Bolt the frame to the foundation through indicated bolt locations on the General Arrangement drawing.

NOTE: Equipment is designed for outdoor installation but should be protected from severe process environments and direct or extreme weather conditions such as heavy snow, rain, or dust.

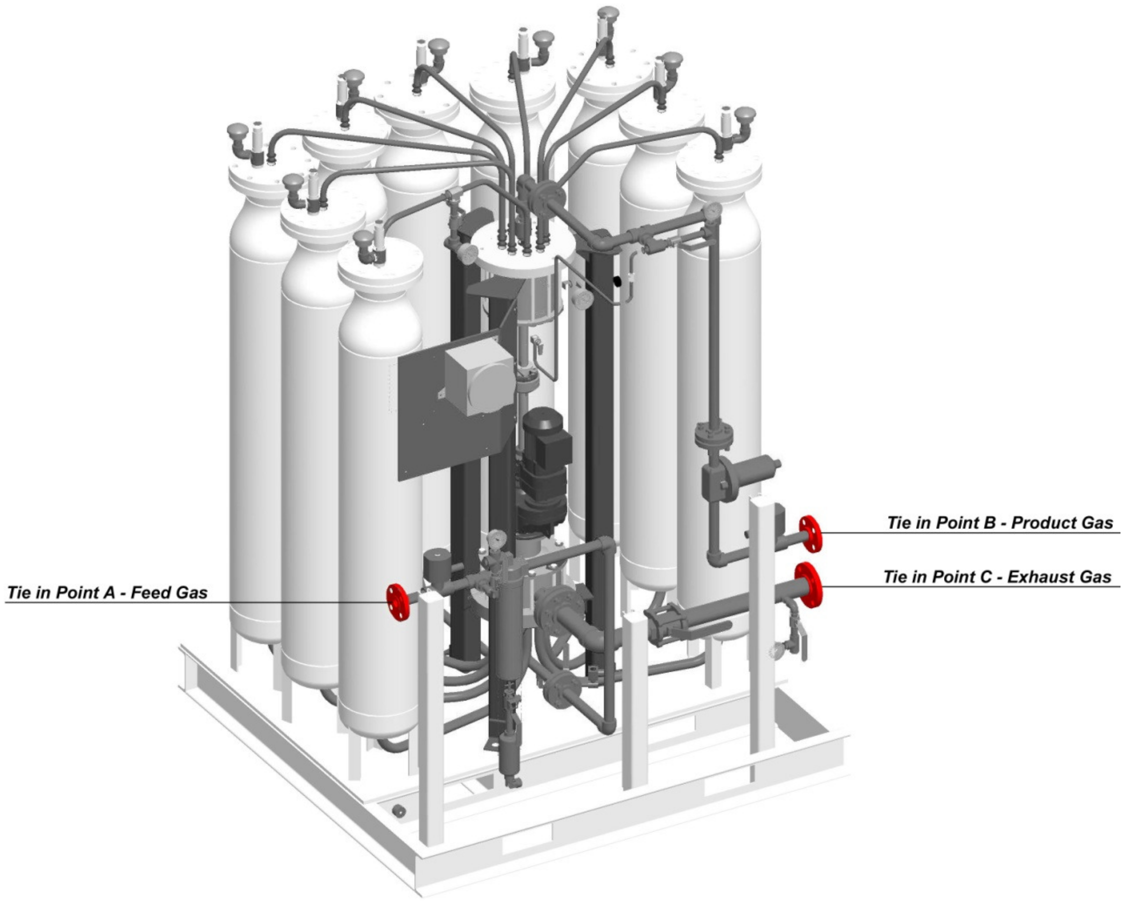
STEP 3: Connect All Tie-in Points

All PSAs include Feed, Exhaust and Product Tie-in Points. Optional lines include drain lines, vent header lines, sampling ports for analyzers, pressure relief vent header, etc. **Please refer to your custom P&ID for accurate TIP details.**

NOTE: If Cold Weather Package option is included, ensure that heat tracing is provided downstream of exhaust (TPC). Otherwise, condensate may accumulate and work back into the exhaust line and eventually enter the PSA unit to deactivate the beds.

Gaskets – proper gaskets for TPs should be used. SS Spiral wound are preferred.

Standard Tie-in Points are illustrated below. Note that on the 2’’ to 6’’ (51 to 152mm) PSA models, the standard tie-in points are in the front of the unit.

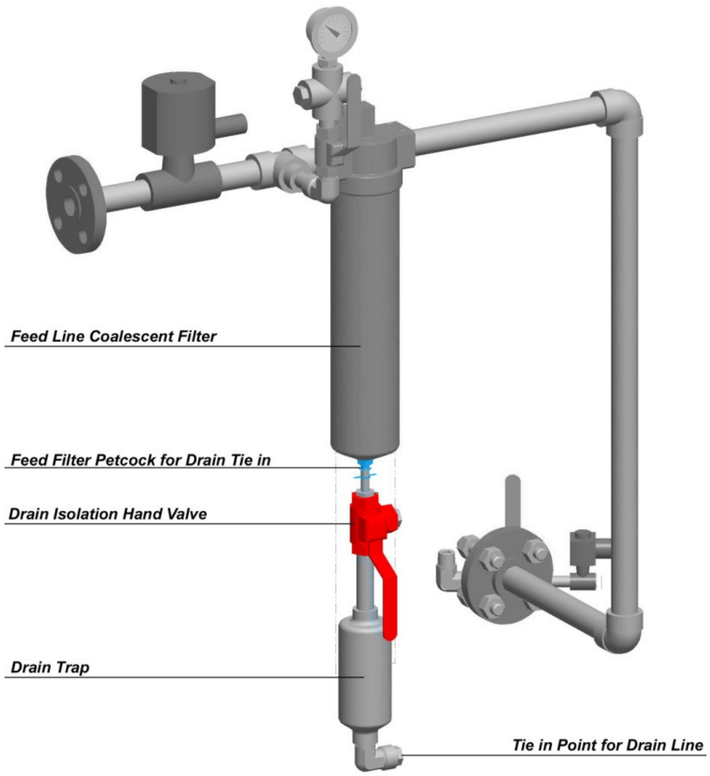


Standard Tie-in Point Locations (Feed, Product, Exhaust)

For a Rapid-Cycle PSA supplied with a manual drain trap on the feed filter:

Remove the plug from the line and connect a drain line (minimum DN8 or ¼” tubing) that leads to a safe location. Maintain slope for proper drainage. Install an isolation ball valve at the end of the drain line. If the PSA skid includes multiple drain lines, they should not be tied together. In the event of a failure, backpressure could cause liquid to enter the PSA.

NOTE: If Cold Weather Package option is included, ensure that heat tracing is provided downstream of the drain tie-in(s). Otherwise, the drain line may freeze such that condensate cannot drain properly from the system.



Feed Line Drain TP and Isolation Valve

WARNING:

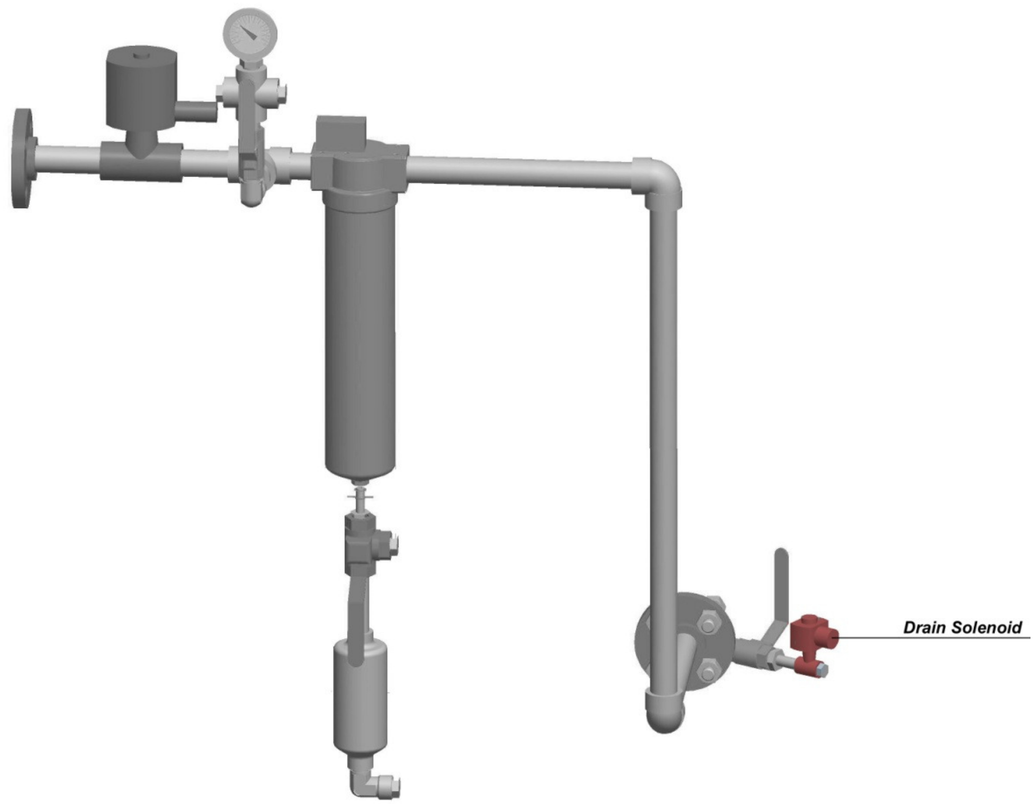
Failure to install an isolation valve will cause feed gas to escape through the drain line at full pressure.

Do not tie the drain lines from the filter/drain trap and the drain solenoid valve together. Both drain lines need to tie into a safe location independently. If tied together, the drain line from the solenoid may backflow into the filter/drain trap check valve.

For Rapid-Cycle PSA supplied with an automatic condensate removal system:

Connect condensate drains (filter/drain trap and feed low-point drain solenoid) lines as required.

The drain line should be rated for the design pressure of the PSA (see Application Specification sheet). Note that the condensate drain line may see the entire feed inlet pressure.



Automatic Drain Solenoid

WARNING:

Do not tie the drain lines from the filter/drain trap and the drain solenoid valve together. Both drain lines need to tie into a safe location independently. If tied together, the drain line from the solenoid may backflow into the filter/drain trap check valve.

Condensate entering the adsorbent beds may lead to partial de-activation.

The filter connection is DN6 or 1/8" FNPT. It is recommended to use at least DN8 or ¼" tube for drainage.

The control panel for CP1 is supplied in its own enclosure.

Please allow the following dimensions for control panel CP1: height 50.8 cm (20 inches), width 50.8 cm (20 inches), depth 20.3 cm (8 inches).

Maximum wiring length of 50 metres (164 feet) between inverter and gearmotor.

Wiring enters terminals along the right side of the control panel "CP1". Please allow space for these cables and fittings.

Ensure sufficient distance between cables of dissimilar signals as per your local electrical code to prevent electrical interference. e.g. motor conductors should be separated from the cable carrying the 4-20mA speed reference.

Use shielded cables for all control signals. For European installations, EMC compliance may dictate the use of "screened" cables and special EMC connectors for all connections.

Additionally, please observe the environmental conditions in Table 2.

Table 2: CP1 Installation Environmental Requirements

Environment	Ambient temperature	25°C +/ 0°C (to maintain highest speed regulation rating)
	Ambient humidity	90%RH (non-condensing)
	Storage temperature	-25°C to 60°C (-4°F to 140°F)*
	Ambience	Indoors (free from corrosive gas, flammable gas, oil mist, dust and dirt)
	Altitude, vibration	Maximum 1000m (3280.80 feet) above sea level for standard operation. Derate the amperage outputs of the motor drive controller by 3% for every additional 500m up to 2500m. (5.9m/s^2 or less conforming to JIS C0911)
	* Temperatures applicable for a short time, e.g. in transit.	

STEP 7: Connect Wiring

Wire motor, proximity switch, and instrument junction box (if fitted) as per your job specific electrical drawings.

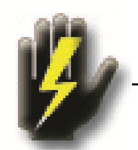
Generally, for installations to North American standards, a hazardous-location-approved connector, conduit seal and appropriate sealing material will be required to complete the installation at the site. European installations will require the appropriate hazardous-location-approved cable glands.

All cables should be tested with an insulation tester before being connected to any device.

STEP 8: Skid Grounding

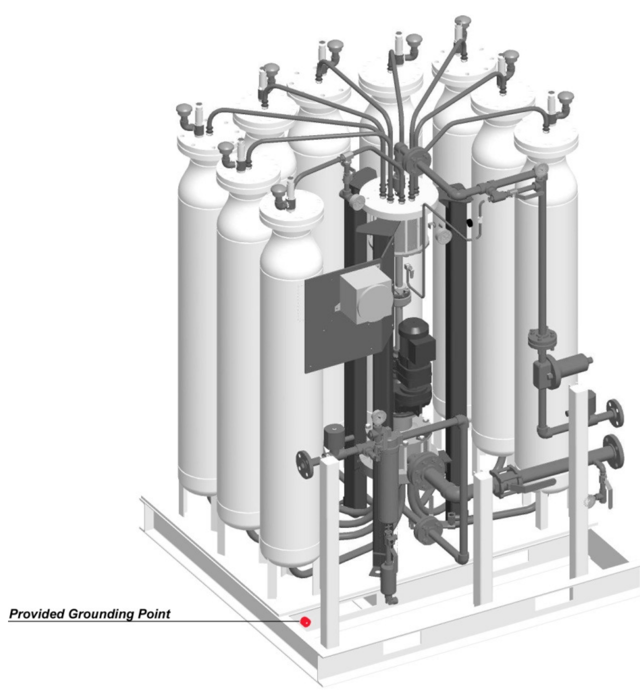
Skid will require a ground connection of not less than #10AWG or 2.59 mm. A 0.635 mm (¼") **tapped hole is provided on the frame for this purpose.**

NOTE: If the grounding point provided cannot be used, drill and tap an accessible point on the PSA frame. Ensure the paint is removed from the grounding point for proper connection.



ELECTRICAL WARNING:

Ensure that adequate **electrical grounding** of the PSA system is provided before start up in accordance with local regulations and using a dedicated grounding boss.



Grounding Point Location

STEP 9: Bonding Jumpers

Bonding jumpers should be placed across all process connections at skid battery limits. **Failure to do so can allow differences of potential between skid and connected equipment by buildup of static charges or electrical fault conditions!**

A static-control wrist strap should be used for all maintenance personnel working around the skid when it contains flammable gas. There are three small 3/8”-16 (0.826 cm-16) threaded “bosses” located on the frame for this purpose: 2 on motor side at bottom, on side of channel; and 1 on square tubing by exhaust isolation valve.

STEP 10: Leak Test Connections

Follow procedure in section 11.3 “Leak/Pressure Test” to confirm all connections are tight.

NOTE: DO NOT exceed System Design Pressure as stated in the Application Specification.

STEP 11 (if equipped): Automatic Drain Check

For PSAs provided with auto-controls linked to an owner/operator supplied PLC:

The timer and frequency for the drain solenoid valve are programmed into the Owner/ Operator’s PLC with provision for by-pass per logic.

Speed-feedback for motor control may be required (please see Xebec if this is applicable to your model).

Xebec will provide control and speed feedback logic. Programming of control systems must comply with the provided logic.
Inadequate control may inhibit performance and lead to partial deactivation of the adsorbent beds.

STEP 12 (if equipped): Instrument Air

Connect dry, clean instrument air supply if your **Rapid-Cycle PSA** is equipped with pneumatic valves or controls.

7.3 Mechanical

The PSA has 3 main tie-in points (TPs): feed, product and exhaust. The height and dimensions of these are shown on the General Assembly drawing. When positioning your PSA, these TPs should align perfectly. No stress should be applied to the flanges due to misalignment. If your PSA is equipped with additional TPs they will be indicated on your P&ID and GA drawing.

The PSA should sit perfectly flat on a concrete or steel platform. It should be fastened to this platform with proper bolts.

The flanges and bolts used should connect the PSA TPs with the off-skid piping must be graded high enough for your application (ex. 150lb, 300lb or 600lb rating). Proper drainage lines should also be installed as per your P&ID.

7.4 Electrical

CTRICAL WARNING:

Neutral to Ground Voltage

Neutral to ground voltages should be within 2VRMS when motor is operational. Excessive N-G voltages can indicate excessive load on the distribution panel or improper system grounding.

Powerfactor-correcting Capacitors

Ensure that there are no PF capacitors on the same phase as "CP1". Non-linear loads such as capacitors can cause resonance in the system and produce high currents at certain frequencies.

Voltage Stability

AC supply should be a smooth sinusoidal waveform without spikes or sags. Excessive high-amplitude spikes can affect the lifespan of variable speed drives (VSDs). Other culprits that reduce efficiency are harmonics and high crest factor.

Electrical Noise

Every attempt should be made to minimize the introduction of noise into the VSD. Electrical switching of large loads, other VSD's and even computer power supplies can create intermittent problems with equipment.

Grounding

All grounds on "CP1" should be terminated at the ground lug on the top right of the panel. A single ground wire, connected to the site system ground, should be tied to the ground lug.

The PSA skid must be grounded with a 12Ga copper grounding strap. The grounding point locations are shown in the diagram below. Only one of these needs to be used.

The power source must consist of a separate "hot" and "neutral" from a local distribution panel. The separate neutral is especially important if the "hot" is part of a 3 phase group (ex. L1, L2 and L3). See warning below.

AC voltage source must be within the following limits:

Rated	Low Limit	High Limit
100VAC 50/60Hz (Japan)	90VAC	132VAC
115VAC 60Hz (Americas)	90VAC	132VAC
230VAC 50Hz (Europe) Non-Hazardous	170 VAC	264VAC
220/380VAC 50Hz 3 Phase 4 Wire (Europe) ATEX	200VAC	440VAC (ATEX requirement)
Frequency is +/- 5% of rated		

STEP 1: Connect Proximity Switch

- The proximity switch requires a source of DC power and an appropriate PLC/DCS input to receive its pulsed signal.
- Regulated 10 to 31.5VDC supply.
- Requires 2 to 25mA load on "normally open" output.

STEP 2: Connect Speed Setting Signal

- A 4-20mA signal must be sent to the drive to set the output speed.
- Better than 0.1% accuracy.
- Less than 30VDC.

STEP 3: Connect "Drive Enable" Signal

- A 24VDC signal to energize the "Enable" or "MCR" relay, which energizes the VSD.

STEP 4: Connect "Motor Start" Signal

- A 24VDC signal to energize the "Start" relay, enabling the VSD to energize the valve gearmotor.

- A source of DC power and an appropriate PLC/DCS input to receive the "Drive OK" signal from the VSD when running.
- DC source to be a regulated 24VDC supply.

8.0 Commissioning

The initial startup or commissioning of the Rapid-Cycle PSA must be done by a Xebec representative or other certified personnel. If the system is started without supervision, the warranty can be void and you risk personal injury and damage to the equipment.

To schedule commissioning of your PSA, please contact service@xebeinc.com or 1-800-GO XEBEC

8.1 Commissioning Safety Precautions

During commissioning, all electrical and mechanical connections are checked and the PSA will be demonstrated at full operating pressure and flow. The safety precautions discussed in previous sections apply. Because this is the first time the PSA will be started at full pressure/flow, extra caution should be taken to ensure the system is leak free.

8.2 Training & Certification

Xebec offers a training and certification program for companies that may be purchasing more than one PSA. This would allow the operators to conduct their own startup and maintenance on the PSA, potentially allowing them to save money in the long run.

If you are interested in finding out more information, please contact service@xebeinc.com or 1-800-GO XEBEC.

8.3 Requirements

After contacting Xebec to schedule your PSA commissioning, we will provide a quote for our services to be reviewed by your company. Once approved, a purchase order must be supplied to Xebec before any work can be performed or travel booked.

In addition to the purchase order, there are several items that you will need to provide in order to make the commissioning go as smoothly as possible. Depending on your application, this may include:

- Feed gas flow measurement
- Product gas flow measurement
- Feed gas composition analysis
- Product gas composition analysis
- Exhaust gas composition analysis (optional – used to confirm recovery)
- Ability to gradually pressurize the PSA
- Feed flow control OR feed pressure control + back pressure regulator on product line
- Ability to adjust the motor rotation speed
- Ability to control the load (amount of gas) being sent to the PSA system
- Purge gas (N_2) for leak testing or padding if PSA is in storage

A complete pre-commissioning checklist will be provided prior to startup.

8.4 Commissioning overview

During commissioning, the goal will be to fine tune the PSA using your exact operating conditions and demonstrate that the performance meets expectations. In order for complete commissioning, all other components surrounding the PSA must be operational, adequate gas must be available for extended PSA run times (8+ hours) and support staff must be available.

The overview of what we will do is below.

Pre-Gas Checks & Reviews (duration –½ day)

- Safety Review w/ Customer
- Review Customer Acceptance Form
- Review Work Completed Documents
- Review Customer Facilities
- Mechanical Valve Inspection
- Electrical JB Inspection/Control Panel
- P&ID Review
- Grounding Points Located
- Confirm I/O Signals
- Confirm Motor Rotation
- Initial Tuning Valve Positions
- Confirm Gradual Pressurization
- PLC Logic Review (if required)
- Map Valve Speed vs HMI Reading
- Confirm Trip Logic
- Connect Pressure Transmitters
- Ensure Readiness of Downstream
- Ensure Readiness of Upstream

Initial Start-Up (duration – ½ day)

- Remove All Lockouts
- Confirm Operating Conditions
- Drain All Lines of Condensate
- Configure ROP for Operation
- Start Rotary Valve
- Time Gradual Pressurization
- Leak Test Connection Points

Tuning / Performance Testing (1-2 days)

- Confirm 110% valve speed
- Confirm Product Pressure
- Adjust Tuning Valves
- Observe Impurity Performance
- Demonstrate Recovery (both methods)
- Provide Turndown Table
- Operate at Steady State for 3-8 hours
- System Sign-off & Turn over

8.5 Process Performance Test

In order to prove that the PSA is working as designed, part of the commissioning is to calculate the recovery based on the gas compositions and flow rates. There are two different methods for making this calculation.

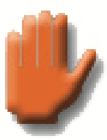
Method 1: Requires feed flow & compositions and product flow & composition

$$\boldsymbol{Recovery} = (\boldsymbol{Product\ Flow} \times \%\, \boldsymbol{Yinproduct})/(\boldsymbol{FeedFlow} \times \%\, \boldsymbol{Yinfeed})$$

Method 2: Requires feed, product and exhaust compositions

$$\boldsymbol{Recovery} = (\boldsymbol{1} - \boldsymbol{Y_{Exhaust}}/\boldsymbol{Y_{Feed}})/(\boldsymbol{1} - \boldsymbol{Y_{Exhaust}}/\boldsymbol{Y_{Product}})$$

For these calculations, stable or average flow rates must be available as well as gas chromatograph or sample bottle readings of gas composition.



ATTENTION:

For these calculations, stable or average flow rates must be available as well as gas chromatograph or sample bottle readings. Without these two measurements, the performance of the PSA cannot be demonstrated.

8.6 Equipment Turnover

Once the PSA has been demonstrated to perform as designed, the equipment will be signed off by the proper customer representative and the commissioning is complete. The warranty will last 1 year from this point or 16 months from the date of shipping, whichever comes first.

Support during your first year of operation is free of charge. We can be reached 24 hours a day at service@xebeinc.com or 1-800-GO XEBEC.

9.0 Operation

ATTENTION:

1. The PSA should not be operated prior to initial start up after installation by qualified Xebec personnel. Failure to do so will void process and performance warranties/guarantees.
2. System pressurization should be performed gradually over a period of 15 minutes at a constant pressurization rate. Failure to do so may result in partial deactivation or upset of the adsorbent beds and other components.

WARNING:

1. **DO NOT** start the system unless the no rotation alarm/trip is proved functional. Failure to do so may result in de-activation of beds.
2. **DO NOT** operate the rotary valves in reverse. Failure to do so may result in damage to the rotary valve seals and/or drive train.
3. **DO NOT** operate the rotary valves without gas flow for more than one hour. Failure to do so may result in damage to the rotary valve seals.
4. **DO NOT** start up and operate the system when the ambient temperature is below 4°C (unless protected with heat tracing and insulation). Failure to do so may result in damage to the rotary valves, beds, filter, and other components.

9.1 Pre-startup Checklist

Prior to startup, there are several items that need to be checked for proper operation and safety:

- Ensure all sample valves and manual drain valves are closed (except for the inline valve upstream of the automatic drain trap which must be in the open position for condensate removal).
- Ensure all Pressure Gauge and Pressure Transmitter isolation valves are open.
- If freezing temperatures are suspected to have occurred during a shutdown of the equipment, inspect for signs of frozen or damaged drain lines and filter element. Replace if necessary. Allow for thorough thawing before attempting to operate the **Rapid-Cycle PSA**.
- If the equipment is expected to be shut down for an extended period (more than 1 day), it is recommended to keep the PSA slightly pressurized with dry inert gas to avoid ingress of moisture.
- If during the shutdown period freezing temperatures are expected, ensure that the filter and all process lines are fully drained. Failure to do so may result in damage to the piping, tubing, and filter element.
- Ensure the entire system is above 4°C at start up. Failure to do so may result in damage to the rotary valve seal faces and other components.
- In the event that reverse flow may have occurred through the feed filter, check the Filter element for rupture or cracks before starting operation.

9.2 Start-up

ATTENTION:

1. Do not attempt to install, operate, inspect or maintain the system until you have read through this instruction manual carefully, have full understanding of all hazards indicated, and can use the system correctly.
2. High condensate loads may be experienced during start-up. Ensure that the solenoid or toggling of the drain valve on the filter (if an automatic drain trap is not provided) frequency are adequate for condensate removal. **Check all low point drains on the feed and exhaust lines.**
3. For systems with Optional cold weather heat tracing and insulation, ensure that the power to the heat tracing is on and has remained on to bring the piping up to process temperatures to minimize condensate formation.

To start the Rapid-Cycle PSA from 0 PSI to full operation:

STEP 1: Lockouts

- Remove all lock-outs from the system including electrical, isolation valves, caps and blind flanges.

STEP 2: Ambient temperature & Heat Trace

- Ensure that the PSA is above 4°C and is free of any frozen condensate.
- If the PSA is supplied with cold weather protection, ensure that the heat-tracing is active and warmed up. Keep heat tracing energized until nominal stable process temperature is achieved.

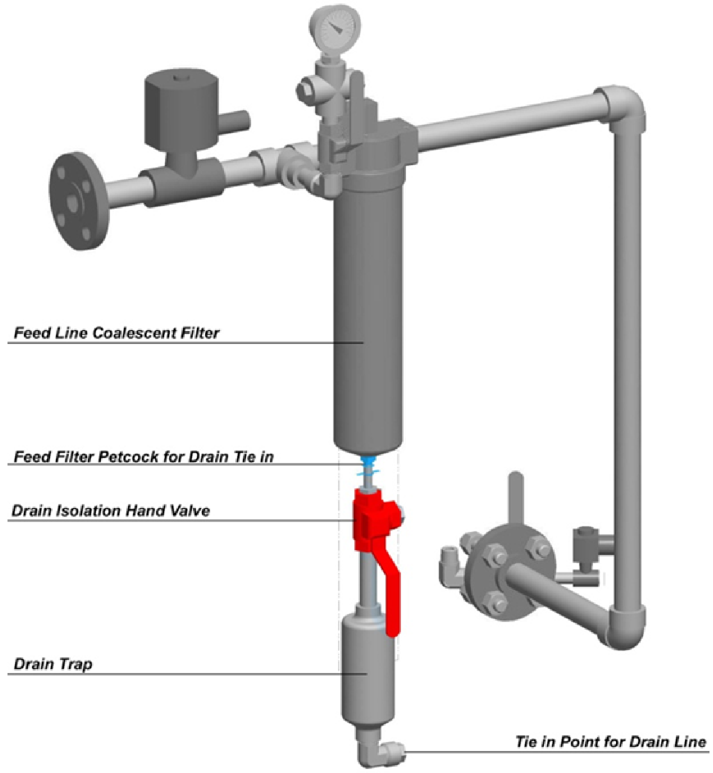
STEP 3: Isolation Valves

- Ensure that feed, product, and exhaust isolation valves are in the closed position (XV-090, XV-507, and XV-606). The valve numbering could be different if it was requested (during the design phase) by the customer to follow their own numbering method. Refer to your P&ID for exact valve numbering and additional isolation points (eg. light recycle line).There should be no pressure indicated at the feed PI (downstream of feed isolation valve)

NOTE: If feed pressure is indicated this may be a sign of a leaking feed isolation valve which should be verified. A leaking feed isolation may cause contamination of the adsorbent. Feed line should be de-pressurized before proceeding.

STEP 4: Coalescing Filter

Ensure coalescing filter is clean and in place. Filter isolation valves (XV-404 AND XV-410) are open, and the filter bypass valve (if applicable) is closed. Valve numbering could be different, refer to your P&ID for exact numbering.



STEP 5: Feed (Upstream)

- Ensure feed pressure to the **Rapid-Cycle PSA** is stable and within +- 5% of Application Specification value.
- Ensure feed temperature to the **Rapid-Cycle PSA** is stable and within +- 5% of Application Specification Value.

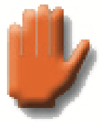
  NOTE: Over temperature may damage equipment. Temperature MUST be below 60°C.

STEP 5: Exhaust Isolation Valve

- Open the exhaust isolation valve

STEP 5: Start PSA Motor

- Start the PSA motor VFD at minimum speed according to your Application Specification and ramp up to start-up speed (30% of full load speed) over a period of no less than 5 seconds. Excessive acceleration of the motor controller can lead to reduced life of the motor gearbox.



ATTENTION:

Do not run the motor without gas flow for more than 1 hour. Operating the rotary valves without process gas flow may lead to accelerated wear of the sealing surfaces.

9.3 Pressurization with Automatic Controls

With the feed line closed and the PSA rotary valve running in auto-control according to the turndown recommendations provided during commissioning, it is now time to pressurize the PSA.

If your system is equipped with an automatic pressurization control on the feed line, open this valve and close the main feed line. With the product line and exhaust line open, allow the PSA to pressurize to full operating pressure and open the main feed line; close the gradual pressurization line.

WARNING:

Pressurizing too quickly may result in excessive attrition (vibration) on the adsorbent beds, partial deactivation due to contamination, and excessive rotary valve seal wear due to adsorbent dust between the seal/rotor interface. A pressurization rate of 1 bar/minute (15 psi) is recommended.

At this time all inlet gas will be exiting through the exhaust line.

Once full pressure and purity have been reached the system will be in automatic operation. Refer to your job specific Control Narrative for details of automation.

9.4 Pressurization with Manual Controls

With the PSA motor rotating at nominal speed according to the turndown conditions provided during commissioning, gradually open the feed isolation hand valve. Observe the pressure increase on the manual gauge on the feed line. Pressurization should be 1 bar (15 psi) per minute.

Once full pressure and purity have been reached and are stable (10-15 minutes), open the product line.

9.5 Normal Operation

Once the PSA has been commissioned and you have started it for the first time, there are several simple tasks that need to be completed periodically during regular operation.

Under MANUAL OPERATION the PSA speed is controlled manually based on varying load conditions. The back pressure regulator on the product line is used maintain a constant pressure drop across the PSA while the speed is adjusted to fine tune the purity and recovery. As the load (feed flow) changes, the speed is adjusted according to the Turndown Performance Parameters given during commissioning.

Under AUTOMATIC OPERATION the PSA speed is controlled by a PLC or customer interface. The product pressure is constant while small adjustments to the speed fine tune purity and recovery. As the load (feed flow) changes, the speed is adjusted automatically to the Turndown Performance Parameters given during commissioning. Refer to your plant specific control narrative for details.

9.6 Routine Inspections

Feed Filter Inspection

- Periodically check the Pressure Differential Indicator (PDI) of the online coalescing filter on the feed line. When indicating in the "mid-green" region shown in the figure below, the filter should be replaced. There will be small fluctuations during operation, but the indicating needle should not remain in the mid-green or yellow zone.
- Under normal operation, the filter is expected to last for 1 year.

WARNING:

The contents of the filter may contain flammable and noxious gases at elevated pressures.

Furthermore, entrained condensate may spray and any particulate matter may be projected from the discharge point.

Condensate Removal

Manual Drain System – If your system includes a manual drain system on the Feed Filter, check frequently for condensate by slowly cracking open the manual drain valve. Frequency of checks will depend on ambient conditions.

Automatic Drain System - If your system includes an automatic drain on the Feed Filter, check periodically for proper functioning of the drain trap:

1. To check for proper automated drain operation: remove plug on XV-408; toggle the 3-way manual by-pass valve (XV 408) to the plug position, no condensate should come out.
2. To check for proper isolation: at the vent (at customer's safe location), there should not be a continuous gas stream.

If during a manual check, larger quantities of condensate are discharged, the drain trap may be stuck in the closed position. It should be replaced or serviced during the next available down time. Note that condensate entering the adsorbent beds can lead to partial deactivation.

If under normal operation there is a continuous flow of process gas at the discharge point, the drain trap seal seat may be stuck open. The drain trap should be replaced or serviced during the next available down time. Normal operation can continue; however, the manual isolation valve should be used to inhibit the continuous discharge of process gas. The procedure for manual removal of condensate (see above) should be followed instead.

Feed Line LOW POINT DRAIN

Check periodically for the opening time and frequency of the solenoid actuation (Auto Control and Advance Control option only):

1. If the opening timing of the solenoid valve is too long, excess feed gas would be passing to the drain line.
2. If the opening timing of the solenoid valve is too short, insufficient water is drained.

The required frequency will depend on the quantity of condensable and temperature of feed gas, combined with ambient temperature conditions.

If the condensate is drained and a prolonged noise of gas passing is heard, shorten opening time or decrease frequency of opening.

If under normal operation, there is a continuous flow of process gas at the discharge point, the seal seat may be stuck open. The drain valve (XV-411) should be replaced or serviced during the next available down time. Normal operation can continue; however, the manual isolation valve should be used to inhibit the continuous discharge of process gas. The procedure for manual removal of condensate (see above) should be followed instead.

If the solenoid is functioning correctly yet there is no sign of condensate, it may be possible to reduce the frequency at which the valve is opened.

Control Valve Actuation (Option)

Occasionally ensure that the flow or pressure control valve on the product line is modulating the flow and pressure as designed. The product pressure should remain constant. The actuation pressure with respect to the valve position is given on the valve nameplate. (e.g. 0 – 30 psig is equal to 0 – 100% open.)

Periodic Leak Check

Periodically check flanges, fittings and piping for leaks using Snoop or a gas sniffer. Tighten bolts and fittings as necessary.

Lights Valve Casing Pressure

Install a 0-1 bar (0-15 psi) pressure gauge onto the XP ports on the lights valve casing. This gauge should show a nominal casing pressure of 0.2 bar (3 psig) during normal operation.

For other routine inspections, see the preventative maintenance chart in this manual.

9.7 Gas Sampling

Gas sampling should be part of your routine checks on the PSA performance. If your system isn't equipped with real time analysis with a Gas Chromatograph then manual samples must be taken. These samples can be taken at various points on the PSA skid or up/downstream of the PSA itself. The gas samples should only be taken after the system has been running and stable for at least 4 hours for methane systems and 8 hours for hydrogen systems. After each change in operating conditions (valve speed, pressure, flow) the same amount of time should be allotted before sampling.

The main sample points of interest are the feed (raw) gas going into the PSA, the product (purified) gas and the exhaust (waste) gas coming out. All three of these streams will have very different gas compositions and the sampling procedure changes slightly for each. Although it is recommended that a qualified expert take and analyze the sample gas, below is a brief description on how to take the samples.

Feed Gas:

To properly sample the feed gas, find a port upstream of the PSA or take a sample directly from the source. Since this gas will likely be at high pressure, the sample can be taken using a "sample bomb". These are typically stainless steel, high pressure vessels with valves on either end. Connect the bomb to the port and purge it several times by pressurizing and depressurizing with the feed gas.

Product Gas:

To properly sample the product gas, find a port on the product line coming out of the PSA, preferably coming off of a surge tank so the sample is averaged. If there is no surge tank, take the sample upstream of the back pressure regulator or control valve on the product line. The process is the same as for the feed gas since it will be at pressure.

Exhaust Gas:

To properly sample the exhaust gas, find a port downstream of the PSA on the exhaust line. On many systems, especially methane, this line will be at negative pressure due to the vacuum pump. For this reason, the sample must be taken using a vacuum pump or on the outlet side of the vacuum blower. Since this gas will likely be at a lower pressure, the sample can be taken using a sample bag. The sampling procedure is similar to that of the bomb.

9.8 Adjusting External Purge Settings (Option)

Your PSA may have come equipped with the “product purge” option. This line takes a slip stream of clean product gas and flushes out the PSA vessel that is on its exhaust step. This clean gas pushes any impurities that are not vented through exhaust normally, thus giving the regeneration stage some additional help.

The downside to this is that a small amount of product gas is actually wasted out the exhaust line. For this reason, product purge is not always used or necessary. Very often the PSA can regenerate itself without the use of product purge and this line remains closed.

During commissioning, it will be determined if product purge is necessary by the commissioning technician. If it is required, the regulator on the line will be set to a specific value and left alone. This typically will not have to be adjusted again.

9.9 Adjusting for Loads

“Turndown” is a term used by Xebec for the load range the PSA is capable of handling; often a limitation of the PSA motor/gearbox arrangement. The typical turndown for a PSA is 40%. This means that the PSA can run from 40 to 100% of the design feed flow rate.

During commissioning, one of the steps is to determine the exact speed range in which the PSA should operate. Given a constant feed gas composition and pressure, the PSA speed can be adjusted for a change in flow rate to give the design purity and recovery. The range for this is 100% to the turndown value (typically 40%). Once the minimum and maximum speed is determined, the relationship for any load in between is linear. The commissioning technician will provide you with a “turndown table” showing the relationship.

If your PSA is equipped with automatic controls, the speed change will occur automatically with a change in the flow rate to the system.

9.10 Shutdown

ATTENTION:

1. If the equipment is expected to be shut down for a considerable time (days or weeks), then it is recommended to keep the PSA slightly pressurized with dry purge gas in order to avoid ingress of moisture. See "Purge for Maintenance Procedure" and "Shipping, Storage and Preservation".

2. If during the shutdown period freezing temperatures are expected, make sure that the filter and any low points are fully drained. Failure to do so may result in damage to the filter element or piping.

9.10.1 Short term (<24 hours)

WARNING:

A special caveat for systems running the kPSA (kinetic) cycle for O_2 and N_2 removal. The media in your PSA is sensitive to high concentrations of methane at high pressures. For this reason, the PSA must be depressurized for any stop that will last more than 1 hour.

A short term stop is defined as a PSA shutdown that will last less than 24 hours. Typically this would be a stop if there was a minor problem that had to be corrected, a minor maintenance task or other unknown issue. Because the stop will be brief, the PSA can be left at pressure thus allowing a direct re-start (no need for gradual pressurization).

If your PSA operates in manual mode, the following steps should be done for a shutdown.

NOTE: In an automatic controls or advanced automatic controls of the Rapid-Cycle PSA, steps 1 and 2 below are performed by the PLC.

1. Close product, feed and exhaust isolation valves within 30 seconds
2. Stop the PSA motor
3. Drain all lines of condensate by opening the hand valves on all low point drains

If your PSA is running the kPSA cycle for O_2 and N_2 removal, follow the steps below for a short term stop.

1. Close product, feed and exhaust isolation valves
2. Allow the PSA to rotate until the pressure in all vessels is < 10psig
3. Stop the PSA motor
4. Close the exhaust isolation valve
5. Drain all lines of condensate by opening the hand valves on all low point drains

9.10.2 Long term (>24hours)

WARNING:

A special caveat for systems running the kPSA (kinetic) cycle for O_2 and N_2 removal. The media in your PSA is sensitive to high concentrations of methane at high pressures. For this reason, a special extended depressurization and purging is required for a long term stop. See below.

A long term stop is defined as a PSA shutdown that will last more than 24 hours. Typically this would be a scheduled shutdown for a major maintenance task, adsorbent change or unknown equipment failure. Because the length of stop is unknown, the system must be depressurized and padded with an inert gas, typically nitrogen. The padding gas will ensure that no air or moisture is allowed into the PSA vessels causing damage to the adsorbent media.

If your PSA operates in manual mode, the following steps should be done for a long term shutdown.

These procedures will be review in detail with the Xebec commissioning Engineer and customer Engineer during the commissioning of the PSA.

NOTE: In an automatic controls or advanced automatic controls of the Rapid-Cycle PSA, steps 1 - 3 below are performed by the PLC.

1. Close product and feed isolation valves
2. Allow the PSA to rotate and vent the internal gas out the exhaust line until the PSA is depressurized.
3. Close the exhaust isolation valve
4. Drain all lines of condensate
5. Connect the source of dry inert gas to an available port on the feed line and allow to rotate for 3 cycles or until a padding of 0.7 bar (10 psi) is shown
6. Stop the PSA motor

For systems running the kPSA cycle, the following steps must be performed if the shutdown will last longer than 1 hour.

1. Close product and feed isolation valves
2. Drain all lines of condensate
3. Connect the source of dry inert gas to an available port on the product line (upstream of the isolation valve)
4. Allow the PSA to rotate for 1 hour while purging through the product line with inert gas
5. Close the exhaust isolation valve
6. Allow PSA to rotate for 3 complete cycles or until a padding of 0.7 bar (10 psi) is shown
7. Stop the PSA motor

9.10.3 Emergency Shutdown

ATTENTION:

When investigating root cause following emergency shutdown, exercise caution and show due consideration for hot surfaces, energized equipment, and pressurized gas containment.

DANGER WARNING:

Personnel should not attempt to approach equipment to close manual isolation valves if it is not safe to do so. The equipment is designed to relieve itself in the case of fire.

An emergency shutdown stop is defined as a PSA shutdown that occurred by pressing one of the available E-stop buttons, a power failure or similar abrupt controlled or uncontrolled shutdown. This is a similar shutdown to the Short Term Shutdown discuss previously.

If your PSA operates in manual mode, the following steps should be done for an emergency shutdown.

NOTE: In automatic control or advanced automatic control of the Rapid-Cycle PSA, steps 1 and 2 below are performed by the PLC.

1. Immediately close product, feed and exhaust isolation valves in that order
2. Stop the PSA motor
3. If safe to do so, assess the immediate danger or cause of the shutdown
4. Resolve the issue

9.10.4 Isolation

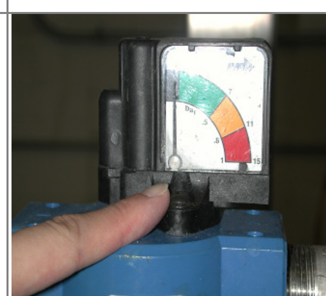
If maintenance is being conducted on the PSA, the unit should be completely isolated from any upstream and downstream components.

Valves, motors and breakers should be properly locked off using local safety practices.

The PSA and any upstream/downstream equipment that will be opened to atmosphere should be purged with dry inert gas to ensure no process gas remains in the system.

10.0 Troubleshooting

10.1 Process

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Flow is Lower than Expected, Dropping, or there is No Product Flow	Product Flowmeter reading incorrectly	Check Calibration of Product Flowmeter
	System leakage to atmosphere: vents or by-passes are open	Check all drains, pressure safety valves, relief devices, sample plugs, bypasses, etc., for presence of process gas.
	System capacity set point higher than actual (lower feed flow than expected)	Check feed flow or secondary capacity indicator. If feed flow is lower than expected for current PSA capacity, reduce PSA capacity by reducing Rotary Valve speed.
	Lower feed pressure than required for PSA operation.	Check feed flow, feed pressure, and losses from feed (by-passes, etc.). Verify upstream equipment is operating at expected capacity. PSA requires feed pressure to be within ±5% of feed pressure as stated in the Application Specification.
	Blocked line/filter or closed valve.	 Check Feed Filter Differential Pressure indicator. If the needle is in the yellow /red area, replace the Feed Filter as per Section 11.4. Check for any closed valves that should be open during operation, and for any blockages in the feed line.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Flow is Lower than expected, Dropping, or there is No Product Flow **AND** **Exhaust Flow is Higher than Expected**	Product purge flowrate set to high	Check Product Purge flow indicator, and ensure product purge flow has not been changed.
	Product purity too high (Rotary Valve speed too fast)	Check Product Purity. If too high, reduce Rotary Valve speed by 0.1 CPM and wait until steady-state conditions are reached. Check Product Purity again. If still too high, repeat until desired Product Purity is achieved. If Product is at required Purity, check that the Rotary Valve speed is properly set as a function of capacity, as per the Turndown Table determined during Commissioning, and correct valve speed set point in the PLC if necessary.
	Rotary Valves do not turn at the specified set point	Measure time for one complete revolution of the Rotary Valve, and compare to the speed set point. If the Rotary Valve is rotating *below* the set point, <u>increase</u> the Rotary Valve speed in 0.02 CPM increments, waiting 5 minutes between each change in Rotary Valve speed. Measure Rotary Valve speed after each change. Repeat until desired Rotary Valve speed is achieved. If the Rotary Valve is rotating *above* the set point, <u>decrease</u> the Rotary Valve speed in 0.02 CPM increments, waiting 5 minutes between each change in Rotary Valve speed. Measure Rotary Valve speed after each change. Repeat until desired Rotary Valve speed is achieved.

Product Flow is Lower than expected, Dropping, or there is No Product Flow **AND** **Exhaust Flow is Higher than Expected**	Damage to Rotary Valve system	Check for signs of mechanical damage to the Rotary Valve, shaft, gearbox, motor, and/or Proximity Switch and associated components. If there is obvious physical damage to any of these, shut down the PSA as per Section 9.10 Contact Xebec and provide the following information: • Location, nature, and severity of damage • Time between inspections, when Rotary Valve was last inspected and undamaged, and when damage was noticed. • Process conditions (feed flow and composition, product flow and composition, etc.) at the time damage was observed. • PSA Pressure Trace (if available) around the time damage was observed. • Any process upsets upstream and/or downstream of the PSA around the time damage was observed.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Flow is Lower than expected, Dropping, or there is No Product Flow **AND** **Exhaust Flow is Higher than Expected (continued – 3)**	Leaking Rotary Valve seals	Any leakage in either Heavies or Lights Valves will end up pressurizing the casing of either valve. If site procedures allow, with the **Rapid-Cycle PSA** operating, connect pressure indicators to the casing of either valve. High leakage at any Rotary Valve will be indicated by a casing pressure that is consistently above the Exhaust pressure, even as the Exhaust pressure rises and falls in step with the PSA cycle. If this is so, the **Rapid-Cycle PSA** should be shut down as per Section 9.10 The PSA should then be purged as per Section 11.2, “Purge for Maintenance” procedure. Next, a pressure test on the **Rapid-Cycle PSA** should be performed as per Section 11.3, “Pressure Test”. This Pressure Test will have been performed at Commissioning, with the leak rate quantified and recorded in Commissioning documentation. Measure the current leak rate of the **Rapid-Cycle PSA**. Contact Xebec with the following information, which will help in determining the cause of the Rotary Valve leak: • Time between inspections, and the time and date suspected leakage was observed. • Information on any maintenance performed on the Rotary Valves and associated components. • Process conditions (feed, product, exhaust flow and composition, etc.) at the time leakage was observed. • PSA Pressure Trace (if available) at the time leakage was observed. • Leak rate as measured in the Pressure Test.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Purity Too High	Rotary Valve Speed Too Fast	Check Product Purity/composition by an analytical method different from the regular measurement method. If Product Purity is confirmed as being too high, reduce Rotary Valve speed set point by 0.02 CPM in the PLC and wait until steady-state is established. Measure Product Purity again. If Product Purity is still too high, repeat the reduction in Rotary Valve speed and wait time until desired Product Purity is achieved. If after three adjustments Product Purity has not been reduced to desired levels, reduce Product Purge flow rate by 10%. Wait for steady-state conditions, and re-check Product Purity. If still too high, repeat reduction of Product Purge, waiting for steady-state between each step, until desired Product Purity is achieved. If neither of these steps has altered the Product Purity sufficiently, contact Xebec and provide the following information: • Time between inspections, and the time, date, and nature of the most recent maintenance on the **Rapid-Cycle PSA**. • Process conditions (feed, product, exhaust flows and compositions, etc.) at the time the off-specification condition was noticed. • PSA Pressure Trace (if available) at the time the off-specification condition was noticed. • Any process upsets upstream and/or downstream of the **Rapid-Cycle PSA**.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Intermittent Product Flow	Product Check Valve sticking	Tapping of the Product Check Valve may temporarily fix the problem. If the Product Check Valve continues to stick, safely shut down and purge the **Rapid-Cycle PSA**, as per Section 9.10 Repair or Replace the Check Valve, leak-check system as per Section 11.3 and re-start system as per section 9.2
Intermittent Product Flow **AND** **Surging Feed Flow**	Low Feed Pressure	Check that any processes upstream of the **Rapid-Cycle PSA** are working properly, and are producing sufficient Feed gas flow and pressure as listed in the Application Specification. Tolerances of ±5% are acceptable. Any Feed gas flow and pressure variances beyond this limit will require corrective actions by the Owner/Operator on upstream equipment as appropriate.
Product Purity is Lower than Expected, or is Dropping	Pressure Transducer Problem (for systems equipped with Pressure Control Valve on Product)	Isolate Pressure Transducer by closing isolation valve. If local procedures allow while **Rapid-Cycle PSA** is in operation, remove the Pressure Transducer, and confirm calibration. If the Pressure Transducer is out of acceptable calibration, replace with a calibrated Pressure Transducer of the same range.
	Improper instrument air pressure supplied to Pressure Control Valve (where equipped)	Check that the required instrument air pressure, as specified in the Product Control Valve specifications in Appendix, are provided to allow for proper operation.
	Increased or Increasing Product Flow	Reduced Product Flow by increase the speed set point of the Rotary Valve (thus reducing Product Flow) until the desired Product Flow is achieved.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Purity is Lower than Expected, or is Dropping (continued)	Faulty Pressure Control Valve (where equipped; unable to control Product Flow)	Safely shut down and purge the PSA, as per Section 9.10 “Normal Shutdown” and Section 11.2 “Purge for Maintenance” procedure. Repair or Replace the Product Control Valve, leak-check system as per Section 11.3 and re-start system as per section 9.2
	Product gas loss to vents or by-passes	Check all drains, pressure safety valves, relief devices, by-passes, etc., for presence of process gas.
	Partially-contaminated adsorbent	Check for signs of water ingress into beds by slowly opening and closing low-point drain valves on the feed line, exhaust line, and each adsorber bed. Check that all condensate drain systems are working properly. If water or condensate is found, and/or any drain systems are improperly working, contact Xebec for instructions, and provide the following information: • Time between inspections, and the time, date, and nature of the most recent maintenance on the **Rapid-Cycle PSA**. • Process conditions (feed, product, exhaust flows and compositions, etc.) at the time the off-specification condition was noticed. • PSA Pressure Trace (if available) at the time the off-specification condition was noticed. • Any process upsets upstream and/or downstream of the **Rapid-Cycle PSA**. • Ambient temperatures around the time water or condensate is believed to have entered the adsorber beds. • If equipped, power draw readings of the heat trace in the Cold Weather Package around the time water or condensate is believed to have entered the adsorber beds.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Product Purity is Lower than Expected, or is Dropping **AND** **Higher than Expected Product Recovery**	Feed gas composition off specification	Check that Rotary Valve speed is correct for current capacity, as per table determined during Commissioning. If adjustment is required, increase or decrease (as the situation requires) the Rotary Valve set point in 0.02 CPM increments, then wait for steady-state conditions. Re-check Product Flow and Purity.
	Feed gas temperature off specification	
	Unusually cold or hot ambient temperatures	
	Rotary Valve speed slower than required at specified load	
Hydrogen Recovery is Lower than Expected, or Dropping	See "Product Flow is Lower than Expected" for causes and corrective actions.	
System Cannot be Pressurized During Normal Start-Up	Feed Isolation Valve Closed or Partially Closed	Open Feed Isolation Valve
	Feed Filter upstream and/or downstream Isolation Valves and Bypass Valve are Closed or Partially Closed	If running <u>with</u> Feed Filter in-line with **Rapid-Cycle PSA**, open manual or automatic isolation valves. If running <u>without</u> Feed Filter in-line, open filter bypass manual or automatic isolation valves.
	Feed Filter is Blocked or Damaged.	Safely shut down and purge the **Rapid-Cycle PSA**, as per Section 9.10 and Section 11.2 Repair or Replace the Feed Filter, leak-check system as per Section 11.3 and re-start system as per section 9.2
	Product Check Valve stuck open.	See "Intermittent Product Flow" for corrective actions.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Rotary Valves will not Rotate	Power is not provided to Motor, Variable Speed Drive, and/or Control Panel	Check integrity of all electrical power connections to all devices, and check that power to the system is available.
	Alarm or Trip condition has locked out system	Check all Alarm and Trip conditions to determine which one(s) may have been activated. Correct the condition(s) before attempting to re-start.
	Run Enable not received from supervisory PLC	Check to see if the PLC "run enable" digital input is "on" and that the signal can be measured at the corresponding input on the drive controller.
	Variable Speed Drive has been placed in Stand-by mode, or VSD software has been corrupted	Check that VSD is in Operating Mode. If so, re-load VSD software and operating parameters.
	Variable Speed Drive has been damaged	Safely shut down and purge the PSA, as per Section 9.10 "Normal Shutdown" and Section 11.2 "Purge for Maintenance" procedure.
		Lock out System. Attempt to re-start the System to check lock-out effectiveness
		Repair or Replace the Variable Speed Drive.
		Reload VSD operating parameters.
		Leak-check system as per Section 11.3 and re-start system as per section 9.2
	Motor has become mechanically decoupled from the gear box (via broken coupling, missing or damagedkeyway)	Safely shut down and lock-out system. Always test lock-out effectiveness by trying to re-start the system. Check that motor has not come loose from the adapter to the gearbox by re-tightening bolts used to connect the motor to the gearbox.

FAULT or SYMPTOM	POTENTIAL CAUSES	ACTIONS REQUIRED
Rotary Valves will not Rotate **(continued)**	Motor and/or gearbox is damaged	Safely shut down and purge the **Rapid-Cycle PSA**, as per section 9.10 and 11.2
		Lock out System. Attempt to re-start the System to check lock-out effectiveness
		Repair or Replace the Gearbox and/or Motor as per manufacturers recommended procedure.
		Leak-check system as per Section 11.3 and re-start system as per section 9.2
	Rotary Valve Shaft has become decoupled from motor	Contact Xebec for specific assembly drawings
	Internal or External Damage to Rotary Valves	From the Variable Speed Drive, obtain current draw information for the motor, and compare to the current draw measured at Commissioning. Contact Xebec and provide the following information to assist in the troubleshooting process: • Time between inspections, and the time, date, and nature of the most recent maintenance on the **Rapid-Cycle PSA**. • Process conditions upstream and downstream of the **Rapid-Cycle PSA** at the time the Rotary Valves failed to rotate, including upsets. • PSA Pressure Traces taken up to 1 month before failure of the Rotary Valves to Rotate. • Current draw information.

10.2 System (motor, electrical, other)

DANGER WARNING:

Failure to disconnect the Variable Speed Drive before performing the insulation test can permanently damage the drive controller!

ELECTRICAL WARNING:

Hazardous voltage will be developed across the terminals of the motor. Only qualified personnel should attempt to perform these tests. Variable Speed/Frequency Drives contain capacitors that discharge over a 20 minute period. **No work should be done on these drives until this time has passed and proper electrical testing has been done to confirm no voltage.**

REMARK:

On a “No Rotation Alarm”, immediately isolate the feed gas to the **Rapid-Cycle PSA**. Failure of valve rotation with continuing feed gas flow will saturate the adsorbent beds and may lead to partial deactivation.

	STEP	DESCRIPTION/TEST	EXPECTED RESULT
Motor Failing to Start	1	Test for supply power: Is the Variable speed drive display lit?	If “no”, ensure that breaker “CB1” is on, relay MCR is energized from PLC/Emergency stop circuit (auto controls) and incoming voltage is within VSD operating limits. (See VSD manual.) If “yes”, continue to Step 2.
	2	Test for “Run Enable” signal. Is “Run Enable” Relay energized from PLC?	If “no”, determine fault in PLC logic, wiring, or if the relay is faulty, replace. If “yes”, continue to Step 3.

	STEP	DESCRIPTION/TEST	EXPECTED RESULT
Motor Failing to Start (continued)	3	Is “Run” LED lit on VSD?	If “no”, programming in drive has been changed. Please consult “Programmed Parameter” document and VSD manual to verify correct values of programmed parameters. If “yes”, continue to Step 4.
	4	Using a multimeter scaled for at least 500VAC, measure the voltage at the motor terminals on CP1 at the external overload block. Ensure that the overload has not been “tripped” before performing this test. Is there an AC voltage between any two terminals that increases and decreases with the speed setting of the VSD?	If “no”, remove the bottom cover from the variable speed drive and **carefully** test for AC voltage at the drive output terminals. The voltage will increase/decrease depending on the speed setting of the VSD. If there is no voltage at these terminals, the VSD is damaged and must be replaced. If “yes”, continue to Step 5.
	5	With the power locked out to control panel “CP1”, if the PSA is equipped with a motor connector (CN1) check to see if the two pieces of the connector are aligned and tight. Separate the two halves and check for corrosion on the connecting pins and sockets. Disassemble each of the two halves and ensure wiring connections are tight and correct. Is wiring correct and corrosion-free?	If “no”, make repairs/changes as necessary. If “yes”, continue to Step 6.

Motor Failing to Start (continued)	6	With the power locked out to control panel "CP1", open the motor connection box on the side of the motor and check the connections for tightness and corrosion. If it is safe to do so, ask the operator to initiate "CP1" power, and enable the VSD. **CAUTION: A hazardous voltage will be developed across the terminals of the motor. Exercise extreme caution! Use only approved meter leads.** Using a multimeter, measure between terminals U1-V1, V1-W1 and U1-W1. There should be an AC voltage that varies in proportion to the output frequency of the drive controller. At drive setting of 60Hz approximately 200 to 208VAC should exist across each terminal pair. Disable the power and reassemble all components. Are proper voltages present at the motor terminals?	Does the motor start? If "no", replace the cable between motor connector and motor. If "yes", motor may be damaged or overloaded. Continue to Step 7.
	7	Separate the motor connector (CN1) halves if equipped. If a motor connector is not used, remove the leads from the motor terminals. Use an insulation tester to determine if there is an internal fault in the motor. CAUTION: Failure to disconnect the Variable Speed Drive before performing the insulation test can permanently damage the drive controller!	If there is an internal fault (short to ground or across phases), replace the motor. If motor passes the test, continue to Step 8.
	8	Load is too high for motor to start. Reduce torque by lowering system pressure.	If "no", contact Xebec for advanced troubleshooting. If "yes", contact Xebec to determine cause of over-torque condition.

	STEP	DESCRIPTION/TEST	EXPECTED RESULT
Unplanned Motor Stoppage	1	Does motor start then stop after 5 to 10 seconds?	If "yes", ensure that the "Drive OK" contact on the VSD is energizing and/or that the Customer PLC "Drive OK" input is being energized. 24VDC should be measured between "Drive OK" and "DC Com". If "no", continue to Step 2.
	2	Does motor start then stop after approximately 30 seconds?	If "yes", ensure that the PLC is receiving the proximity switch pulses. Continue troubleshooting at Step 1 of "Troubleshooting for Motor Failing to Start". If "no", continue to Step 3.
	3	Does motor start then stop at random times?	Some units are equipped with over-temperature sensors within the motor windings; a high motor temperature may cause the system to trip. An over-temperature setpoint controller located on "CP1" will indicate "Tripped" if the controller is not used to initiate a full emergency stop. The motor must be allowed to cool before attempting to re-start the system. If the controller is not present or not indicating a tripped condition, continue to Step 1 of "Troubleshooting for Intermittent Failure of Proximity Switch".
	4	When valve is turning, is the PLC input turning on and off with every pulse of the proximity switch?	If "no", ensure that the proximity switch is wired correctly and has the correct voltage feeding it. See proximity switch specification sheet. Continue at Step 1 of "Troubleshooting for Complete Failure of Proximity Switch". If "yes", continue to Step 5.

Unplanned Motor Stoppage **(continued)**	5	If proximity switch seems to be working but is causing occasional system trips, check the following: - Mounting of proximity switch bracket, proximity switch and sensing gear. - Alignments of proximity switch in relation to "target." - Distance between proximity switch sensing face and "target" should be approximately 1.5 to 2.5mm.	Was everything correctly adjusted? If "no", perform needed adjustments, as per Section 11.7, "Proximity Switch Adjustment Procedure". If "yes", continue to Step 3.
	6	Is the proximity switch cable routed beside any cables carrying high current/voltages?	If "yes", reroute cables to maintain appropriate separation (follow local codes and practices). Dissimilar signals should only cross at right angles. If "no", continue to Step 4.
	7	Is the shield/screen of the cable that extends from JB1 to the PLC connected to earth/ground at the PLC end?	If "no", connect the shield/screen to earth/ground to minimize any inductively or capacitive coupled noise that can affect the signal quality from the proximity switch. If "yes", continue to Step 5.
	8	Does the signal from the proximity switch match the characteristics of the PLC input card? Some PLC inputs require a very low "zero" point before they switch off. Measure the "on" and "off" voltage at the PLC input and check with manufacturers recommendations.	If "no", connect a variable resistor from the signal return lead of the proximity switch to DC Com. Starting with a high resistance while measuring the voltage swings from "on" to "off" states, turn the resistance down slowly until the "on" and "off" levels are well within the thresholds recommended by the PLC manufacturer. Replace the variable resistor with a fixed resistor of the nearest value and re-test.

11.0 Inspection and Maintenance

11.1 Recommended Service Schedule

In order to keep your **Rapid-Cycle PSA** in good working order, the following preventative checks are recommended. The time intervals are recommended and may be adjusted over time by the customer based on experience with the machine.

DANGER WARNING:

Before opening any valves or devices that would release the contents of **Rapid-Cycle PSA** or would allow ingress of air into **Rapid-Cycle PSA**, ensure that the system is purged with inert gas.

If Cold Weather Package is provided, the heat trace may be hot during maintenance. Ensure all protective gloves and attire are worn when doing maintenance work.

(Refer to section 11.2: "Purge for Maintenance Procedure")

ATTENTION:

Do not attempt to install, operate, inspect or maintain the system until you have read through this instruction manual carefully, have full understanding of all hazards indicated, and can use the system correctly.

Special tools and utilities required

Standard Imperial measurement tools	
Gearbox Lubricant	(see vendor information).
Inert Purge Gas	Extra-Dry Nitrogen, regulated to 345 kPa (50 psig)
Pressure Test Gas	Extra-Dry Helium, regulated to 345 kPa (50 psig)

	PERIODIC INSPECTIONS (weekly)	MAINTENANCE IF REQUIRED
Rotary Valves and Drive	Inspect motor/gearbox for any unusual regular or irregular running noise.	If present, follow instruction in the Troubleshooting Section of the motor/gearbox Manufacturer's Operating Instructions in the Appendix.
	Inspect for excessive leaking oil from the motor or gearbox.	If present, follow instruction in the Troubleshooting Section of the motor/gearbox Manufacturer's Operating Instructions in the Appendix.
	Inspect for the presence of oil on the couplings on the PSA	Oil coupling and chain as required.
	Using suitable gas detector, check for gas leakage at Bed connections between the Bed Casing and the Bed Endcap, and between the Bed Endcap and the Valve Distributor.	**Caution: All fasteners are factory pre-set to critical torque values. Visually inspect fasteners only. Do not attempt to verify or adjust torque settings.** If excessive leakage is detected, contact Xebec for technical support.
	Using a suitable gas detector, check for gas leakage on the Heavies and Lights Valves at the shaft seal.	**Caution: All fasteners are factory pre-set to critical torque values. Visually inspect fasteners only. Do not attempt to verify or adjust torque settings.** If excessive leakage is detected, contact Xebec for technical support.
	Visually inspect nuts on beds for obvious backing off or loosening.	**Caution: All fasteners are factory pre-set to critical torque values. Visually inspect fasteners only. Do not attempt to verify or adjust torque settings.** If obvious loosening is present, contact Xebec for technical support.
	Check for any obvious damage to or degradation of electrical wires, cables and connections that may potentially cause sparks	See "Electrical Component Maintenance Procedure"

	PERIODIC INSPECTIONS (weekly)	**MAINTENANCE IF REQUIRED**
Rotary Valves and Drive (continued)	Check proximity switch mounting bracket and locknuts. Do not over tighten. If loose, check the proximity switch gap.	See "Electrical Component Maintenance Procedure".
	Check grounding and bonding connections for tightness and corrosion.	
Rest of Plant	Check Pressure Differential Indicator (PDI) on all in line filters.	See "Change Feed Filter Procedure".
	Check liquid drains for presence of water. 1. By-pass the PLC control of all automatic drains and check for functionality. 2. Open all manual drains and check for signs of condensate. 3. Using suitable gas detector, check for gas leakage at drains.	See "Change Drain Trap Procedure".
	Check if pressure gauges are reading accurately. Close the pressure gauge isolation valve, unbolt a sample plug and depressurize the pressure gauge tee. The pressure gauge needle should now drop down to 0 barg (0 psig). If not, replace the gauge.	See "Change Pressure Gauge Procedure".

In addition to regular maintenance inspections, major inspections and parts replacements are recommended every 2.5 years of operation. Below you will see the maintenance required after 2.5 years and 5 years of operation. This maintenance is to be done only by a Xebec trained technician; failure to comply can result in machine damage and void warranty.

ATTENTION:

Protect beds from ingress of moisture during shutdown by maintaining slight positive pressure with dry inert gas.

	2.5 YEAR INSPECTIONS	**MAINTENANCE REQUIRED** **(provided by Xebec Certified Technician only)**
ROTARY VALVES AND DRIVE	Rotary Valve	Internal components of Rotary Valve inspected; parts are replaced as required.
	Motor, Gearbox, and Mechanical Couplings	Lubricated and/or Oil Changed as Required.
REST OF PLANT	Condensate and particulate filter inspection	Check filter elements for damage. If the filter element appears torn, replace the element. If not, re-install the filter. Inspect housing internal side for corrosion. See “Change Feed Filter Procedure”.
	Instruments and Equipment	Tested and Checked for accuracy.

	5 YEAR INSPECTION	**MAINTENANCE REQUIRED** **(provided by Xebec Certified Technician only)**
ROTARY VALVES AND REST OF PLANT	A major valve overhaul is recommended.	Contact Xebec to arrange for service.

11.2 Purge for Maintenance Procedure

STEP	ACTION	RESPONSE/DISCUSSION
1	Connect inert purge gas supply to available port + hand valve fitting on PSA feed/inlet line	See Section 11.1 “Special tools and utilities required” for acceptable purge gases.
2	Record purge inert gas bottle pressure.	
3	Close all Feed and Product isolation valves. Open Exhaust valves to allow system to vent.	System drains to low pressure through exhaust port.
4	Ensure Vent location is on-line. Start motor and ensure that rotary valves are turning.	All process gases in system will be exhausted to vent. Do not run the Rotary Valves for more than 1 hour without gas flow.
5	Open hand valve next to inert gas supply connection Isolation Valve and nitrogen supply valve.	
6	Purge system for 15 minutes, feeding into system at about 3.4 barg (50 psig). Procedure will require approximately 1.7 to 17 standard cubic metres (60 to 600 standard cubic feet) of inert gas depending on bed diameter. Close Exhaust isolation valve.	This represents a pressure drop of approximately 2758 kPa (400 psi) on the recommended gas cylinder. Rapid-Cycle PSA bed diameters range from 2” to 60” (51 to 1524 mm). Any maintenance work that involves opening process connection requires extra care that all pressure is relieved and gas concentrations are at safe levels.
7	Close hand valve. Allow purge gas to vent until no pressure is indicated on either the bed or product pressure gauges.	
8	Close all Isolation Valves	Ensure that no moisture from air will enter the adsorbent beds.
9	Turn off motor	
10	Record final inert gas bottle pressure.	Deduct original pressure (see step 2) from final pressure to determine amount of inert gas used in purge. This procedure will facilitate planning the amount of gas required in future maintenance.

11.3 Leak/Pressure Test

Pressure testing should be performed at installation and following any pipe disconnections to ensure system integrity. Leak testing can be done at lower pressures if you suspect a leaky flange or fitting.

ATTENTION:

Ensure all personnel are aware that the system will be under pressure during the test. Posting a warning sign by the **Rapid-Cycle PSA** is recommended.

STEP	ACTION	RESPONSE/DISCUSSION
1	Ensure that the isolation valves on the feed, product and exhaust are in the closed position. Refer to your P&ID for specific valve numbers	Contact Xebec Service if you question which valves should be closed. The goal is to isolate the PSA from all upstream and downstream components.
2	If equipped, open Feed Filter isolation valves and bypass valves	
3	Connect inert gas to an available port + hand valve location, ensure hand valve is still closed.	See Section 11.1 “Special tools and utilities required” for recommended pressure test gas. Helium is recommended because the helium molecular size is similar to typical product gas and will leak more noticeably that would nitrogen. Alternatively, nitrogen gas can be used initially to pressurize up to 70% of the test pressure. Helium can then be used to make-up the remaining pressure to achieve desired test pressure (70% nitrogen and 30% helium).
4	Close the hand valve near the exhaust line pressure indicator to isolate it from the pressure test.	Note that this is the only instrument within the system that is not rated at design pressure.
5	Turn on the rotary valve and run at nominal speed.	In the automatic and advanced control options, the rotary valve will be prevented from running without starting the system. The rotary valve interlock will need to be bypassed to turn the valve.

6	Slowly and continuously over a 30 min period, increase system pressure at a 7% of test pressure/minute. Introduce the inert gas into the system by throttling the cylinder or hand valve.	The feed pressure step rate is necessary to avoid fluidization of the beds. Amount of inert gas depends on the bed size of your system.
7	Stop the rotary valve and isolate the inert gas source	
8	Record the starting test pressure.	Consider using a leak detector to check for any leaks before starting the test pressure.
9	Wait for 3 hours and then record the ending test pressure.	
10	If the change in pressure is less than 1% of the test pressure, then the system has passed the pressure test. If not, use a leak detector on the lines and connections. Tighten connections as required.	Gas may leak through control valves as these are not an isolation valve. If gas is leaking, install blind flange, then go back to Step 8 of this procedure.
11	Depressurize the system by running the rotary valve and opening the exhaust valve to flare or other safe location.	Note that the product line will not depressurize because of the product check valve.

11.4 Filter Changing

DANGER WARNING:

1. It is strongly recommended that the filter element be changed out only when the system is not operating and has been purged.
2. If the procedure must be performed while the system is in operation, be aware that hazardous gas is present at system pressure and that minor leaks may occur through isolation valves in the filter by-pass system. If any of the isolation valves appear to be leaking heavily, do not proceed with filter change without first shutting down and purging the system.

ATTENTION:

Operating time with the filter bypass valve open must be minimized. Particles and water may cause damage to the rotary valves or partial deactivation of adsorbent beds.

Filter changing with the system NOT running:

Although it is always best to change filters with the system depressurized, if there is a bypass line available then it can be used during a filter change. The steps below outline the procedure for changing the filter with the system off and depressurized.

STEP	ACTION	RESPONSE/DISCUSSION
1	Shut down, purge, and isolate the system.	See "Shut Down" procedure Section 9.7
2	Close valves upstream and downstream to minimize exposed volume. Remove filter housing, replace filter element, and reinstall housing. Open isolation valves. Check for leaks around filter housing.	Inspect O-ring inside head element to be certain it has not been damaged or dislodged from seat.
3	**Rapid-Cycle PSA** is ready to start-up	Refer to "Start Up" procedure in section 9.2
4	Log filter replacement into Maintenance Records or log book indicating old and new differential pressure indications.	

DANGER WARNING:

When the procedure is performed while the system is in operation, be aware that hazardous gas is present at system pressure and that any isolation valve may have some leakage.

Filter changing with the system running:

Although it is always best to change filters with the system depressurized, if there is a bypass line available then it can be used during a filter change. The steps below outline the procedure for changing the filter with the system ON and at pressure.

STEP	ACTION	RESPONSE/DISCUSSION
1	Open filter by-pass valve if available	This is the line that goes around the filter housing. It should have isolations valves to direct flow away from the filter.
2	Isolate filter to be changed by closing the immediate upstream and downstream valves.	
3	Release any remaining pressure in filter housing by opening the auto drain by-pass valve. Remove condensate drain piping (and drain trap if applicable) to allow removal of filter housing.	
4	Remove filter housing, replace filter element, and reinstall housing.	Inspect O-ring inside head element to be certain it has not been dislodged from seat or damaged.
5	Open upstream valve to the newly changed filter – close and check for leaks using industrial leak detection fluid.	If leaks are found, tighten connections as required.
6	After no leaks are detected, bring filter back online by slowly opening the remaining isolation valves and then closing the by-pass valves	
7	Record filter replacement into Maintenance Records or log book indicating old and new differential pressure indications.	

11.5 Pressure Gauge Maintenance

DANGER WARNING:

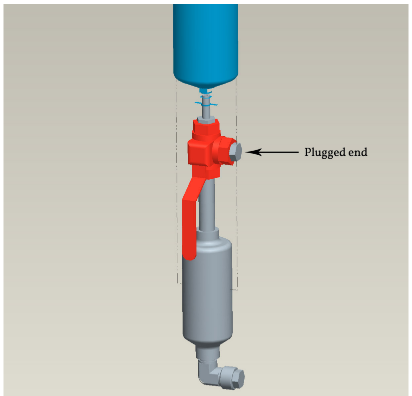
When the procedure is performed while the system is in operation, be aware that hazardous gas is present at system pressure and that any isolation valve may have some leakage.

STEP	ACTION	RESPONSE/DISCUSSION
1	Close Isolation Hand Valve(s) for gauge to be changed out.	
2	Unscrew gauge carefully.	
3	Remove any Teflon tape from the female port.	
4	Install new gauge, ensuring correct Teflon taping procedures are followed.	
5	Open Isolation Hand Valve slowly and test for leakage with commercial quality leak test solution.	
6	Ensure Isolation Hand Valve(s) is left in proper position as per the P&ID.	
7	Record gauge replacement into Maintenance Records or log book.	

11.6 Drain Trap Maintenance

DANGER WARNING:

When the procedure is performed while the system is in operation, be aware that hazardous gas is present at system pressure and that any isolation valve may have some leakage.

STEP	ACTION	RESPONSE/DISCUSSION
1	Divert condensate flow from the drain trap to the plugged end of the 3-way valve. The drain trap should be isolated from the system.	**WARNING:** Ensure that the plug is tightly bolted into the valve. 
2	Unscrew piping and drain trap carefully. Install new drain trap. Re-divert condensate flow from the filter to the drain trap.	
3	Record gauge replacement into Maintenance Records or log book.	

11.7 Proximity Switch Adjustments

DANGER WARNING:

Power should be disconnected and locked out when working around the rotating drive components.
Always attempt to start the system after locking out in order to test lockout effectiveness.

STEP	ACTION	RESPONSE/DISCUSSION
1	Shut down, purge, depressurize and isolate the system as per procedures. Disconnect electrical power to the gearmotor and VSD.	See "Shut Down" and "Purge for Maintenance" procedures. Attempt to start the motor to test lockout of system.
2	Gain access to the proximity switch by removing shaft safety screen. Ensure that proximity switch bracket is tight and that the proximity switch is within 1.5 to 2.5mm of the sprocket. Carefully tighten the locknuts on the proximity switch.	Do not over tighten, as it is possible to strip the threads on the proximity switch or the locknuts.
3	Reconnect power and test proximity switch. Ensure proximity switch is not "missing" any pulses in a revolution of the target sprocket.	
4	Record inspections into Maintenance records or log book.	

12.0 Xebec Support

12.1 Onsite

The Xebec service group is at your service 24 hours a day, 365 days a year. Because our customers are spread out around the globe, we maintain at least one of our technicians on call at all times.

To request a site visit for troubleshooting or maintenance of your PSA system, simply call or email for a quote. You will be required to provide a purchase order and may be asked to do some pre-visit troubleshooting and preparation working. To avoid paying travel premiums, please schedule maintenance at least 4 weeks in advance.

Phone:

1-800-GO-XEBEC

(1-800-469-3232)

Email:

service@xebecinc.com

12.2 Phone

Because our client base is spread out over the globe, we offer 24 hr phone support for emergency service. The initial 1 hr of troubleshooting is included but subsequent hours of support are charged at our normal hourly rate.

13.0 Spare Parts

13.1 Recommended Spare Parts

Xebec highly recommends keeping a small list of critical spare parts on hand in case of emergencies. Some of these parts have unavoidable lead times of 4-6 weeks and if the Xebec PSA is critical to your process, the cost of downtime can be substantial.

NOTE: DO VERIFY YOUR PSA COMPONENTS IN THE APPENDICES.

13.2 Storage of Spare Parts

Spare parts should be stored in a cool, dry place free of dust and in their original shipping packaging. Adsorbent should be kept under a padding of dry nitrogen gas until used. Failure to do so can result in deactivation of certain types of materials.

13.3 Ordering Procedure

Spare parts can be ordered by contacting the Xebec Service department at the number or email given below. In some cases (ie. large orders) , a deposit payment may be required.

Phone (Worldwide):

1-800-GO-XEBEC

(1-800-469-3232)

Phone (Local):

1-450-979-8700

Email:

service@xebecinc.com

Fax:

1-450-979-7869

Mail:

Xebec Adsorption Inc.
c/o Product Support Department
730 Boul. Industriel
Blainville, QC
J7C 3V4

Appendix 1 - Drawings and Engineering Documents

A complete set of drawings and engineering documents can be found in your documentation package. These drawings provide a high level overview of your system as well as detailed drawings for piping and design basis. Contents include:

1. Application Specification
2. P&ID (Piping and Instrumentation Diagram)
3. GA (General Assembly)
4. Electrical schematic

Xebec Adsorption Inc

Xebec PSA UNIT		Application Specification
SERIAL NO		PROJECT NAME (LOCATION):
NAME:	PSA3H-G2-330A-480	Inentec

ISSUED	APRVD	CKD	DATE:	REV.:
ARO			2017.10.27	0A
PMO			2017.12.11	
	JLA		2017.12.14	

PROJECT N0.: 30784 | REV.: 0A
DOCUMENT NO.: | DCN NO.:

GENERAL

APPLICATION	H2 Purification	
REFERENCES	H17-1016	Xebec Quotation
		Purchase Order
	Y1212034	P&ID

PROCESS PERFORMANCE

FEED CONDITIONS

			COMPOSITION	
	11.1	bar(g)	H2	29.60%
	160	psi(g)	CH4	0.00%
Temperature:	40	° C	CO2	21.20%
Flow:	22	NCMH	N2	19.00%
			CO	29.60%
			H2O	0.60%

EXHAUST CONDITIONS

Pressure:	2	psig nominal

PRODUCT CONDITIONS

			COMPOSITION	
Recovery:	68%	guaranteed		
	70%	expected	H2	99.999%
Pressure:	10.0	bar(g)	CH4	Balanced
	145	psi(g)	CO2	Balanced
Temperature:	40	° C	N2	Balanced
Flow:	5	NCMH	CO	Balanced
			H2O	Balanced

MECHANICAL DESIGN DATA

PRESSURE AND TEMPERATURE

Max Operating Pressure	11.1	bar(g)	Storage Temp.	-20 / 48	°C
	160	psi(g)	Minimum Operating Temp, with Cold Weather Package.	4	°C
System Design Pressure	22.8	bar(g)			
	330	psi(g)			
Test Pressure (Vessels)	25	bar(g)			
	363	psig	TEMPERATURE - CONTROL PANEL		
Test Pressure (Pressure Piping)	34.1	bar(g)	Operating Temp.	4 / 40	°C
	495	psig			
Relief Set Pressure	22.8	bar(g)			
	330	psi(g)	Minimum Storage Temp.	-20	°C

SEISMIC: N/A
WINDLOAD: N/A
GENERAL NOTES:

VESSELS

	ADSORBENT BEDS [X]	FEED SURGE [ ]	EXHAUST SURGE [ ]	PRODUCT SURGE [ ]
QTY	9			
DIMENSIONS	3" x 72"			
VOLUME				
GENERAL NOTES				

ELECTRICAL AND CONTROL

HAZARDOUS AREA CLASSIFICATION

[X] Hydrogen	Class 1, Div 2, Group B, T3 / Class 1, Zone 2, Group IIC, T3
[ ] Methane	Class 1, Div 2, Group D, T3 / Class 1, Zone 2, Group IIA, T3
[ ] Helium	None
[ ] Special Gas	None

PROTECTION METHOD

Motor	[ ] AEX-nA	[ ]AEX-ia	[X]AEX-e	[ ]AEX-d	[ ] N/A
Junction Box "JB1"	[ ] AEX-nA	[ ]EEX-iaII	[ ]AEX-e	[ ]AEX-d	[X] N/A
Junction Box "JB2"	[ ] AEX-nA	[ ]AEX-ia	[ ]AEX-e	[ ]AEX-d	[X] N/A
Control Panel "CP1"	[ ] AEX-nA	[ ]AEX-ia	[ ]AEX-e	[ ]AEX-d	[ ] Z-Purge
Instruments	[ ] AEX-nA	[ ]AEX-ia	[ ]AEX-e	[ ]AEX-d	[ ] N/A

Clarification: nA = non sparking, ia = intrinsically safe, e = extra safety, d = flame proof

CONTROLS

[ ] Manual [X] Auto [ ] Advanced Controls [ ] Custom

Include control pannel CP1 to be installed in safe area; with VFD, thermistor and interface relay for start/stop, Faulted signal.
JB1 for proximity switch connection to customer controller
JB2 for SV connection to customer controller (120VAC coil)
HT-JB3 for heat tracing customer connection to pannel (120VAC, 30 ft of 5w/ft = 150W)

OPERATING SPEED: Nominal (cpm): 0.40 Turndown: 40% Minimum (cpm): 0.16
DRIVE SPEED: Maximum (RPM/Hz) : 0.8 rpm/60hz Minimum (RPM/Hz) : 0.08 rpm/6hz
SOLENOID POWER: [X] 120 VAC/1ph/60Hz [ ] 220 VAC/1ph/50Hz [ ] Other (24VDC)
MOTOR SPEC.: 0.16HP, 230-460VAC/3ph/60Hz
DRIVE POWER: 0.5HP, 120VAC/1ph/60Hz
GENERAL NOTES:

	FEED	PRODUCT	EXHAUST	BY-PASS	PROD-PURGE	LIGHTS
Air actuated isolation valve						
Manual isolation valve			X		X	
Solenoid isolation valve	X	X				
Pressure Gauge assembly, including:	X	X	X		X	X
(2) 1/2" Sample/purge/PT ports	X	X	X		X	
1/2" Pressure gauge isolation valve	X	X	X		X	X
Pressure gauge	X	X	X		X	X
Filter assembly, including:						
Filter	X	X				
(2) filter isolation valves	X	X				
Filter by-pass valve						
Filter drain valve	X					

VALVES AND INSTRUMENTS	Auto drain assembly, including	X					
	Drain trap	X					
	Drain trap isolation valve	X					
	Drain trap by-pass valve						
	Check Valve		X				
	Pressure Safety Valve						
	Pressure Regulator		X			X	
	Pressure Control Valve, including:						
	Pressure Transmitter						
	Drain solenoid with isolation valve						
	Low Point Drain with valve	X		X			
	Needle Valve					X	
	Multi-Variable Flowmeter						
GENERAL NOTES							

SPARE PARTS	Qty	Rotary Valves and Drive	Qty	Rest of Plant
	[ ]		[ ]	Variable Frequency Drive (VFD)
	[ ]	Motor/Gearbox and coupling	[ ]	Pressure Transmitter
	[ ]		[ ]	Pressure Control Valve repair kit
	[ ]	HV Feed Seal	[ ]	Pressure Regulator repair kit
	[ ]	HV Seal Assembly	[ ]	Solenoid Valve repair kit
	[ ]	LV Seal Assembly	[ ]	
	[ ]	HV & LV O-rings (complete set) + lube	[ ]	Filter Element
	[ ]	HV & LV U-cups (full set)	[ ]	Pressure Gauge Feed/Product/Bed
	[ ]		[ ]	Pressure Gauge Exhaust
	[ ]		[ ]	Drain Trap Service Kit
	[ ]		[ ]	Hand valve service kits (all sizes, 1 each)
	[ ]		[ ]	Pressure Relief Valve
GENERAL NOTES	A list of recommanded spare parts is available in the project book			

APPLICABLE STANDARDS		
		[X] ASME Section VIII Div 1 — [X] ANSI B31.3; Piping
		[ ] Fatigue analysis according to ASME Section VIII Div 2 (Cyclic)
		[ ] International Building Code, IBC2009, ASCE 7-05
		[ ] IEC 60529 — Degrees of Protection Provided by Enclosures
		[ ] NEC
	CE (EUROPE)	[ ] 97/23/EC — Pressure Equipment Directive
		[ ] 89/336/EEC — EMC Directive; [ ] 73/23/EEC — Low Voltage Directive
		[ ] 98/37/EEC — Machinery Directive; [ ] 94/9/EC — ATEX Directive
	GENERAL NOTES	

PAINT		
	FRAME	[X] RAL 7011 GREY – SILICONE ACRYLIC PAINT
	BEDS/SURGE TANK	[X] RAL 9016 WHITE – SILICONE ACRYLIC PAINT
	VALVE TOWER	[X] RAL 5005 BLUE – SILICONE ACRYLIC PAINT
	GENERAL NOTES	

LIFTING		
	FORKLIFT	[X] Provision for lifting by forklift
	CRANE	[X] Provision for lifting by crane
	GENERAL NOTES	

NOTES		
	IOM MANUALS	[X] 3 Copies of IOM given
	DOCUMENTATION	[ ] Extended documentation
	GENERAL NOTES	

VALVES SYMBOLS

	NORMALLY OPEN	NORMALLY CLOSED
BALL VALVE		
SOLENOID VALVE	S	S
NEEDLE VALVE		
3-WAY VALVE		
PRESSURE SAFETY VALVE (PSV)		
PSV WITH VENT PROTECTOR		
PRESSURE REGULATING VALVE		
PRESSURE CONTROL VALVE		
CHECK VALVE		

INSTRUMENTS & MISC

REDUCER	
PRESSURE INDICATOR	PI XXX
PRESSURE TRANSDUCER	PT XXX
DIFF. PRESSSURE INDICATOR	DPI XXX
INSULATE & HEAT TRACE	
FILTER & FILTER ELEMENT	
DRAIN TRAP	T
TO SAFE DRAIN (BY CUSTOMER)	
MOTOR	

ELECTRICAL/CONTROL SYMBOLS

PSA SHUTDOWN SIGNAL	PSASD
VARIABLE FREQ. DRIVE	SY 400
VFD FAULT CONTACT	XS 400
POSITION PROX. SWITCH	ZSL 904
PRESSURE CONTROLLER	PC 500
MOTOR SPEED CONTROLLER	SC 400
VALVE POSITION OPEN/CLOSED	ZC XXX
VFD FAULT STATUS	XY 400
SHAFT POSITION STATUS	ZY 904

MAJOR COMPONENTS LIST

TAG	COMPONENT DESCRIPTION
06TW01, ...06TW09	CYCLIC PRESSURE VESSELS
04FX01	COALESCING FILTER
06GX01 06GX02	ROTARY VALVE
M400	MOTOR
DT-197	DRAIN TRAP

PIPE SPECIFICATION INDEX

PREFIX		MATERIAL & SERVICE		SUFFIX	
BG	BIOGAS	SS150A	STAINLESS, 150#, SUSTAINED PRESS.	P#	P = PIPING # = DIAMETER
CO	CARBON MONOXIDE	SS150C	STAINLESS, 150#, CYCLIC PRESS.	T#	T = TUBING # = DIAMETER
EX	EXHAUST GAS	CS150A	CARBON, 150#, SUSTAINED PRESS.		
H2	HYDROGEN GAS	CS150C	CARBON, 150#, CYCLIC PRESS.		
PG	BIOMETHANE	CS300C	CARBON, 300#, CYCLIC PRESS.		
		SS300C	STAINLESS, 300#, CYCLIC PRESS.		

REV	Date	Revision Description	DSGN	DWN	APPD
2	7-MAR-2018	AS BUILT	ARO	ACA	
01	15-SEP-2017	FOR CONSTRUCTION	ARO	ARO	ARO
00	25-AUG-2017	INITIAL RELEASE	ARO	ARO	ARO

Xebec Adsorption Serial Number: PSA3H-G2-330A-480

Xebec Adsorption Project No: JOB 30784

H3200 PSA (G2 HIGH PURITY)

Dwg: Y1212034 | Rev: 2 | Sheet: 1 of 2 | Size: B

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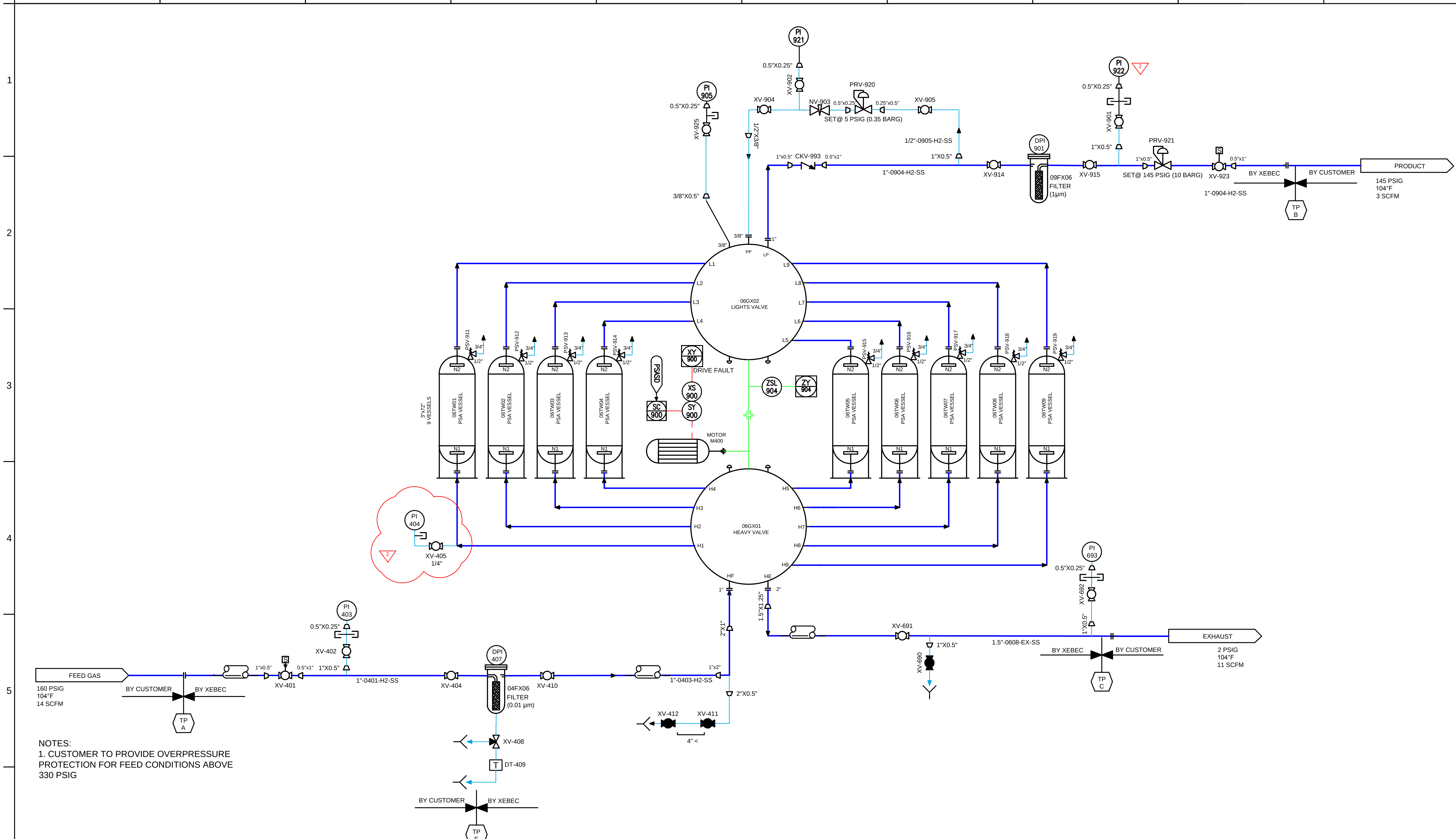

NOTES:
1. CUSTOMER TO PROVIDE OVERPRESSURE PROTECTION FOR FEED CONDITIONS ABOVE 330 PSIG

REV	Date	Revision Description	DSGN	DWN	APPD
2	7-MAR-2018	AS BUILT	ARO	ACA	
01	15-SEP-2017	FOR CONSTRUCTION	ARO	ARO	ARO
00	25-AUG-2017	INITIAL RELEASE	ARO	ARO	ARO

Xebec Adsorption Serial Number: PSA3H-G2-330A-480

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H3200 PSA (G2 HIGH PURITY)

Dwg: Y1212034 | Rev: 2 | Sheet: 2 of 2 | Size: B

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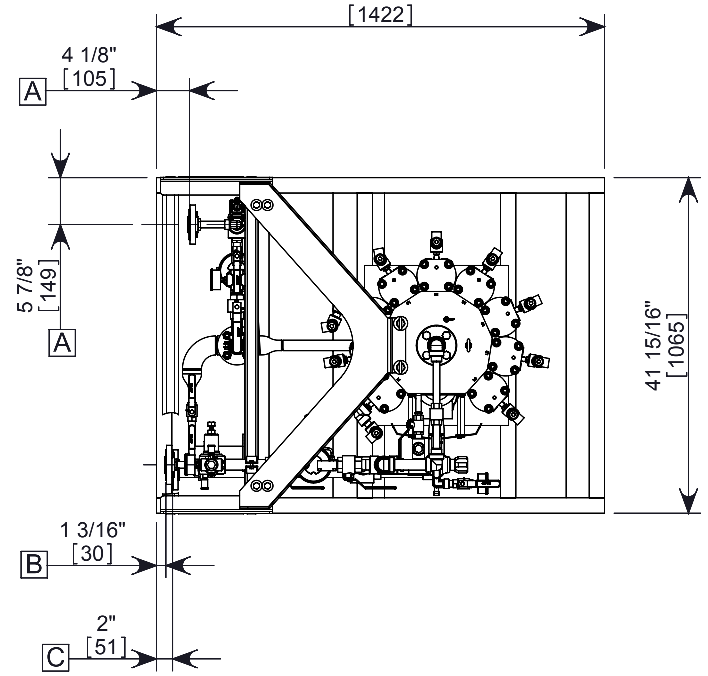
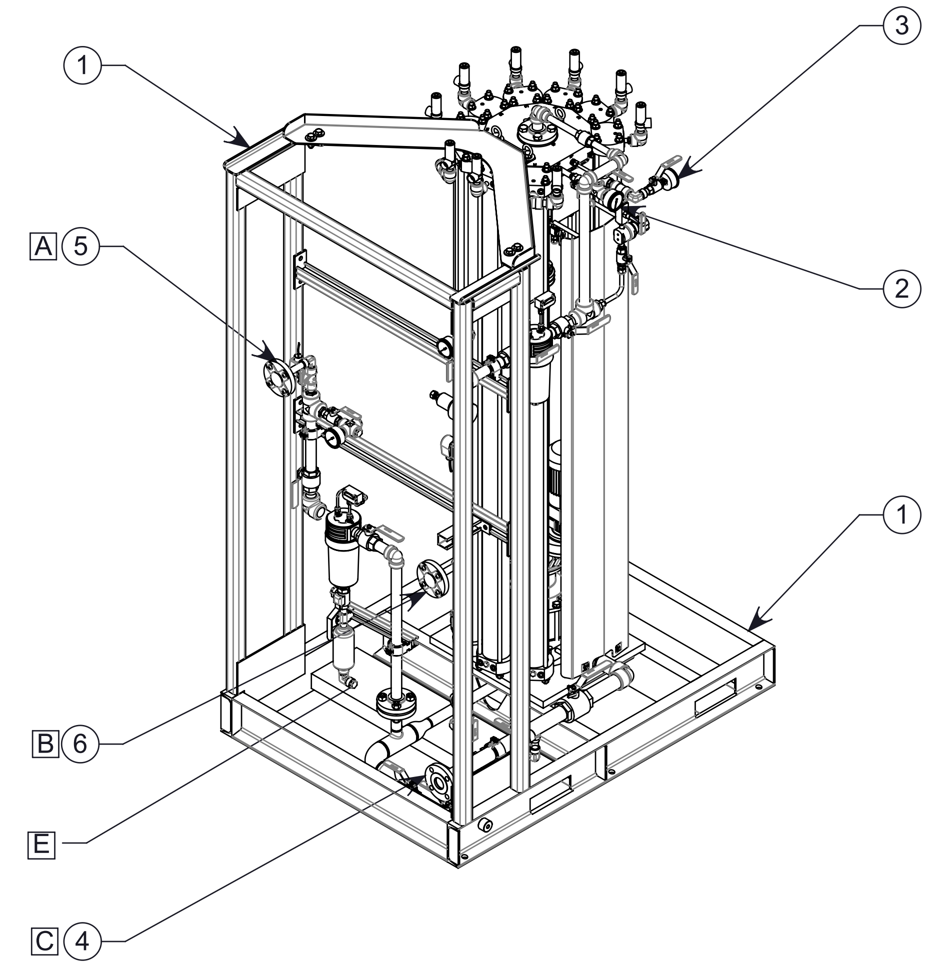
TIP IDENT.	SERVICE (DESC)	NPS	CONN. TYPE	CLASS
A	FEED	1"	RF FLANGE	300
B	PRODUCT	1"	RF FLANGE	300
C	EXHAUST	1 1/2"	RF FLANGE	150
E	FILTER DRAIN	1/2"	FNPT	

BILL OF MATERIAL

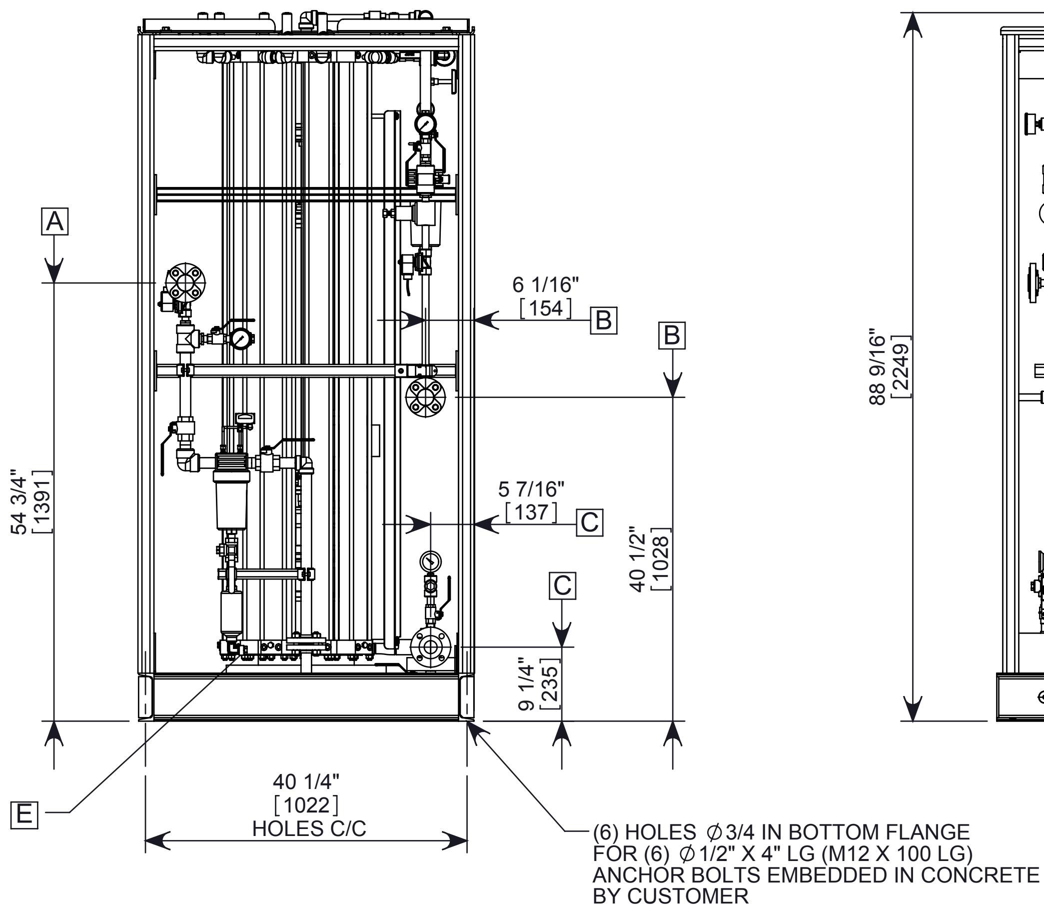
ITEM	QTY	DESCRIPTION	PART NO.
1	1	PSA BASE FABRICATION	A1212008
2	1	PS 1/2" STD BS CL3000	A107438
3	1	PRODUCT PURGE	A1212035
4	1	EXHAUST PIPING ASSEMBLY	A1212013
5	1	FEED PIPING ASSEMBLY	A1212011
6	1	PRODUCT PIPING ASSEMBLY	A1212009



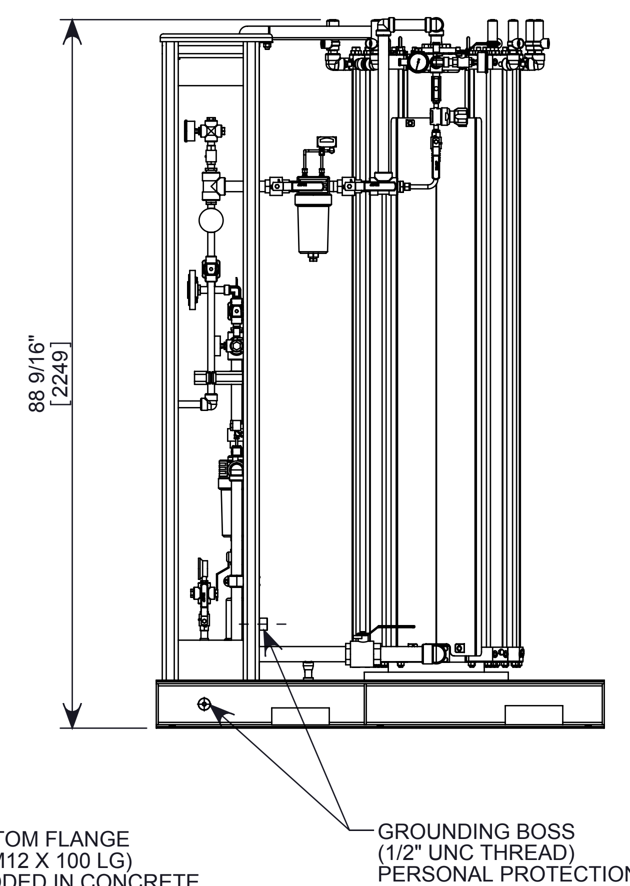
TOP VIEW



ISOMETRIC VIEW



FRONT VIEW



SIDE VIEW

NOTES:

1. ESTIMATED WEIGHT:
   EMPTY: **1908.61** lbs.
   WITH DESICCANT: **2173.61** lbs.
2. ALLOW MAINTENANCE AREA AROUND SKID: 3'-4" (1m) FRONT, 2' (0.6m) SIDES AND BACK MINIMUM
3. PRIMARY DIMENSIONS ON THE DRAWING ARE INCHES, SECONDARY SHOWN IN BRACKETS ARE IN mm
4. FLANGE LOCATIONS ARE APPROXIMATE -ALLOW FOR FIELD FIT
5. **USE P&ID Y1212034.**

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REV.	DESCRIPTION	DATE	APP.	DSS.
00	DRAWING CREATION	2017-08-12	JPL	VTA

	DRAWN	CHECKED	APPROVED
NAME :	VTA	PMO	JPL
DATE :	2017-08-10	2017-08-12	2017-08-12

TOLERANCE (INCHES):
GENERAL: ±3/4"
NOZZLES: ±1"
ANGLE: ±1/2°
(EXCEPT CHAMFER)

THIRD ANGLE PROJECTION:

SCALE 1:24

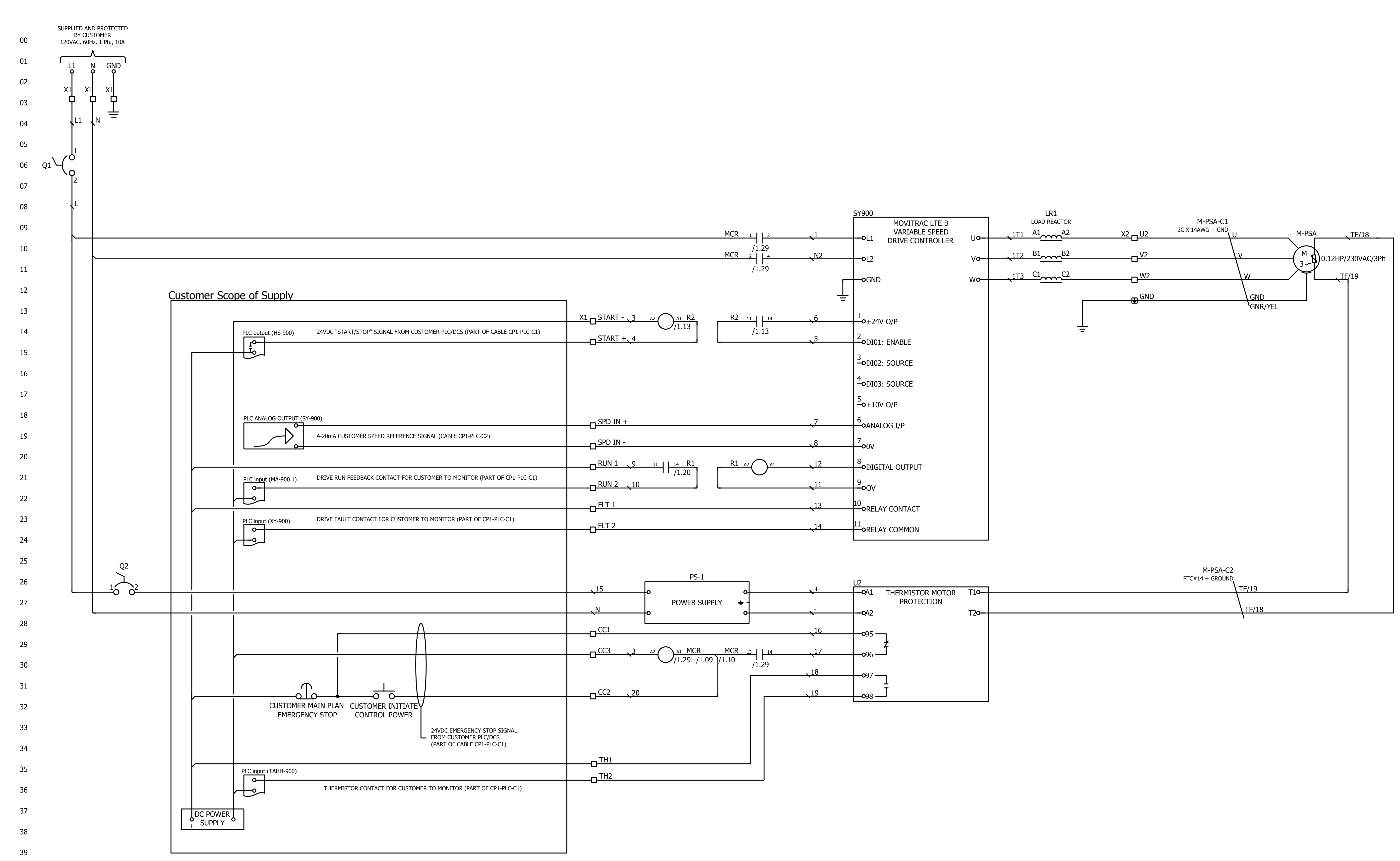
TITLE: **GENERAL ASSEMBLY PSA3H-G2-330A-480**

FORMAT: B

DWG NO. : A1212007

PAGE: 1 de 1

A1212007 - 00-EN TRAVAIL



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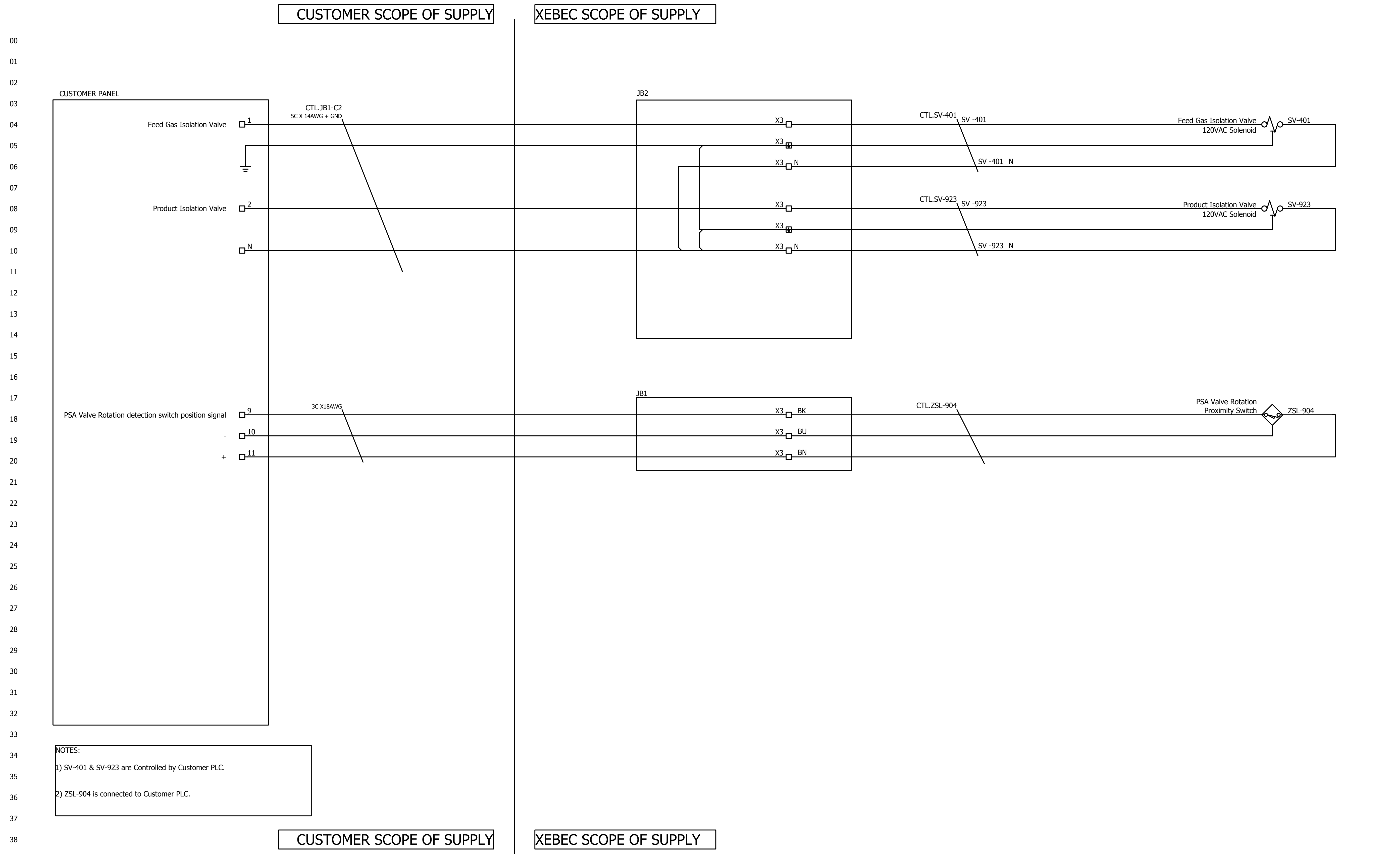
Project name / Titre du projet
Inentec
JOB: 30784

Rev.	Description	Date	Name / Nom		Date	Name / Nom
00	ISSUE FOR CUSTOMER REVIEW	2017/09/19	JLA	Creation	2017/09/19	JLA
01	ISSUE FOR FABRICATION	2017/10/18	JLA	Chk / Vér.	2017/10/18	DBA
				Appr	2017/10/19	JLA

Title / Titre
CONTROL PANEL WIRING DIAGRAM
120V/1PH

Drawing number / Numéro de dessin
E1212081

Page
1
Of/De
5



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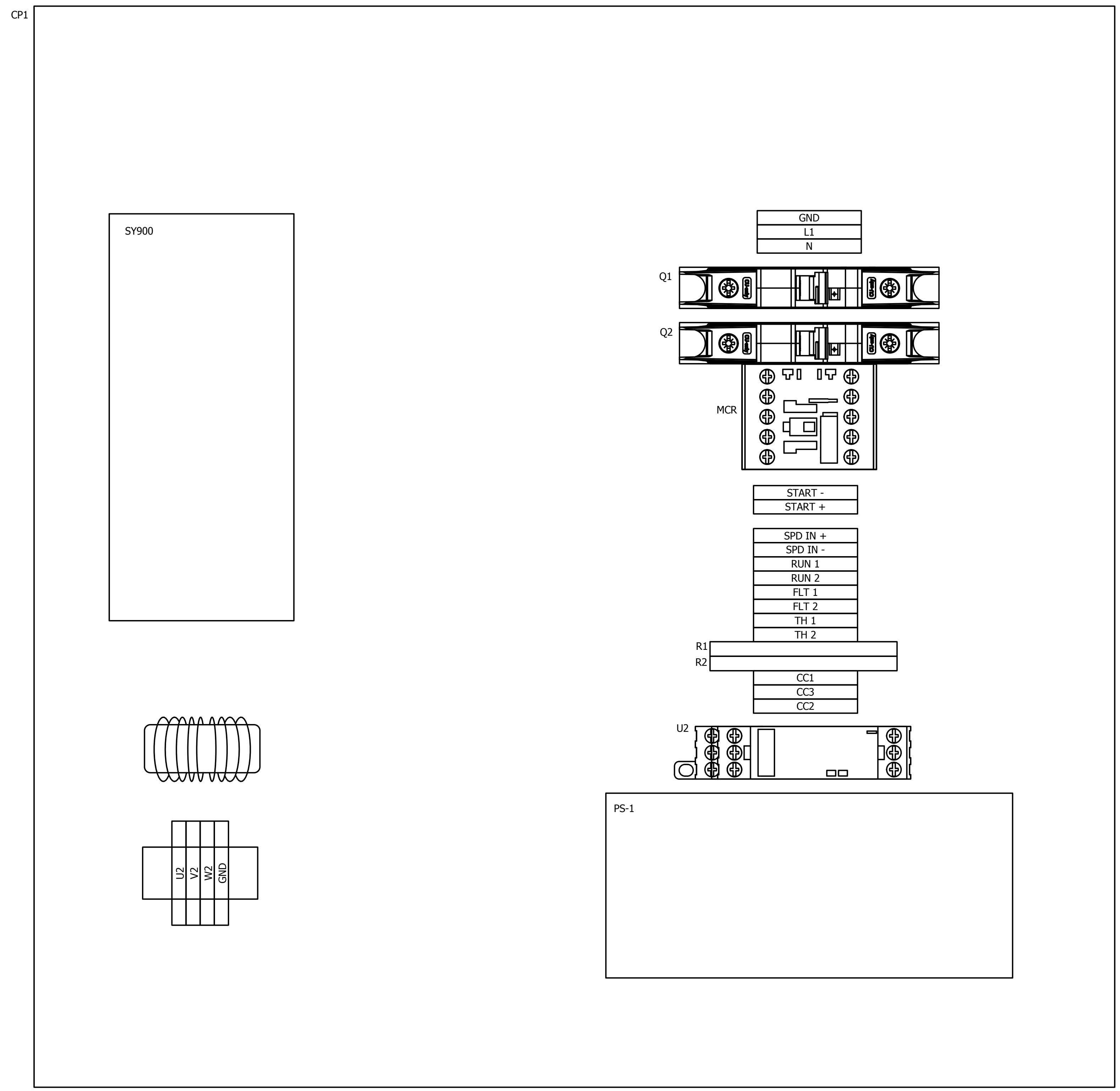
01	ISSUE FOR FABRICATION	2017/10/18		Chk / Vér.	2017/10/18	DBA
				Appr	2017/10/19	JLA

Title / Titre
JUNCTION BOXES WIRING DIAGRAM

Drawing number / Numéro de dessin
E1212081



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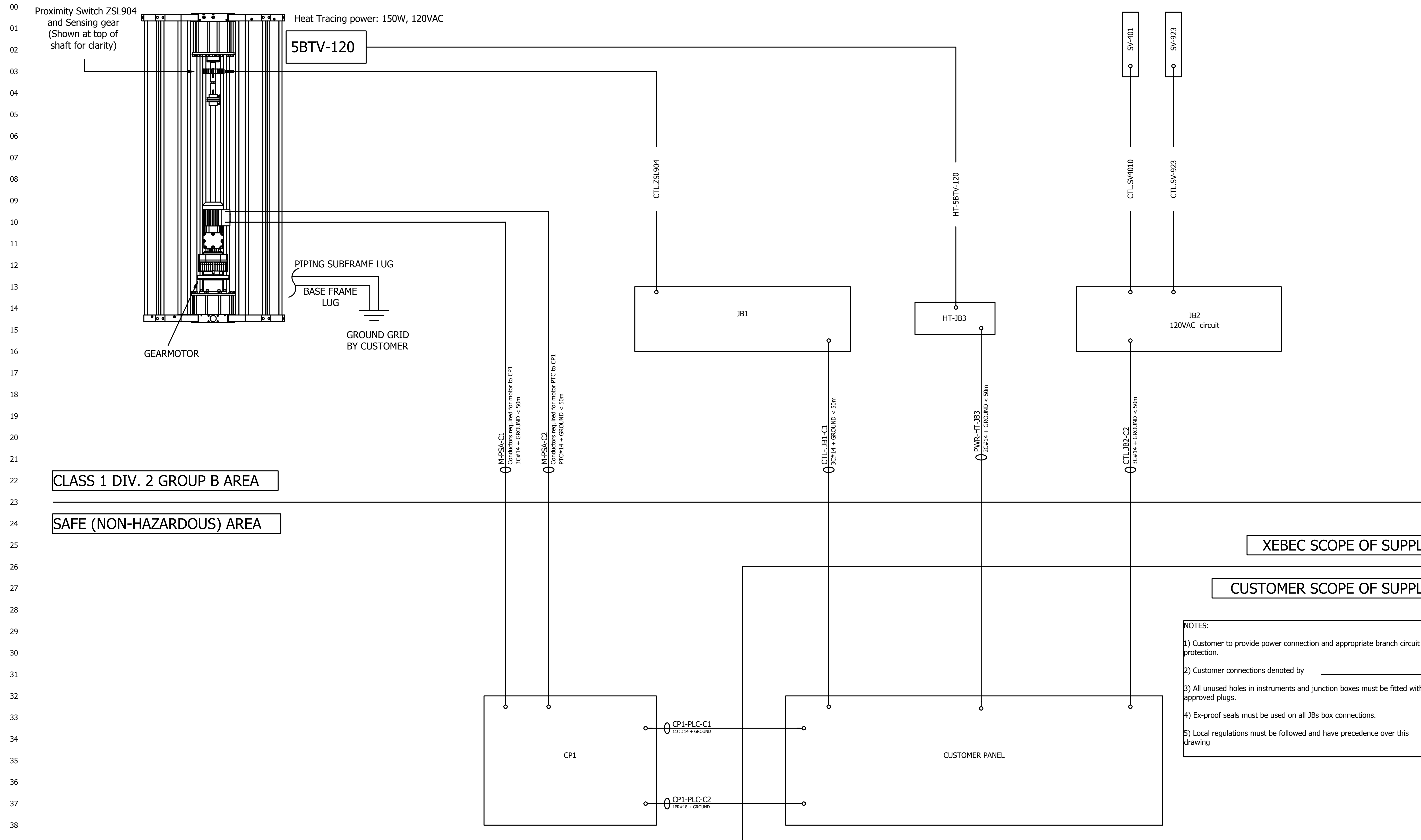
01	ISSUE FOR FABRICATION	2017/10/18		Chk / Vér.	2017/10/18	DBA
				Appr	2017/10/19	JLA

Title / Titre
CP1 PANEL LAYOUT

Drawing number / Numéro de dessin
E1212081



NOTES:

1) Customer to provide power connection and appropriate branch circuit protection.

2) Customer connections denoted by

3) All unused holes in instruments and junction boxes must be fitted with approved plugs.

4) Ex-proof seals must be used on all JBs box connections.

5) Local regulations must be followed and have precedence over this drawing

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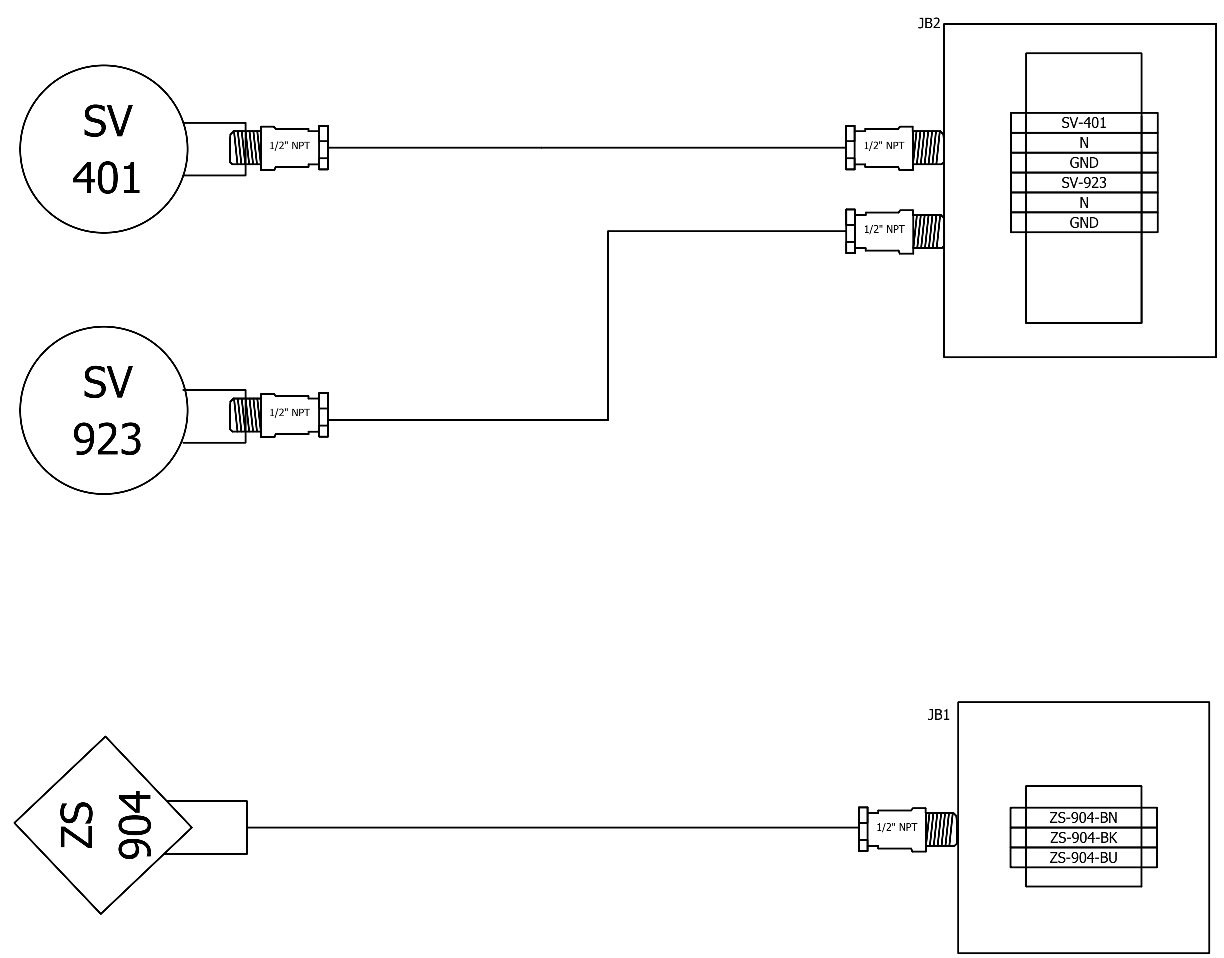
01	ISSUE FOR FABRICATION	2017/10/18		Chk / Vér.	2017/10/18	DBA
				Appr	2017/10/19	JLA

Title / Titre
INTERCONNECTION DIAGRAM

Drawing number / Numéro de dessin
E1212081



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				Appr	2017/10/19	JLA

Title / Titre
JUNCTION BOXES LAYOUT

Drawing number / Numéro de dessin
E1212081

Appendix 2 - Safety Data Sheets (SDS)

These documents provide safe handling and emergency practices pertaining to the materials in your PSA vessels and surrounding equipment.

MOLSIV ADSORBENT

XEBEC PART #P400012

1. IDENTIFICATION OF SUBSTANCE/MIXTURE FROM MANUFACTURER

1.1. Product identifier

Product Name	Molsiv Adsorbents 13X HP 10X20

1.2. Details of the supplier of the substance or mixture:

Supplier	**Manufacturer**
Xebec Adsorption Inc	
730, Boulevard Industriel	
Blainville, Québec	
Canada J7C 3V4	
Tel +1 450 979 8700	
Email : sales@xebecinc.com www.xebecinc.com	

1.3. Emergency phone number – 24hours/day, 7 days/week

Medical: 1-800-498-5701 or +1-303-389-1414	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887**

2. HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

Carcinogenicity, Category 1A, Inhalation

2.2. Label elements

GHS Label elements, including precautionary statements

2.3. Other hazards

Signal word:	Danger
Hazard Statement	May cause cancer by inhalation
Precautionary Statement	Prevention: Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Use personal protective equipment as required.
Hazards Not Otherwise Classified:	The product gets hot as it first adsorbs water.
Carcinogenicity	
NTP: Quartz (SiO_2) Known Carcinogen	14808-60-7
IARC: Quartz (SiO_2) Group 1: Carcinogenic to humans	14808-60-7
ACGIH: Quartz (SiO_2) A2: Suspected human carcinogen	14808-60-7

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Mixture

Chemical Name	**CAS #**	**Purity %**
Zeolite, cuboidal, crystalline, synthetic, non-fibrous	1318-02-1	>70.00%
Mineral Binder		<30.00%
Quartz (SiO_2)	14808-60-7	<5.00%
Note: Composition as shown represents the chemical composition of the product and is not representing the composition for inventory purposes. See Section 15 for inventory information.		

4. FIRST AID MEASURES

4.1. Description of first aid

Skin contact	Wash off with soap and plenty of water. If skin irritation persists, call a physician.
Eye Contact	Rinse immediately with plenty of water for at least 15 minutes. If eye irritation persists, consult a physician.
Ingestion	Drink at least 2 glasses of water. Obtain medical attention. Never give anything by mouth to an unconscious person.
Inhalation	Remove to fresh air. If symptoms persist, call a physician.

Indication of the immediate medical attention and special treatment needed	This product is a desiccant and generates heat as it adsorbs water. The used product can contain material of a hazardous nature. Identify that material and treat symptomatically.

5. FIRE-FIGHTING MEASURES

5.1. Extinguishing media

Suitable extinguishing media	Not combustible. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

5.2. Special hazards arising from the substance or mixture

Special danger	The product itself does not burn. The used product can retain material of a hazardous nature. Identify that material and inform the fire fighters.

5.3. Advice for fire-fighters

Special measures for the protection of fire-fighters	In the case of respirable dust and/or fumes, use self-contained breathing apparatus and dust impervious protective suit.

6. ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

IF exposed or concerned: Get medical advice/ attention.

6.2. Environmental precautions

No special environmental precautions required.

6.3. Methods and material for containment and clean up

Sweep, shovel or vacuum spilled product into appropriate containers (do not use a vacuum if material has contacted a hydrocarbon material). Pick up and arrange disposal without creating dust. Never return spills in original containers for re-use. Spilled product should be disposed of in accordance with all applicable government regulations.

6.4. Reference to other sections

See Section 8.

7. HANDLING AND STORAGE

7.1. Precautions for safe handling

Store in original container. Keep in a dry place. Store locked up.

7.2. Conditions for safe storage, including any incompatibilities

Handle and open container with care. Avoid dust formation. Avoid contact with skin and eyes. Provide an electrical ground connection during loading and transfer operations to avoid static discharge in an explosive atmosphere and to prevent persons handling the product from receiving static shocks.

7.3. Specific end use(s)

Exposure	No special requirements
Other information	No information available

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1. Control parameters

Components	**CAS-No.**	**Value**	**Control Parameters**	**Update**	**Basis**
Zeolite, cuboidal, crystalline, synthetic, non-fibrous	1318-02-1	TWA (Time Weighted Average)	1 mg/m3	2008	ACGIH: US. ACGIH Threshold Limit Values
Form of exposure: respirable fraction					
Quartz (SiO2)	14808-60-7	TWA (Time Weighted Average)	0.025 mg/m3	2008	ACGIH: US. ACGIH Threshold Limit Values
Form of exposure: respirable fraction					
Quartz (SiO2)	14808-60-7	REL: Recommended exposure limit	0.05 mg/m3	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards
Form of exposure: respirable dust					
Quartz (SiO2)	14808-60-7	TWA: Time Weighted Average	0.1 mg/m3	1989	Z1A: US. OSHA Table Z-1-A (29 CFR 1910.1000)
Form of exposure: respirable dust					
Quartz (SiO2)	14808-60-7	TWA: Time Weighted Average	2.4 millions of particles per cubic foot of air. The exposure limit is calculated from the equation, 250/(%SiO2+5), using a value of 100% SiO2. Lower percentages of SiO2	2000	Z3:US. OSHA Table Z-3 (29 CFR 1910.1000)

			will yield higher exposure limits.		
Form of exposure: Respirable					
Quartz (SiO_2)	14808-60-7	TWA: Time Weighted Average	0.1 mg/m3. The exposure limit is calculated from the equation, 10/(%SiO2+2), using a value of 100% SiO2. Lower percentages of SiO2 will yield higher exposure limits.	2000	Z3:US. OSHA Table Z-3 (29 CFR 1910.1000)
Form of exposure: Respirable					
Quartz (SiO_2)	14808-60-7	TWA: Time Weighted Average	0.3 mg/m3. The exposure limit is calculated from the equation, 30/(%SiO2+2), using a value of 100% SiO2. Lower values of % SiO2 will give higher exposure limits	2000	Z3:US. OSHA Table Z-3 (29 CFR 1910.1000)
Form of exposure: Total dust					

8.2. Exposure controls

Appropriate engineering controls	Ensure adequate ventilation, especially in confined areas.
Individual protection measures – personal protective equipment	Handle in accordance with good industrial hygiene and safety practice
Eye / Face Protection	Safety Glasses/Goggles
Skin Protection	Work uniform and gloves to prevent prolonged contact.
Respiratory Protection	In case of insufficient ventilation, wear suitable respiratory equipment. Breathing apparatus with filter: NIOSH classification N-100 or if oil/liquid aerosols are present P-100 (42 CFR 84).
Hand Protection	Protective Gloves

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	Beads or Pellets

Colour	White to Tan
Odour	None
pH	N/A
Melting Point	> 400 °C
Boiling Point	N/A
Flash Point	N/A
Density	> 2.0 g/cm3
Specific Gravity	> 1.0
Water Solubility	Insoluble
Solubility in Other Solvents	Insoluble

10. STABILITY AND REACTIVITY

Chemical Stability	Stable
Incompatible Materials	Sudden contact with high concentrations of chemicals having high heats of adsorption such as olefins, HCl, etc. When first wetted, the product can heat up to the boiling point of water. Flood with water to cool material
Hazardous decomposition products	No decomposition if used as directed. Hydrocarbons and other materials that contact the product during normal use can be retained on the product. It is reasonable to expect that decomposition products will come from these retained materials of use.

11. TOXICOLOGICAL INFORMATION

Acute Oral Toxicity	LD50 Oral: > 32,000 mg/kg. Species: Rat
Acute Inhalation Toxicity	No data available
Acute dermal Toxicity	LD50 Dermal: > 2,000 mg/kg. Species: Rabbit
Skin Irritation	Species: Rabbit. Classification: Not classified as a skin irritant in animal testing.

Eye Irritation	Species: Rabbit. Result: mild transient effects
Further Information	Note: The toxicological data has been taken from products of similar composition.

12. ECOLOGICAL INFORMATION

12.1. Eco-toxicity Effects

Toxicity to fish:	No data available
Toxicity to Daphnia and Other Aquatic Invertebrates	No data available
Toxicity to Algae	No data available

12.2. Elimination Information (Persistence and degradability)

Bioaccumulative potential	No data available
Mobility	No data available
Biodegradability	No data available
Other Information	No information on ecology available

13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste Treatment Methods:	Dispose of contents/ container to an approved waste disposal plant.
Additional Information :	This product (in its fresh unused state) is not listed by generic name or trademark name in the U.S. EPA's RCRA regulations and does not possess any of the four identifying characteristics of hazardous waste (ignitability, corrosivity, reactivity or toxicity). Materials of a hazardous nature that contact the product during normal use may be retained on this product. The user of the product must identify the hazards associated with the retained material in order to assess the waste disposal options.

14. TRANSPORT INFORMATION

DOT	Not dangerous goods
TDG	Not dangerous Goods
IMDG	Not dangerous goods

15. REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific to the substance or mixture:

US. Toxic Substances Control Act	On the inventory, or in compliance with the inventory	
Australia. Industrial Chemical (Notification and Assessment) Act	On the inventory, or in compliance with the inventory	
Canada. Canadian Environmental Protection Act (CEPA). Domestic Substances List (DSL)	All components of this product are on the Canadian DSL.	
Japan. Kashin-Hou Law List	On the inventory, or in compliance with the inventory	
Korea. Toxic Chemical Control Law (TCCL) List	On the inventory, or in compliance with the inventory	
Philippines. The Toxic Substances and Hazardous and Nuclear Waste Control Act	On the inventory, or in compliance with the inventory	
China. Inventory of Existing Chemical Substances	On the inventory, or in compliance with the inventory	
New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand	On the inventory, or in compliance with the inventory	
Note: Zeolites are considered Statutory Mixtures for Inventory purposes,		
SARA 302 Components	No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.	Quartz (SiO_2) 14808-60-7
SARA 313 Components	This material does not contain any chemical components with known CAS numbers that exceed the	Quartz (SiO_2) 14808-60-7

	threshold (De Minimis) reporting levels established by SARA Title III, Section 313.	
SARA 311/312 Hazards	Chronic Health Hazard	
California Prop. 65	WARNING! This product contains a chemical known to the State of California to cause cancer.	Quartz (SiO_2) 14808-60-7
Massachusetts RTK		Quartz (SiO_2) 14808-60-7
New Jersey RTK		Quartz (SiO_2) 14808-60-7
Pennsylvania RTK		Quartz (SiO_2) 14808-60-7
WHMIS Classification	D2A: Very Toxic Material Causing Other Toxic Effects. This product has been classified according to the hazard criteria of the CPR and the MSDS contains all of the information required by the CPR	

16. OTHER INFORMATION

Health Hazard	HMIS III: 1Chronic Health Hazard NFPA: 1
Flammability	HMIS III: 0 NFPA: 0
Physical Hazard	HMIS III: 1
Instability	NFPA: 1
Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.	
Abbreviations and Acronyms	
GHS	Globally Harmonized System of Classification and Labelling of Chemicals.
CLP	EU regulation (EC) No 1272/2008 on classification, labelling and packaging of chemical substances and mixtures.
CAS	Chemical Abstracts Service (division of the American Chemical Society).
EINECS	European Inventory of Existing Commercial chemical Substances.
IARC	International agency for research on cancer.

RID	European Rail Transport.
IMDG	International Maritime Code for Dangerous Goods.
IATA	International Air Transport Association.
OSHA	The United States Occupational Safety and Health Administration.
DSD	Dangerous Substance Directive (67/548/EEC).
TSCA	Toxic Substances Control Act, The American chemical inventory.
RCRA	The Resource Conservation and Recovery Act of United States.
CERCLA	The Comprehensive Environmental Response, Compensation, and Liability Act of United States.
SARA/TITLE III	The Emergency Planning and Community Right-to-Know Act.
DSL	Domestic Substances List, The Canadian chemical inventory.
AICS	The Australian Inventory of Chemical Substances.
ECL	Existing Chemicals List, the Korean chemical inventory.
PICCS	Philippine Inventory of Chemicals and Chemical Substances.
MITI	Ministry of International Trade and Industry.
IECSC	Inventory of existing chemical substances in China.

Notice to End User:

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

ACTIVATED ALUMINA

XEBEC PART #P400016 / M001-210 / 213 / 214

1. IDENTIFICATION OF THE SUBSTANCE/MIX FROM THE MANUFACTURER

1.1. Product Identifier

Product Name	Aluminium Oxide
Molecular Formula	$A1_2$-nH_2 (0<n≤ 0.8)
Molecular Weight	102
CAS # / EC #	1344-28-1 / 215-691-6

1.2. Relevant identified uses of the substance or mixture and uses advised against:

Relevant Identified Uses	Industrial abrasive use
Uses Advised Against	No information available

1.3. Details of the supplier of the substance or mixture:

Supplier	**Manufacturer**
Xebec Adsorption Inc	
730, Boulevard Industriel	
Blainville, Québec	
Canada J7C 3V4	
Tel +1 450 979 8700	
Email : sales@xebecinc.com www.xebecinc.com	

1.4. Emergency Phone Number

(86) 131 7327 3855	Only office hours are available

2. HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

The product is not classified as hazardous under Regulation (EC) No 1272/2008[CLP].
The product is not classified as hazardous under Council Directive 67/548/EEC.

2.2. Label Elements

No label information available. The product is not classified as hazardous under Regulation (EC) No 1272/2008 [CLP].

2.3. Other Hazards

No information available

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Mixture

Chemical Name	**EC #**	**CAS #**	**Purity %**	**Synonyms**
Activated Alumina	215-691-6	1344-28-1	≥93%	Alumina Oxide, AA

4. FIRST AID MEASURES

4.1. Description of first aid

Skin contact	Wash skin with soap and copious amounts of water.
Eye Contact	Flush eyes with copious amounts of water.
Ingestion	If swallowed, wash out mouth with water provided person is conscious.
Inhalation	If inhaled, remove to fresh air. If not breathing give artificial respiration.
Most important symptoms and effects, both acute and delayed	May cause minor irritation to lungs & eyes.
Indication of the immediate medical attention and special treatment needed	Persons with pre-existing skin, eye, or respiratory disease may be at increased risk from the irritant or allergic properties of this material. Attending physician should treat exposed patients symptomatically.

5. FIRE-FIGHTING MEASURES

5.1. Extinguishing media

Suitable extinguishing media	Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

5.2. Special hazards arising from the substance or mixture

Special danger	Not combustible

5.3. Advice for fire-fighters

Special measures for the protection of fire-fighters	Wear self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes.

6. ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures.

Avoid inhalation of dusts. Ensure adequate ventilation.

6.2. Environmental precautions

Do not allow material to be released into the environment without the proper governmental permits.

6.3. Methods and material for containment and clean up

Sweep up, place in a bag and hold for waste disposal. Avoid raising dust.

6.4. Reference to other sections

See sections 7, 8, and 13

7. HANDLING AND STORAGE

7.1. Precautions for safe handling

Do not breathe dust. Provide appropriate exhaust ventilation at places where dust is formed.

7.2. Conditions for safe storage, including any incompatibilities

Store in a tightly closed container. Store in a cool, dry, well-ventilated area away from incompatible substances.

7.3. Specific end use(s)

Exposure	No special requirements
Other information	No information available

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1. Control parameters

Exposure limit values: CAS #: 1344-28-1

	Belgium	**Germany (DFG)**	**France**	**Spain**	
Limit value – Eight (8) hours	10 mg/m^3	4 inhalable aerosol mg/m^3 1.5 respirable aerosol mg/m^3	10 respirable aerosol mg/m^3	10 inhalable aerosol mg/m^3 5 respirable aerosol mg/m^3	
Limit Value – Short Term	2 mg/m^3				

	Hungary	**Sweden**	**Switzerland**	**Denmark**	**Poland**
Limit value – Eight (8) hours	6 respirable aerosol mg/m^3	5 inhalable aerosol mg/m^3 2 respirable aerosol mg/m^3	3 respirable aerosol mg/m^3	5 inhalable aerosol mg/m^3 2 respirable aerosol mg/m^3	2
Limit Value – Short Term				10 inhalable aerosol mg/m^3 4 respirable aerosol mg/m^3	16

	USA	OSHA	United Kingdom		
Limit value – Eight (8) hours	15 total dust mg/m^3 5 inhalable dust mg/m^3	15 total dust mg/m^3 5 inhalable dust mg/m^3	10 inhalable aerosol mg/m^3 4 respirable aerosol mg/m^3		
Limit Value – Short Term					

8.2. Exposure controls

Appropriate engineering controls	Use adequate general or local exhaust ventilation to keep airborne concentrations below the permissible exposure limits.
Individual protection measures – personal protective equipment	
Eye / Face Protection	Chemical safety glasses and face protection
Skin Protection	Wear compatible chemical-resistant gloves and clothing to prevent skin exposure.
Respiratory Protection	Use breathing protection with high concentrations.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	Solid
Colour	White
Odour	Odourless
pH	7-8, 100 g/l water, at 20 °C
Melting Point	2050 °C
Boiling Point	2980 °C
Flash Point	Not applicable
Vapour Pressure	1 mmHg at 2158 °C

Density	0.75 g/cm^{3}, at 20 °C
Bulk Density	ca. 750 kg/m^{3}
Water Solubility	Insoluble at 20 °C
Partition Coefficient, n-octanol/water	No information available
Auto-ignition Temperature (^{0}C)	Not applicable
Explosive Properties	Not applicable

9.2. Other information

This SDS covers all sphere sizes and packaged sizes – 1.5 to 3 mm; 4 to 6 mm (50lb box); 4 to 6 mm (330lb drum)

4 to 6 mm (Supersac)

10. STABILITY AND REACTIVITY

Reactivity	No information available
Chemical Stability	Stable under recommended temperatures and pressures.
Possibility of hazardous reaction	Exothermic reaction with water.
Conditions to avoid	Exposure to moisture.
Incompatible Materials	Strong acids; strong bases; chlorine trifluoride; ethylene oxide; halogenated hydrocarbon; oxygen difluoride; sodium nitrate; vinyl compounds.
Hazardous decomposition products	No information available

11. TOXICOLOGICAL INFORMATION

Toxicokinetics, metabolism and distribution:	No information available
Acute Toxicity	No information available
Skin Corrosion / Irritation	Skin, rabbit : no irritation
Serious Eye Damage / Irritation	Eyes, rabbit : no irritation

Respiratory or Skin Sensitization	No information available
CMR Effects (Carcinogenicity, Mutagenicity and Toxicity for Reproduction)	A4: Not classifiable as a human carcinogen.(HSDB)

STOT – Single Exposure and Repeated Exposure	No information available
Additional Information	RTECS #BD1200000

12. ECOLOGICAL INFORMATION

12.1. Toxicity

No information available

12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

No information available

12.4. Mobility in soil

No information available

12.5. Results of PBT et vPvB assessment

No information available

12.6. Other effects

No information available

13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste Treatment Methods:	Observe all federal, state, and local environmental regulations.
Additional Information :	Disposal must be made according to official regulations.

14. TRANSPORT INFORMATION

Land Transport (ADR/RID) :	This product is not regulated as a hazardous material or dangerous goods for transportation.
Sea Transport (IMDG-Code):	This product is not regulated as a hazardous material or dangerous goods for transportation.
Air Transport (ICAO-TI / IATA-DGR):	This product is not regulated as a hazardous material or dangerous goods for transportation.

15. REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific to the substance or mixture:

CAS #	EU-EINECS	US-TSCA	Canada-DSL	Australia-AIC S	Korea-E CL
1344-28-1	Listed	Listed	Listed	Listed	Listed

15.2. Chemical Safety Assessment:

A chemical safety assessment has not been carried out for the substance by the manufacturer.

16. OTHER INFORMATION

Date of this revision	01/15/2016
Abbreviations and Acronyms	
GHS	Globally Harmonized System of Classification and Labelling of Chemicals.
CLP	EU regulation (EC) No 1272/2008 on classification, labelling and packaging of chemical substances and mixtures.
CAS	Chemical Abstracts Service (division of the American Chemical Society).
EINECS	European Inventory of Existing Commercial chemical Substances.
IARC	International agency for research on cancer.
RID	European Rail Transport.

IMDG	International Maritime Code for Dangerous Goods.
IATA	International Air Transport Association.
OSHA	The United States Occupational Safety and Health Administration.
DSD	Dangerous Substance Directive (67/548/EEC).
TSCA	Toxic Substances Control Act, The American chemical inventory.
RCRA	The Resource Conservation and Recovery Act of United States.
CERCLA	The Comprehensive Environmental Response, Compensation, and Liability Act of United States.
SARA/TITLE III	The Emergency Planning and Community Right-to-Know Act.
DSL	Domestic Substances List, The Canadian chemical inventory.
AICS	The Australian Inventory of Chemical Substances.
ECL	Existing Chemicals List, the Korean chemical inventory.
PICCS	Philippine Inventory of Chemicals and Chemical Substances.
MITI	Ministry of International Trade and Industry.
IECSC	Inventory of existing chemical substances in China.

Data Sources:

Merck Index. International Chemical Safety Cards (ICSC)

Notice to End User:

Companies should use this information only as a supplement to other information gathered by them, and should make independent judgment on the suitability of this information to ensure proper use and protect the health and safety of employees. This information is furnished without warranty, and any use of the product not in conformance with this Safety Data Sheet, or in combination with any other product or process, is the responsibility of the user.

ZEOLUM MOLDED (GROUP 1)

XEBEC PART #P400025

1. IDENTIFICATION OF SUBSTANCE/MIXTURE FROM MANUFACTURER

1.1. Product identifier

Product Name	Zeolum Molded (Group 1), Zeolum Beads, Pellets, Zeolites
Molecular Formula	
Molecular Weight	
CAS # / EC #	

1.2. Relevant identified uses of the substance or mixture and uses advised against:

Relevant Identified Uses	Adsorbent, General Industrial Products
Uses Advised Against	No information available

1.3. Details of the supplier of the substance or mixture:

Supplier	**Manufacturer**
Xebec Adsorption Inc	
730, Boulevard Industriel	
Blainville, Québec	
Canada J7C 3V4	
Tel +1 450 979 8700	
Email : sales@xebecinc.com www.xebecinc.com	

1.4. Emergency phone number

CHEMTREC 1-800-424-9300 OR 1-703-527-3887	

2. HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

GHS Classification – not classified	**Not classified**
Hazard Symbol	**None**
Signal Word	**None**
Hazard Statements	**None**

2.2. Label elements

EUH210 – SDS available on demand

2.3. Other hazards

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Mixture

Chemical Name	**OSHA Hazardous**	**CAS #**	**Purity %**	**Synonyms**
Zeolites	Yes	1318-02-1	65-100	Zeolum Pellets, Zeolum Beads
Kaolin	Yes	1332-58-7	0-30	

4. FIRST AID MEASURES

4.1. Description of first aid

Skin contact	Remove contaminated clothing and shoes. Wash with plenty of water, for at least 15 minutes. Seek medical attention if irritation develops or persists. Launder contaminated clothing and shoes before re-use.
Eye Contact	Hold eyelids open and flush with a steady, gentle stream of water for at least 15 minutes. Seek medical attention if irritation develops or persists or if visual changes occur.
Ingestion	Do not induce vomiting. If victim is conscious and alert, give 1-2 glasses of water to drink. Do not give anything by mouth to an unconscious person. Seek immediate medical attention. Do not leave victim unattended.

Inhalation	If respiratory irritation or distress occurs, remove victim to fresh air. Seek medical attention if respiratory irritation or distress continues.
Most important symptoms and effects, both acute and delayed	
Note to Physician	All treatments should be based on observed signs and symptoms of distress in the patient. Consideration should be given to the possibility that overexposure to materials other than this product may have occurred. Treat symptomatically. No specific antidote available.

5. FIRE-FIGHTING MEASURES

5.1. Extinguishing media

Suitable extinguishing media	Water spray, fog, dry chemical, foam, CO2

5.2. Special hazards arising from the substance or mixture

Special danger	Closed containers may rupture due to buildup of pressure when exposed to extreme heat.

5.3. Advice for fire-fighters

Special measures for the protection of fire-fighters	Firefighters should wear NIOSH/MSHA-approved self-contained breathing apparatus and full protective clothing. Cool containers exposed to fire with water.

5.4. Hazardous decomposition materials under fire conditions

	Metallic Oxides

6. ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Wear appropriate protective gear for the situation. (See Personal Protection Information in Section 8).

6.2. Environmental precautions

Do not flush to drain. Spills may be reportable to the National Response Center (800-424-/8802) and to state and/or local agencies.

6.3. Methods and material for containment and clean up

Extinguish or remove all sources of ignition. Sweep up and place in an appropriate closed container. Clean up residual material by washing area with water. Collect washings for disposal. Spills may be reportable to the National Response Center (800-424-8802) and to state and/or local agencies.

6.4. Reference to other sections

7. HANDLING AND STORAGE

7.1. Precautions for safe handling

Handle material with suitable protection (See Section8). Handle with adequate ventilation. Avoid breathing dusts. Avoid contact with eyes, skin and clothing. **Caution: Pressure may build up in drum where severe temperature changes have occurred. Slowly open top bung hole to relieve any pressure before removing drum lid.**

7.2. Conditions for safe storage, including any incompatibilities

General area dilution/exhaust ventilation. Store upright in a cool, dry, well ventilated area out of direct sunlight. Keep away from heat, open flames and ignition sources. Keep container tightly closed. Material will absorb moisture from the air. Do not reuse container.

7.3. Specific end use(s)

Exposure	
Other information	

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1. Engineering measures

Set up hand-wash station and eyewash station near work area. General area dilution/exhaust ventilation.

8.2. Exposure limits

(where: TWA = Time-Weighted Average (8-hour exposure), STEL = Short-Term Exposure Limit (15-minute time-weighted average exposure)

Kaolin, total dust	15mg/M3 – OSHA PEL
Kaolin, respirable fraction	5mg/M3 – OSHA PEL
Kaolin, respirable fraction	3mg/M3 – ACGIH TWA
Particulates, not otherwise classified, inhalable particulate	10 mg/M3 – ACGIH TWA

Particulates, not otherwise regulated, total dust	15mg/M3 – OSHA PEL
Particulates, not otherwise regulated, respirable fraction	5mg/M3 – OSHA PEL 3mg/M3 – ACGIH TWA

8.3. Personal protection measures

Individual protection measures – personal protective equipment	The following general measures should be taken when working or handling this material: 1) Do not store, use, and/or consume foods, beverages, tobacco products, or cosmetics in areas where this material is stored. 2) Wash hands and face carefully before eating, drinking, using tobacco, applying cosmetics, or using the toilet. 3) Wash exposed skin promptly to remove accidental splashes of contact with this material
Eye / Face Protection	Safety glasses with side shields, goggles or face shield are recommended.
Skin Protection	Skin contact should be minimized through the use of chemical-resistant gloves and boots, and suitable protective clothing.
Respiratory Protection	When respirators are required, select NIOSH/MSHA approved equipment based on actual or potential airborne concentrations and in accordance with regulatory standards and/or industrial recommendations. Dust/mist filtering respirator is recommended.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	Beads or pellets
Colour	Beige
Odour	Odourless
pH	9-11.5 (5% aq.slurry)
Melting Point	No information available

Boiling Point	No information available
Flash Point	No information available
Vapour Pressure	Negligible
Density	No information available
Bulk Density	0.5-0.9 g/cm^3
Water Solubility	Insoluble
Partition Coefficient, n-octanol/water	No information available
Auto-ignition Temperature (^{0}C)	No information available
Explosive Properties	No information available

9.2. Other information

10. STABILITY AND REACTIVITY

Reactivity	
Chemical Stability	This material is stable under normal handling and storage conditions described in Section 7.
Possibility of hazardous reaction	
Conditions to avoid	Heat, open flames, sparks
Incompatible Materials	Strong oxidizing agents
Hazardous decomposition products	Metallic oxides

11. TOXICOLOGICAL INFORMATION

Toxicokinetics, metabolism and distribution:	
Acute Toxicity	Oral: LD50 > 5000 mg/kg (rat) (Data for zeolites) Dermal: LD50 > 2000 mg/kg (rabbit) (Data for zeolites).

	Inhalation: LC50 > 3.35 mg/L (4-hour exposure, rat) (Data for zeolites)
Skin Corrosion / Irritation	Non-irritating (rabbit) (Data for zeolites).
Eye corrosion / Irritation	Minimal irritant (rabbit) (Data for zeolites)
Respiratory or Skin Sensitization	Not a sensitizer (guinea pig) (Data for zeolites).
CMR Effects (Carcinogenicity, Mutagenicity and Toxicity for Reproduction)	Genetic: Negative in the Ames test and mouse lymphoma study. (Data for zeolites) Carcinogenicity: This product does not contain any substances that are considered by OSHA, NTP, IARC or ACGIH to be “probable” or “suspected” human carcinogens. No Observed Adverse Effect Level (NOAEL) ≥ 60 mg/kg/bodyweight (rat.) (Data for zeolites). Reproductive: No Observed Adverse Effect Level (NOAEL) ≥ 1251 mg/kg/bodyweight (rat.) (Data for zeolites)

STOT – Single Exposure and Repeated Exposure	No information available
Additional Information	

12. ECOLOGICAL INFORMATION

12.1. Eco-Toxicity

96hr LC50 = 1800-3200 mg/L (guppy) 48hr EC50 > 100 mg/L (daphnia magna) 72hr EC50 > 100 mg/L (algae, growth rate) (All data for zeolites)

12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

12.4. Mobility in soil

No information available

12.5. Results of PBT et vPvB assessment

12.6. Other effects

13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Residual waste:	Chemical additions, processing or otherwise altering this material may make the waste management information presented in this MSDS incomplete, inaccurate or otherwise inappropriate. Please be advised that state and local requirements for waste disposal may be more restrictive or otherwise different from Federal laws and regulations. Consult state and local regulations regarding the proper disposal of this material.
Contaminated vessels and containers:	Rinse containers before disposal. Do not allow rinsate to enter the water systems. EPA Hazardous Waste = No

14. TRANSPORT INFORMATION

Land Transport (ADR/RID) :	
Sea Transport (IMDG-Code):	
Air Transport (ICAO-TI/IATA-DGR):	

15. REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific to the substance or mixture:

JAPAN-MITI	**EU-REACH**	**US- TSCA**	**Canada-DSL**	**Australia-AICS**	**Korea-KECL**
YES	NO	YES	YES	YES	YES

Where: Yes = all ingredients are listed on the inventory, Exempt = All ingredients are either on the inventory or exempt from the requirements of listing, No = Not determined, or one or more ingredients are not on the inventory and are not exempt from listing.

15.2. SARA Title III Hazard Classes:

Fire Hazard	Reactive Hazard	Release of Pressure	Acute Health Hazard	Chronic Health Hazard
NO	NO	NO	YES	NO
SARA Extremely Hazardous Substances/CERCLA Hazardous Substances:				NONE
California Proposition 65: This product does not contain any components that are regulated under Proposition 65.				

16. OTHER INFORMATION

National Fire Protection Association (NFPA) Hazard Ratings	Health: 1 (slight); Flammability: 0 (minimal); Reactivity: 0 (minimal)
National Paint and Coatings Hazardous Materials Identification System (HMIS) Hazard Ratings:	Health: 1 (slight); Flammability: 0 (minimal); Physical hazard: 0 (minimal)
Abbreviations and Acronyms	
GHS	Globally Harmonized System of Classification and Labelling of Chemicals.
CLP	EU regulation (EC) No 1272/2008 on classification, labelling and packaging of chemical substances and mixtures.
CAS	Chemical Abstracts Service (division of the American Chemical Society).
EINECS	European Inventory of Existing Commercial chemical Substances.
IARC	International agency for research on cancer.
RID	European Rail Transport.
IMDG	International Maritime Code for Dangerous Goods.
IATA	International Air Transport Association.
OSHA	The United States Occupational Safety and Health Administration.
DSD	Dangerous Substance Directive (67/548/EEC).
TSCA	Toxic Substances Control Act, The American chemical inventory.

RCRA	The Resource Conservation and Recovery Act of United States.
CERCLA	The Comprehensive Environmental Response, Compensation, and Liability Act of United States.
SARA/TITLE III	The Emergency Planning and Community Right-to-Know Act.
DSL	Domestic Substances List, The Canadian chemical inventory.
AICS	The Australian Inventory of Chemical Substances.
ECL	Existing Chemicals List, the Korean chemical inventory.
PICCS	Philippine Inventory of Chemicals and Chemical Substances.
MITI	Ministry of International Trade and Industry.
IECSC	Inventory of existing chemical substances in China.

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